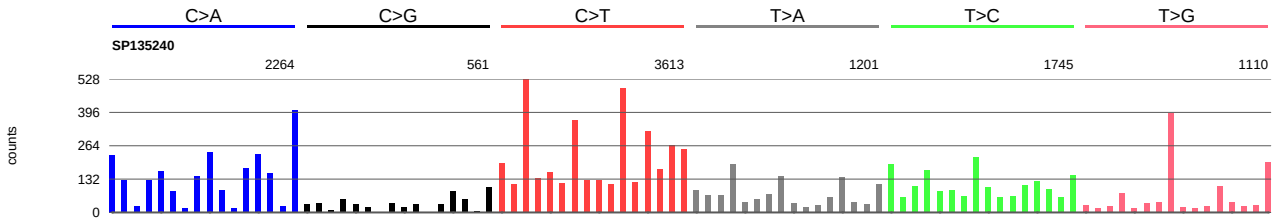


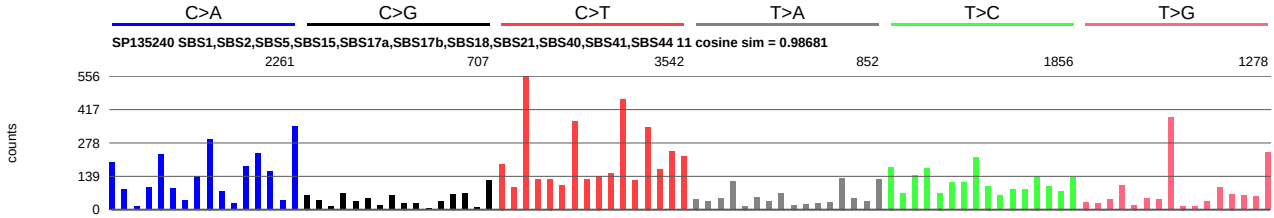


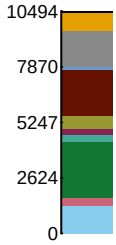


Signature
- SBS44
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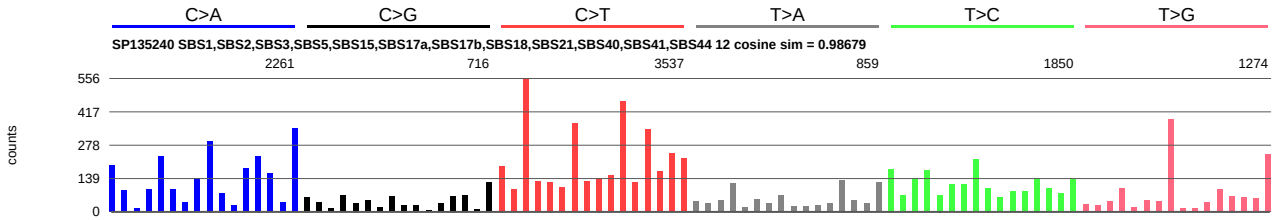
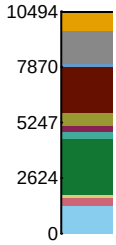
- SBS40
- SBS21
- SBS18
- SBS17b
- SBS17a
- SBS15
- SBS5
- SBS2
- SBS1

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- SBS44
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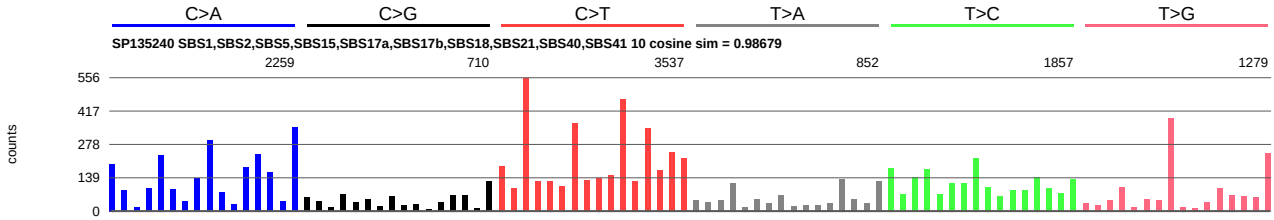
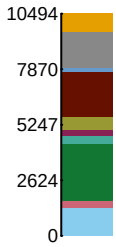
- SBS1




Signature
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- SBS17b
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- SBS15
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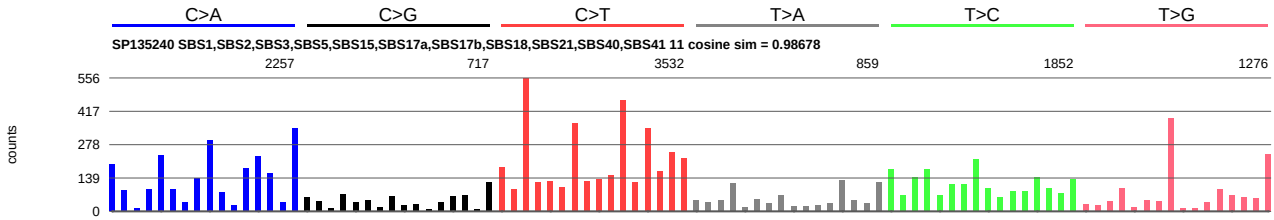
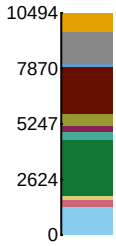
- SBS2
- SBS1




Signature
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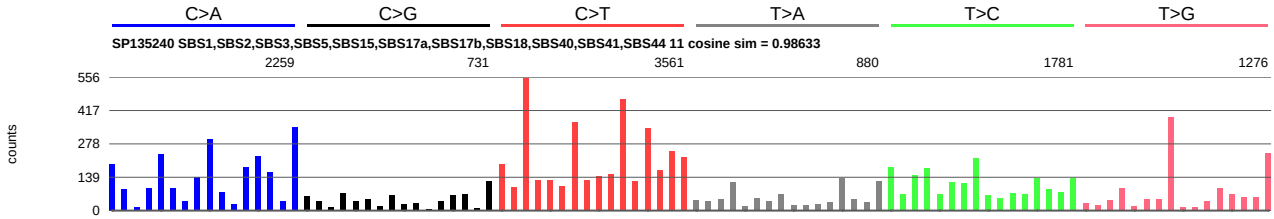
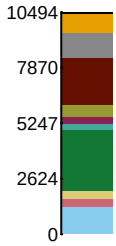
- SBS1




Signature
- SBS44
- SBS41
- SBS40
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- SBS17b
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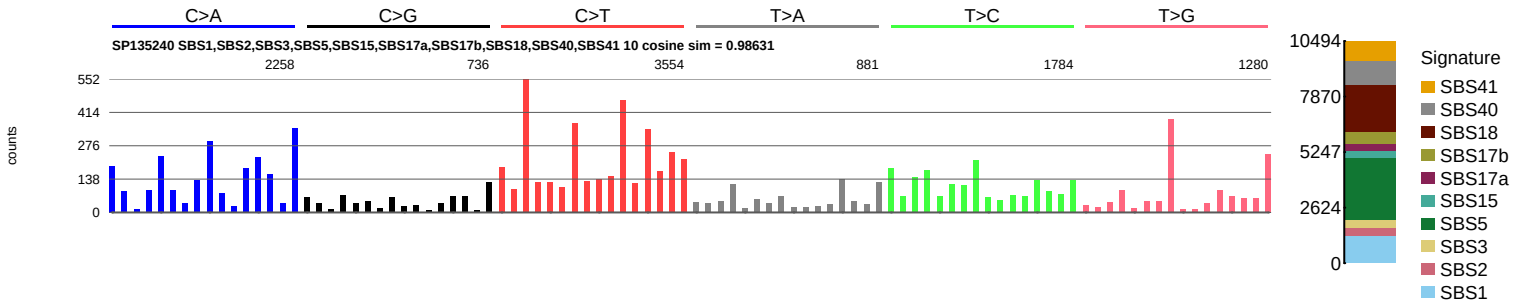
- SBS2
- SBS1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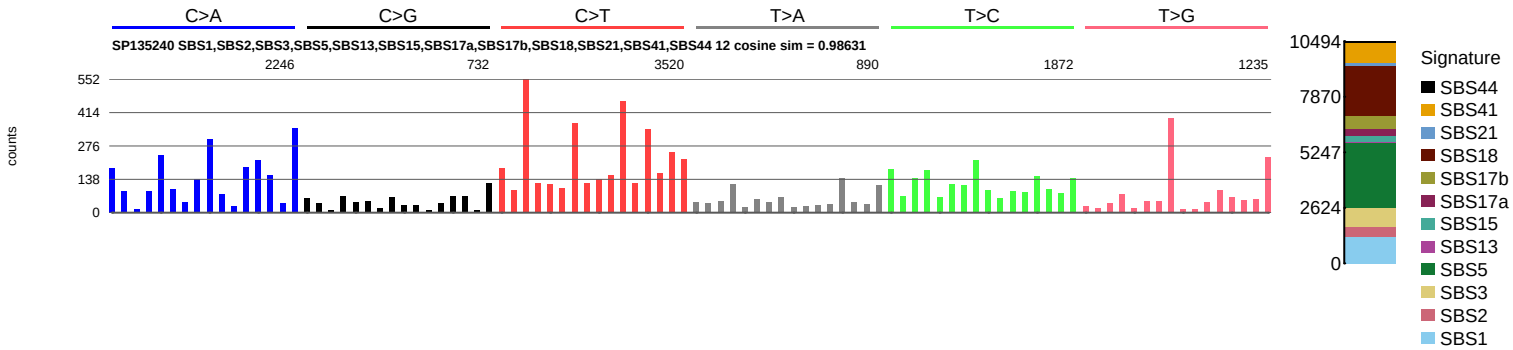
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98631
2258
736
3554
881
1784
1280
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.98631
2246
732
3520
890
1872
1235
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
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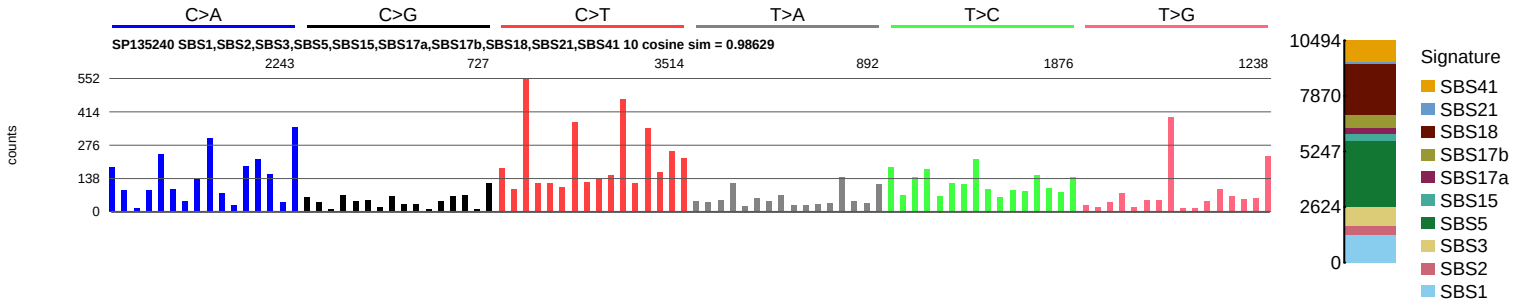
SBS2
SBS1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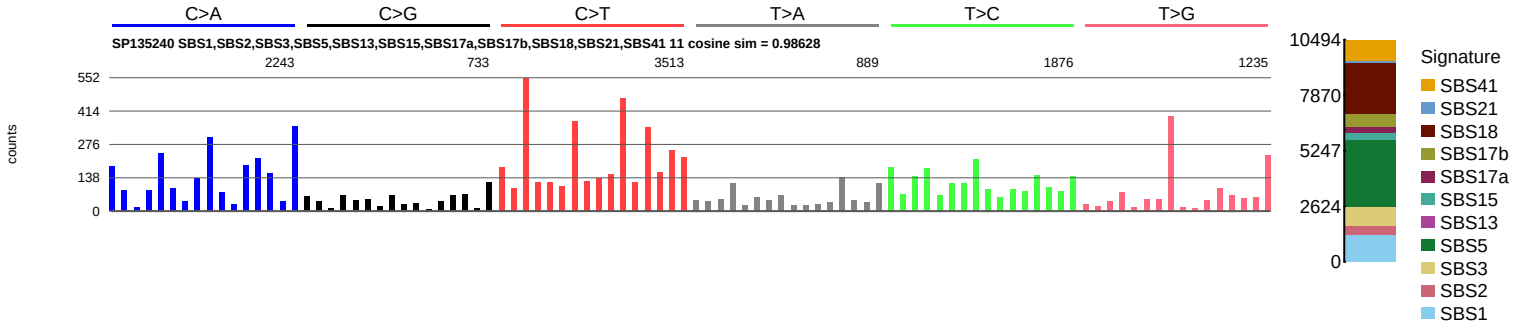
2243
727
3514
892
1876
1238
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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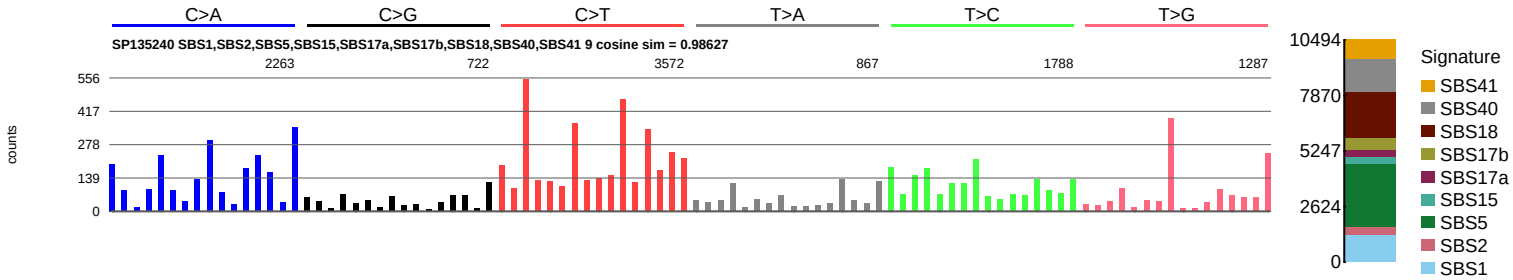
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98628
2243
733
3513
889
1876
1235
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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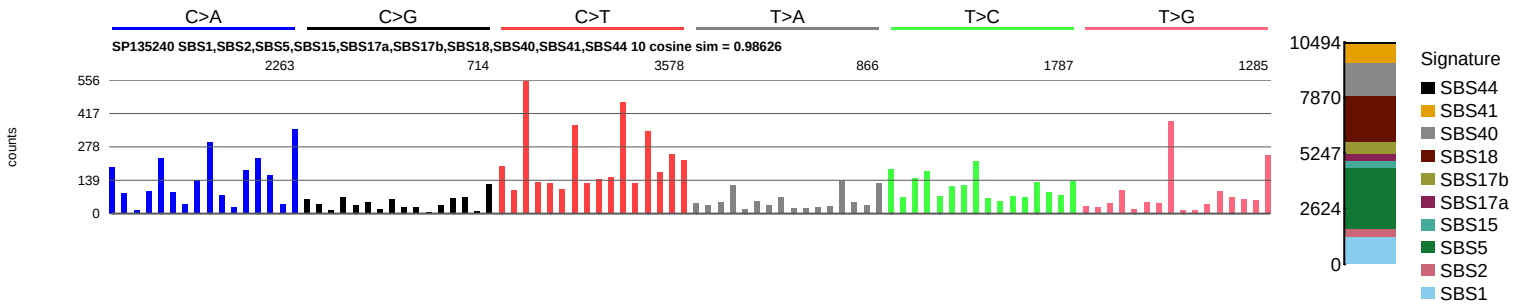
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98627
2263
722
3572
867
1788
1287
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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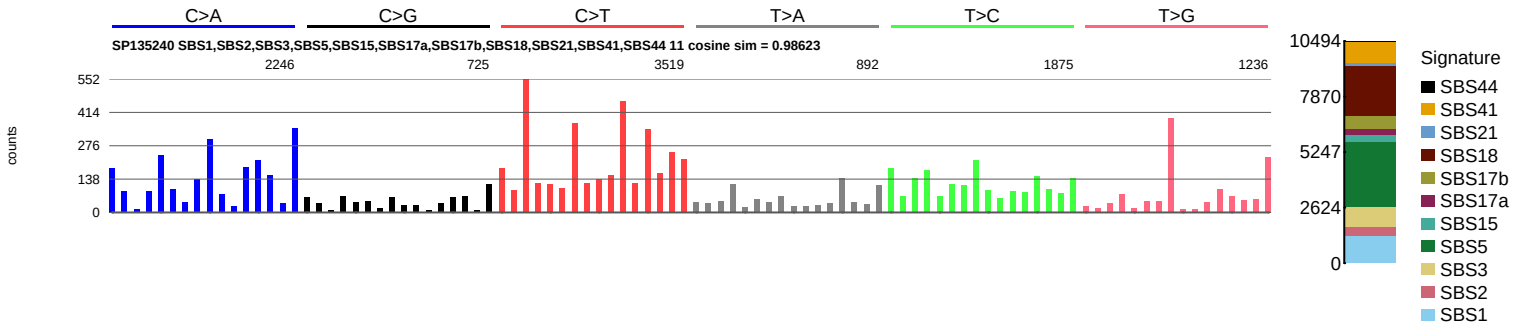
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98626
2263
714
3578
866
1787
1285
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98623
2246
725
3519
892
1875
1236
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
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SBS18
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SBS15
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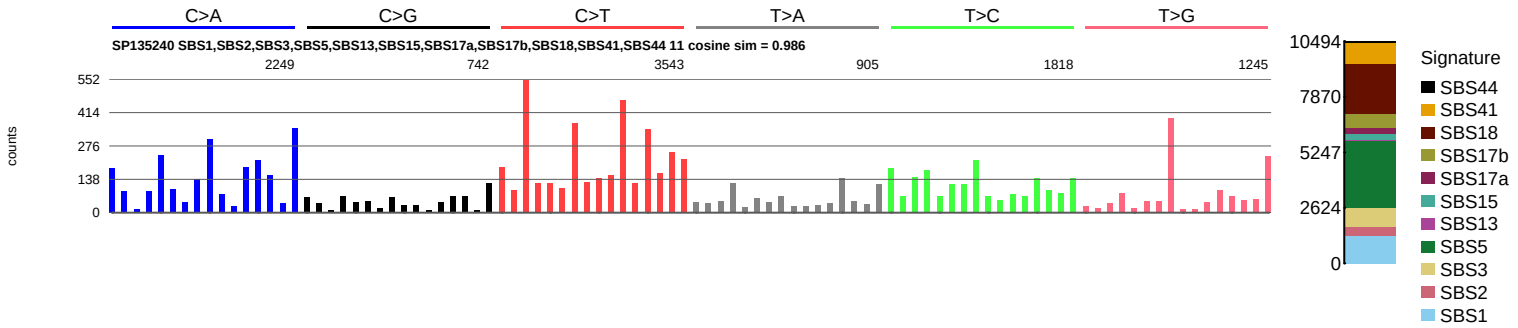
SBS1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C>G
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T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.986
2249
742
3543
905
1818
1245
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
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SBS13
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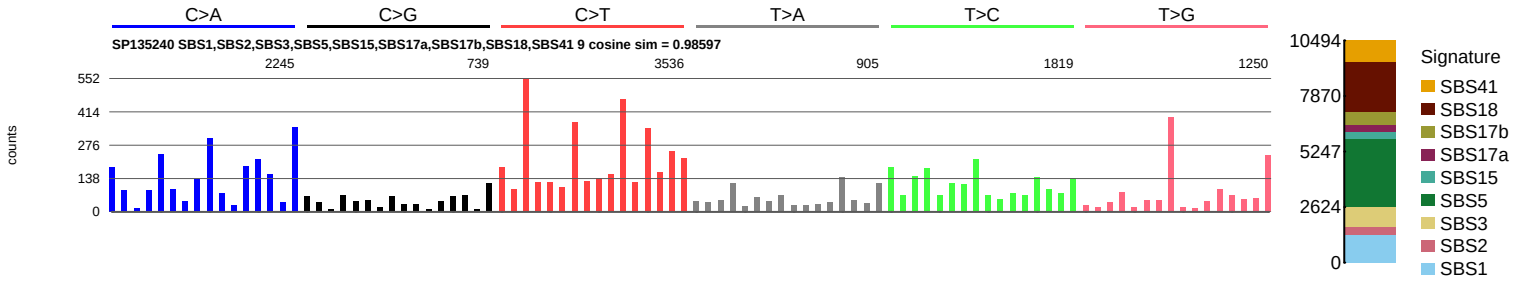
SBS3
SBS2
SBS1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C>T
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T>C
T>G
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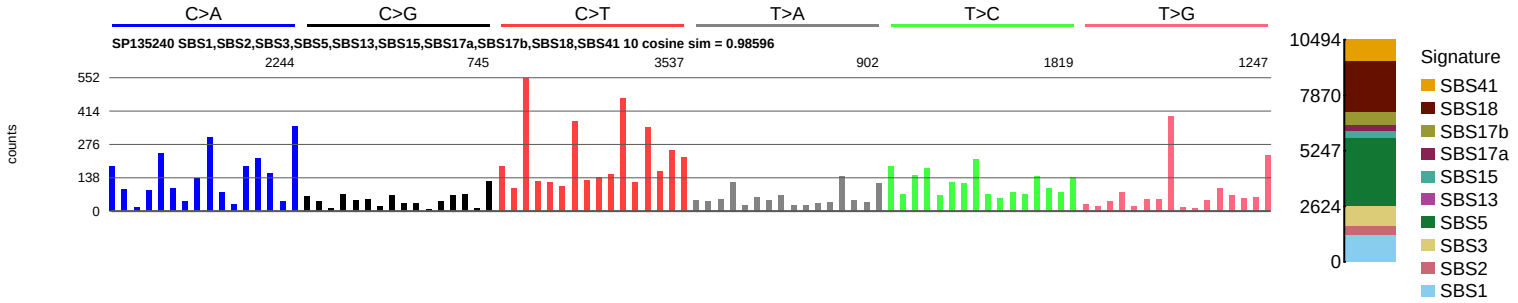
2245
739
3536
905
1819
1250
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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T>C
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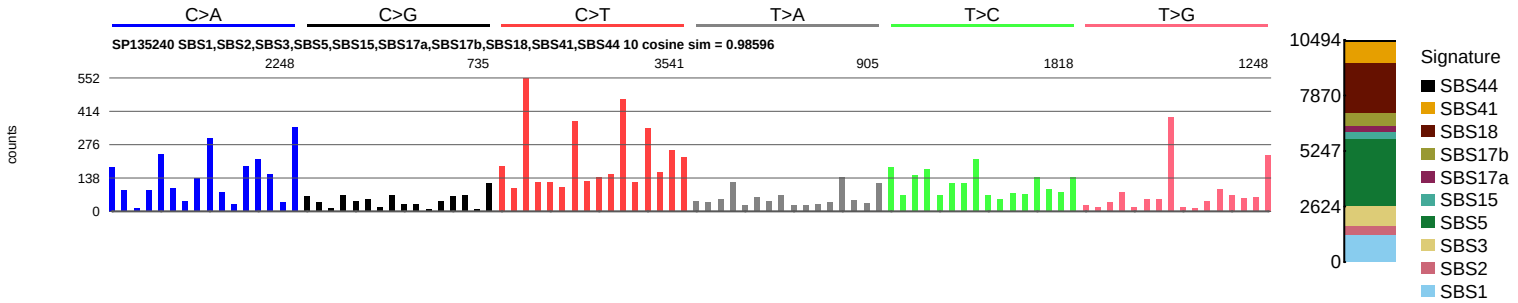
2244
745
3537
902
1819
1247
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98596
2248
735
3541
905
1818
1248
counts
552
414
276
138
0
10494
7870
5247
2624
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Signature
SBS44
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SBS18
SBS17b
SBS17a
SBS15
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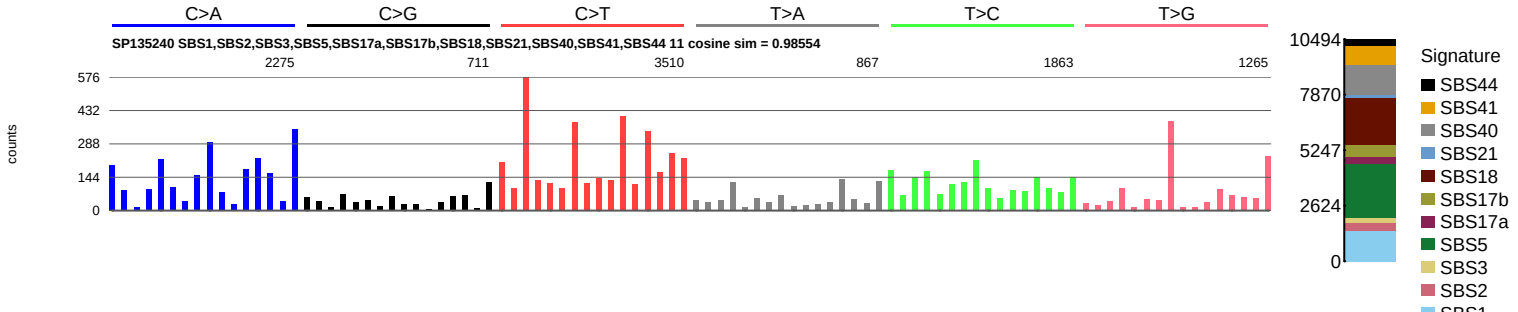
SBS2
SBS1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C>G
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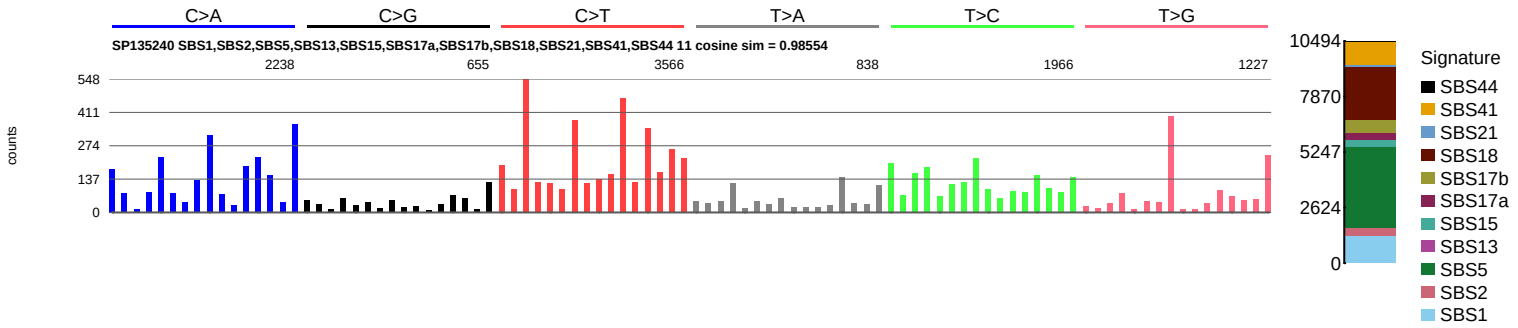
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98554
2275
711
3510
867
1863
1265
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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C>G
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T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98554
2238
655
3566
838
1966
1227
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
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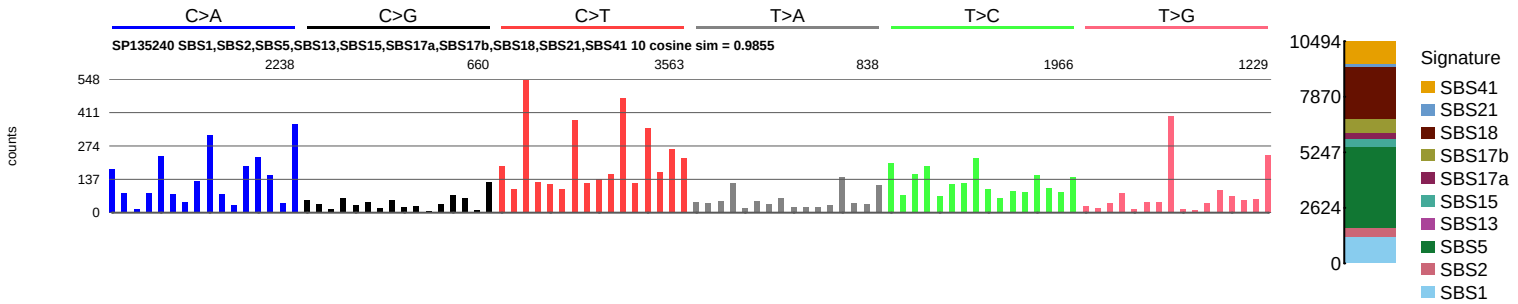
SBS2
SBS1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C>T
T>A
T>C
T>G
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2238
660
3563
838
1966
1229
counts
548
411
274
137
0
10494
7870
5247
2624
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Signature
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SBS2
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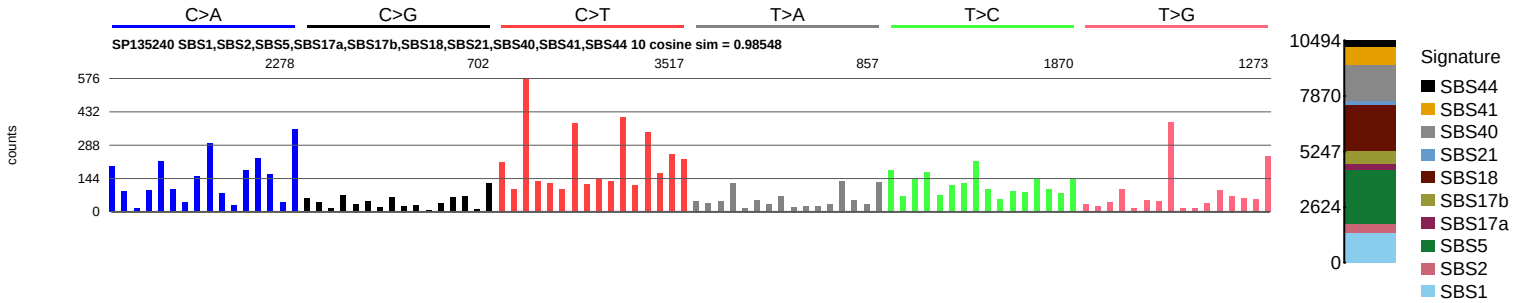
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T>G
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2278
702
3517
857
1870
1273
counts
576
432
288
144
0
10494
7870
5247
2624
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SBS40
SBS21
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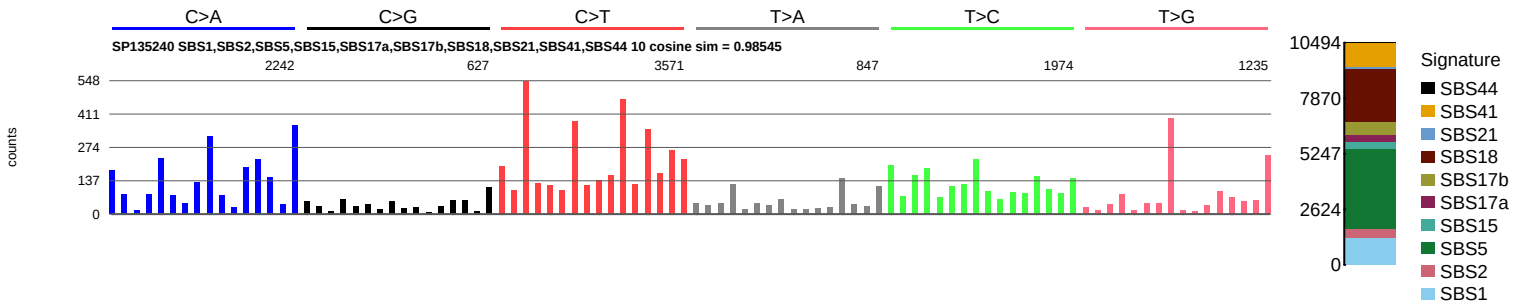
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98545
2242
627
3571
847
1974
1235
counts
548
411
274
137
0
10494
7870
5247
2624
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Signature
SBS44
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SBS21
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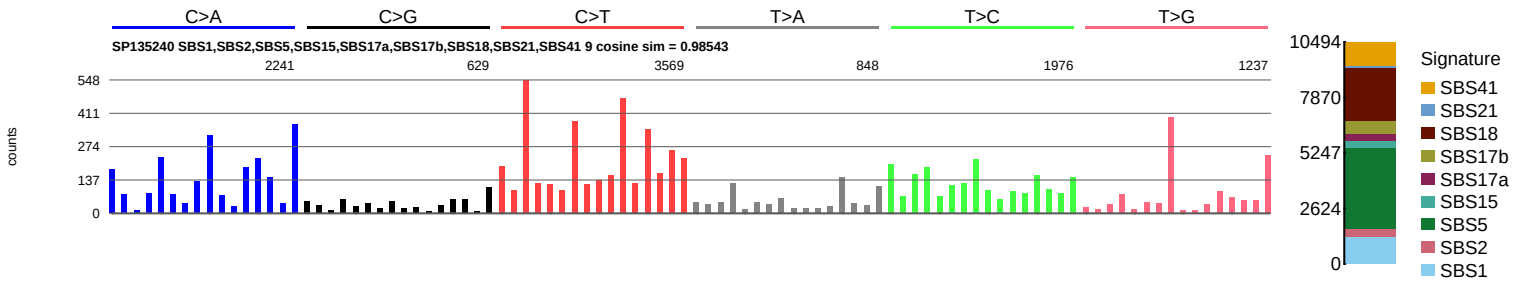
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C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98543
2241
629
3569
848
1976
1237
counts
548
411
274
137
0
10494
7870
5247
2624
0
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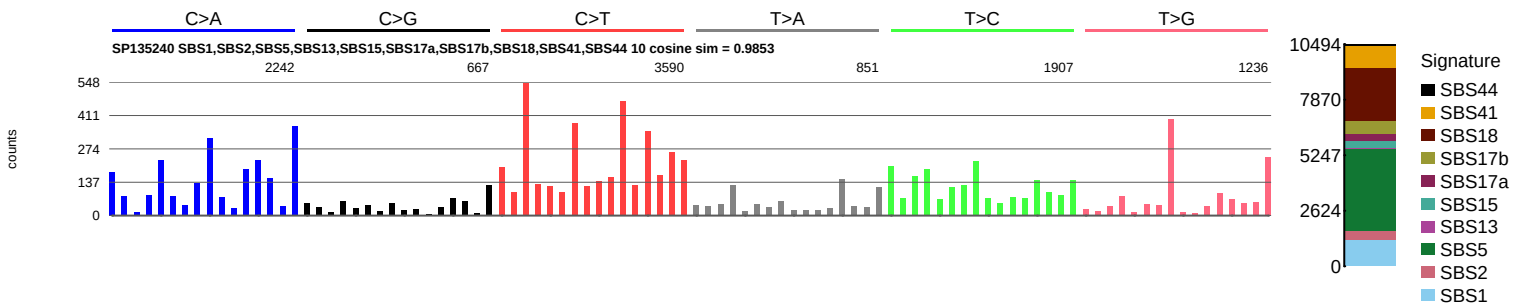
SBS2
SBS1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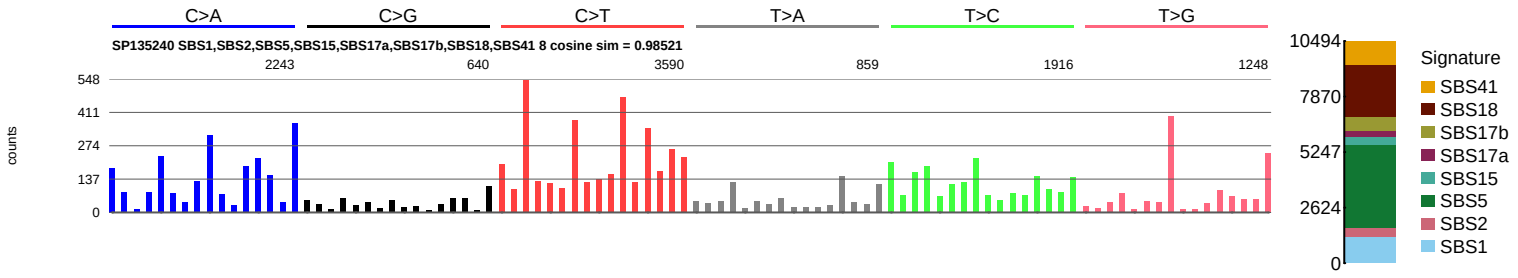
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9853
2242
667
3590
851
1907
1236
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98521
2243
640
3590
859
1916
1248
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
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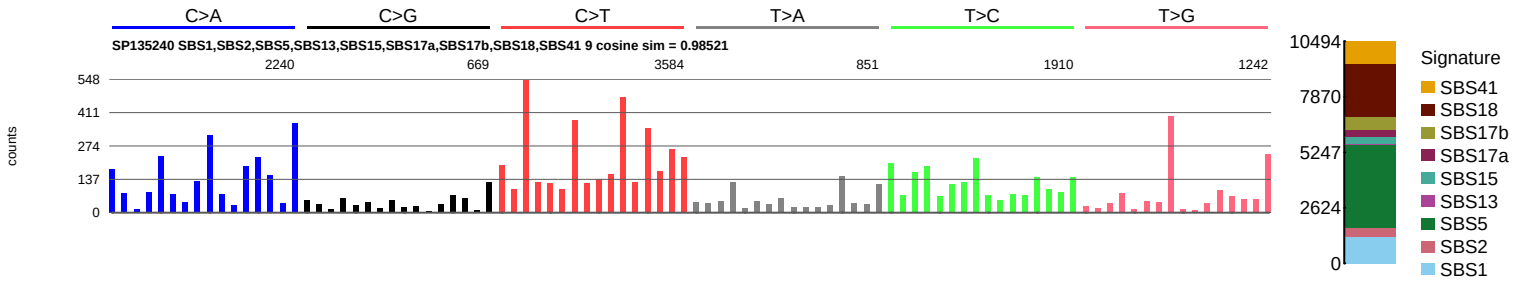
SBS1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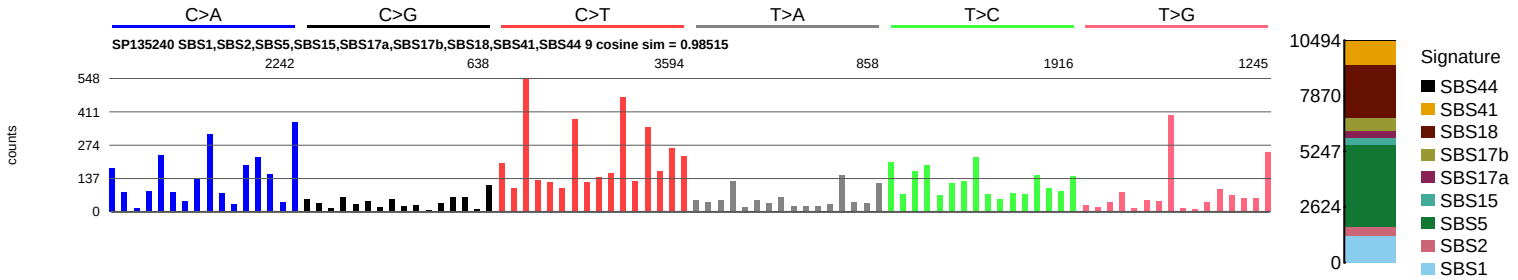
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98521
2240
669
3584
851
1910
1242
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98515
2242
638
3594
858
1916
1245
548
411
274
137
0
counts
10494
7870
5247
2624
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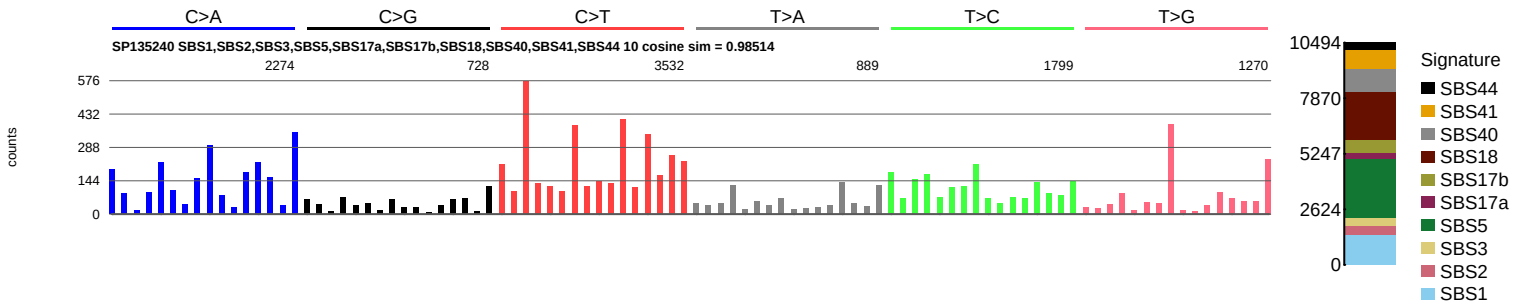
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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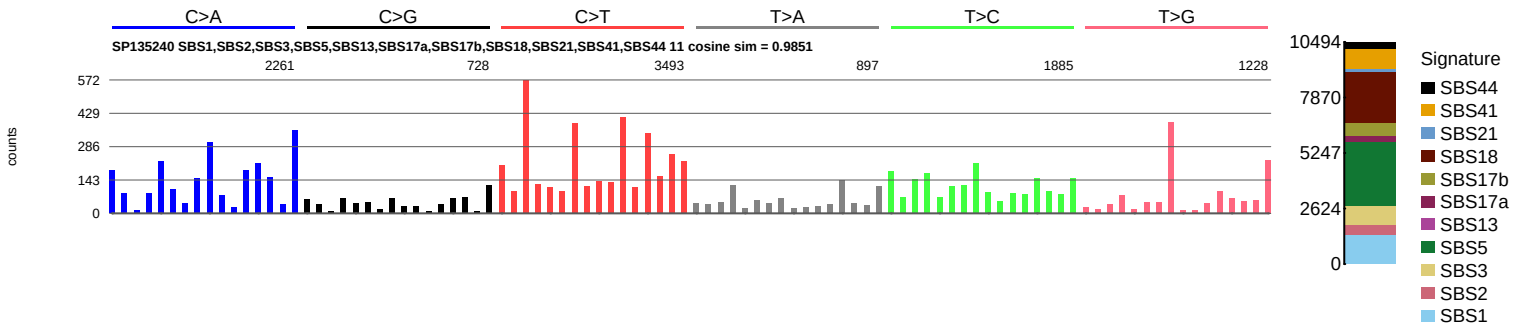
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98514
2274
728
3532
889
1799
1270
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.9851
2261
728
3493
897
1885
1228
572
429
286
143
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
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SBS18
SBS17b
SBS17a
SBS13
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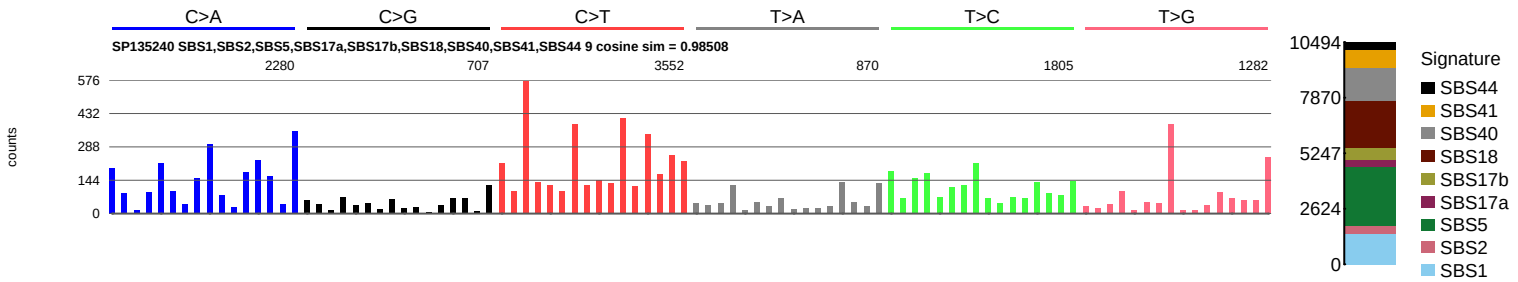
SBS2
SBS1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C>G
C>T
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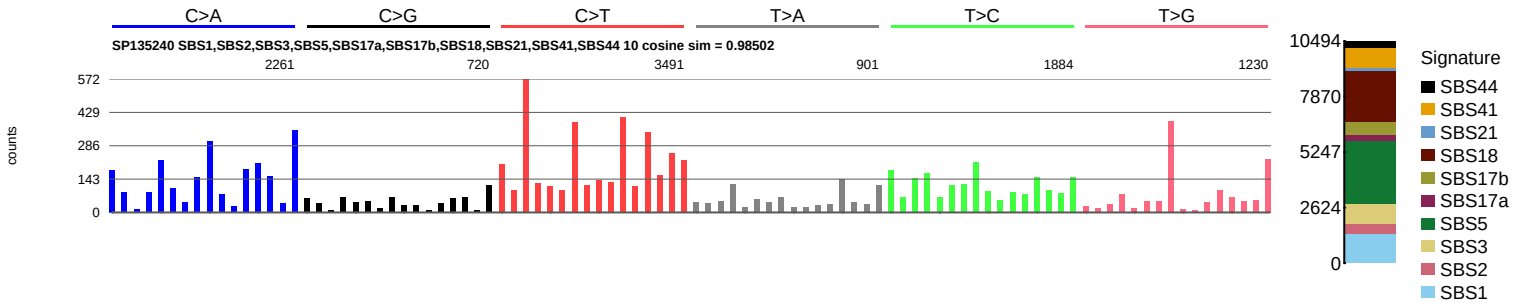
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98508
2280
707
3552
870
1805
1282
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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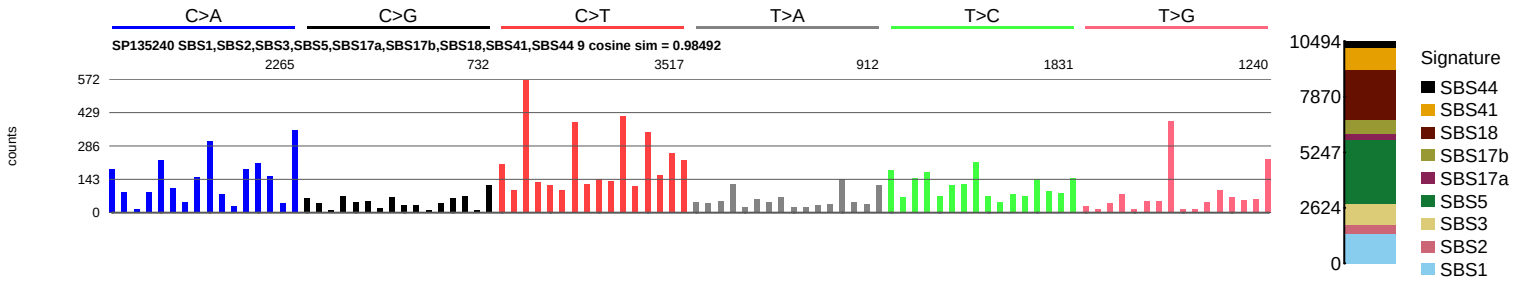
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98502
2261
720
3491
901
1884
1230
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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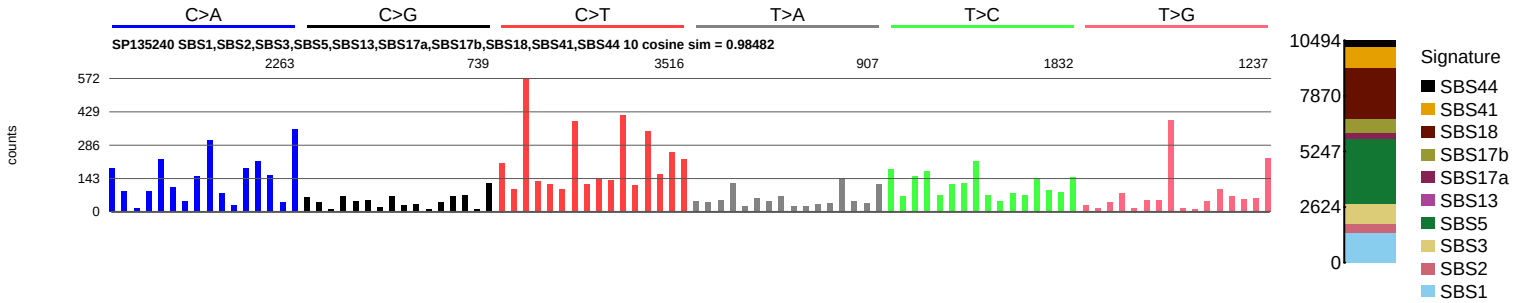
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98492
2265
732
3517
912
1831
1240
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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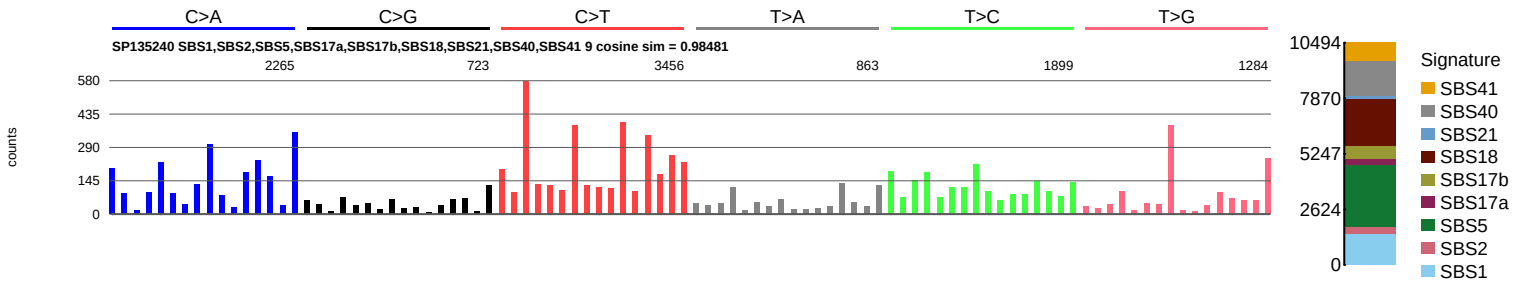
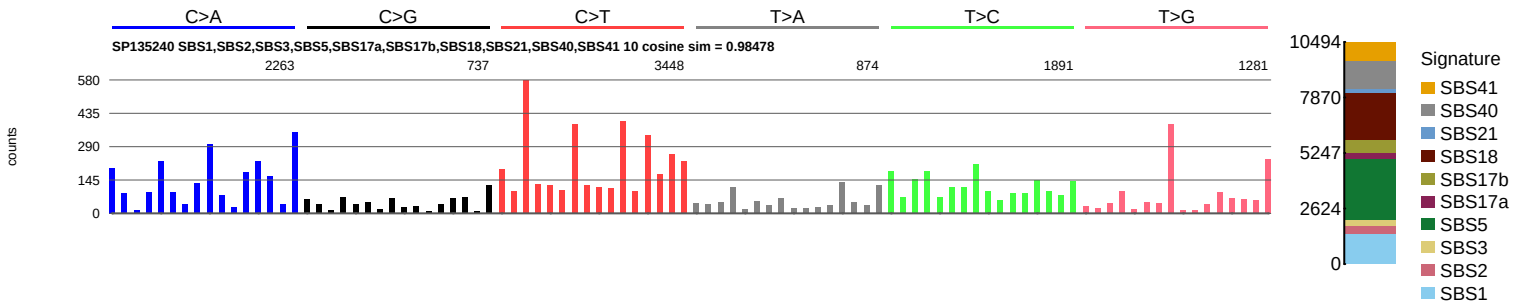
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98482
2263
739
3516
907
1832
1237
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98481
2265
723
3456
863
1899
1284
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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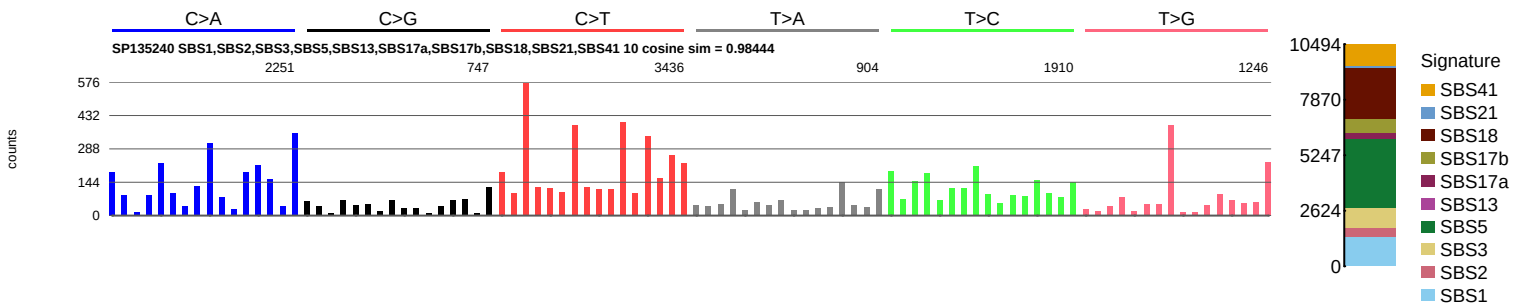
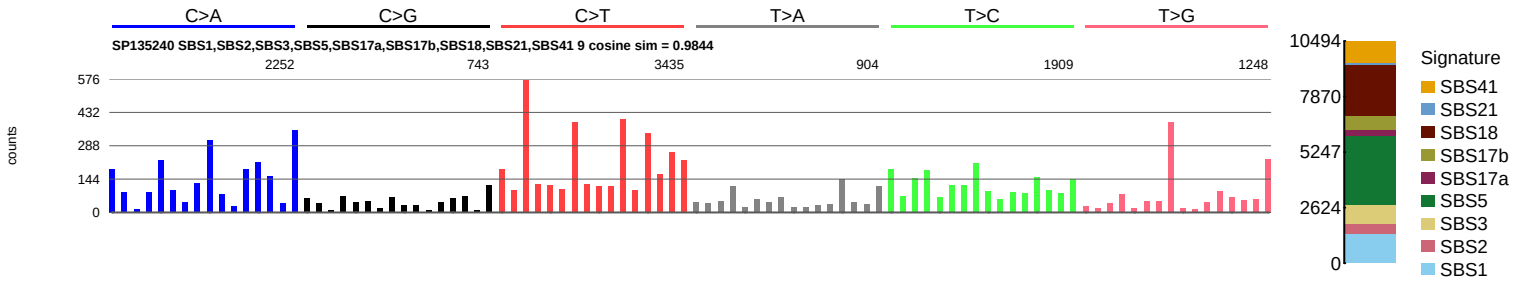
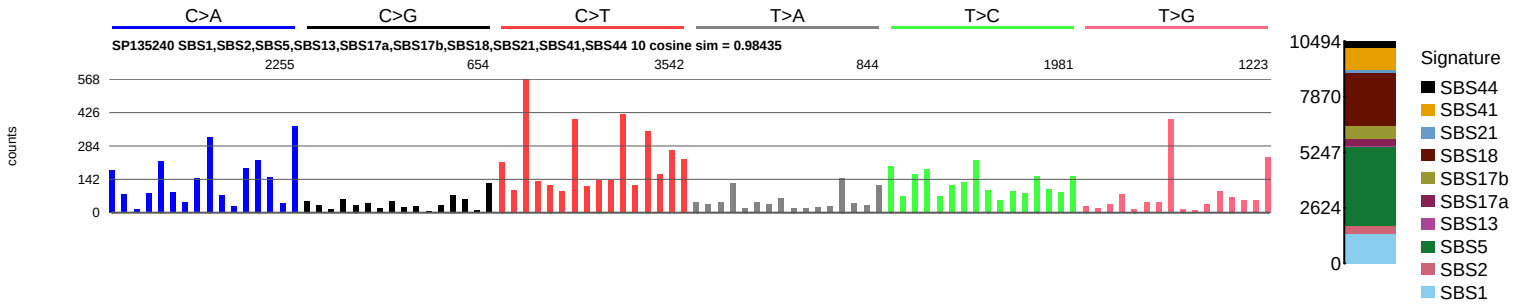
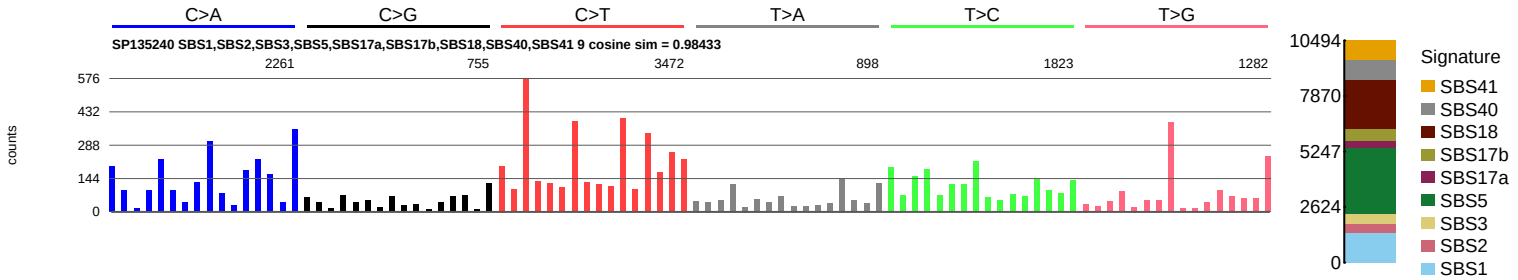
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98478
2263
737
3448
874
1891
1281
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98444
2251
747
3436
904
1910
1246
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9844
2252
743
3435
904
1909
1248
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98435
2255
654
3542
844
1981
1223
568
426
284
142
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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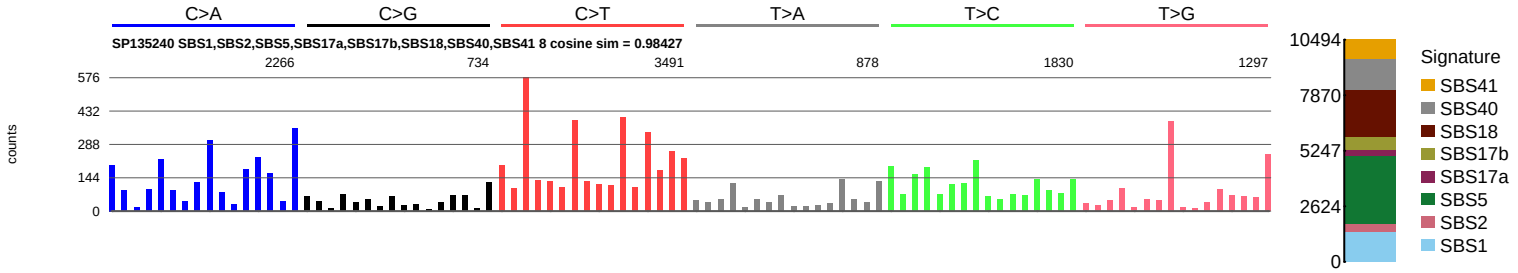
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98433
2261
755
3472
898
1823
1282
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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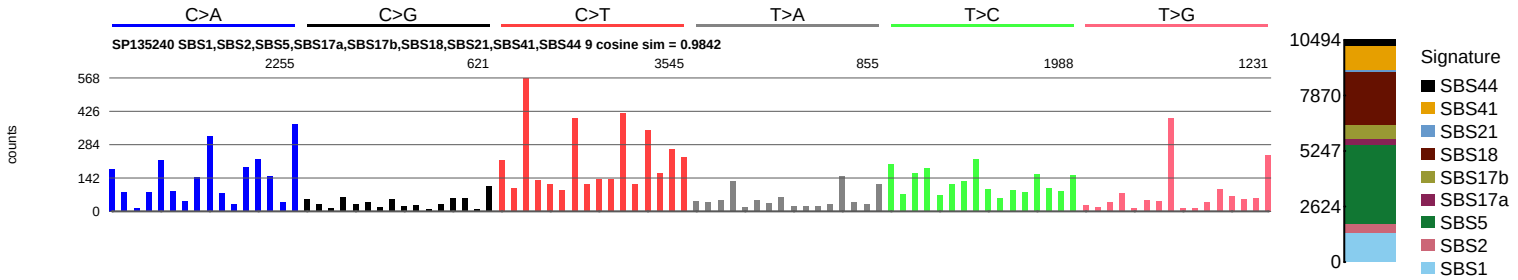
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98427
2266
734
3491
878
1830
1297
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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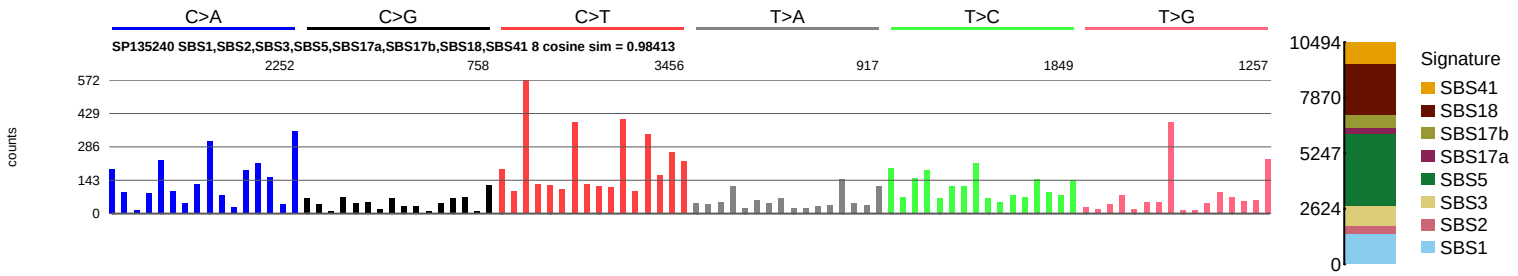
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9842
2255
621
3545
855
1988
1231
568
426
284
142
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98413
2252
758
3456
917
1849
1257
572
429
286
143
0
counts
10494
7870
5247
2624
0
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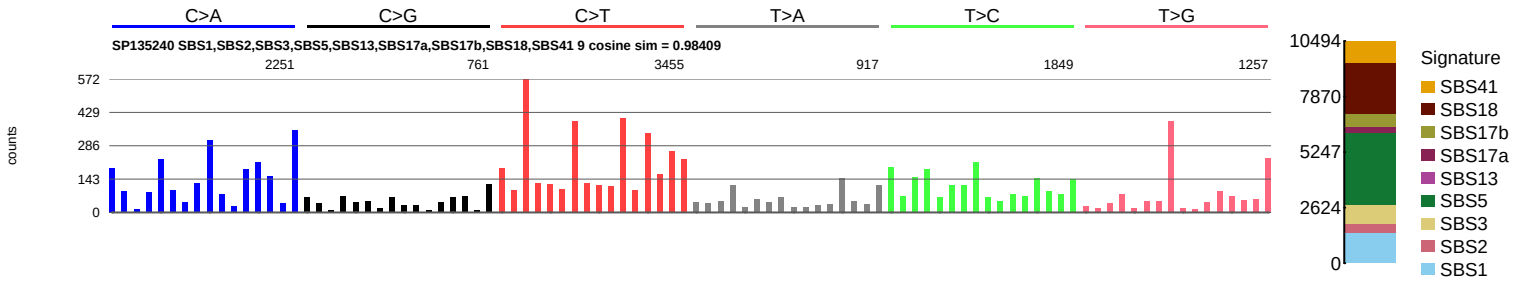
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98409
2251
761
3455
917
1849
1257
counts
572
429
286
143
0
10494
7870
5247
2624
0
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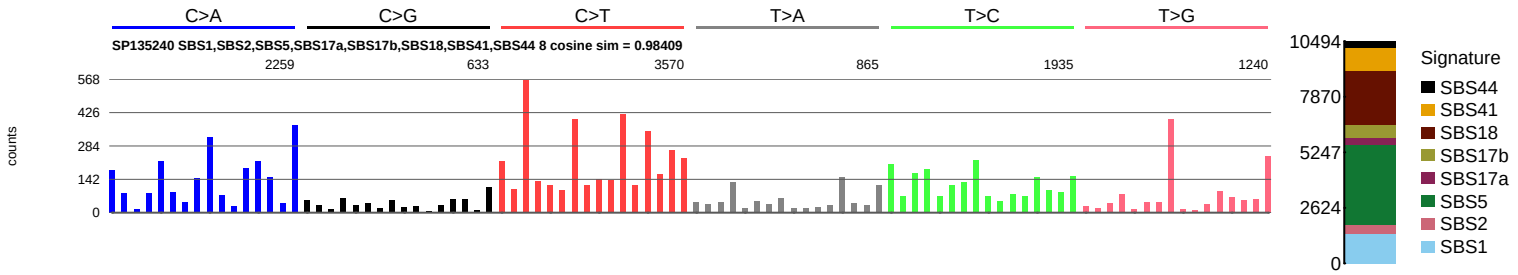
SBS2
SBS1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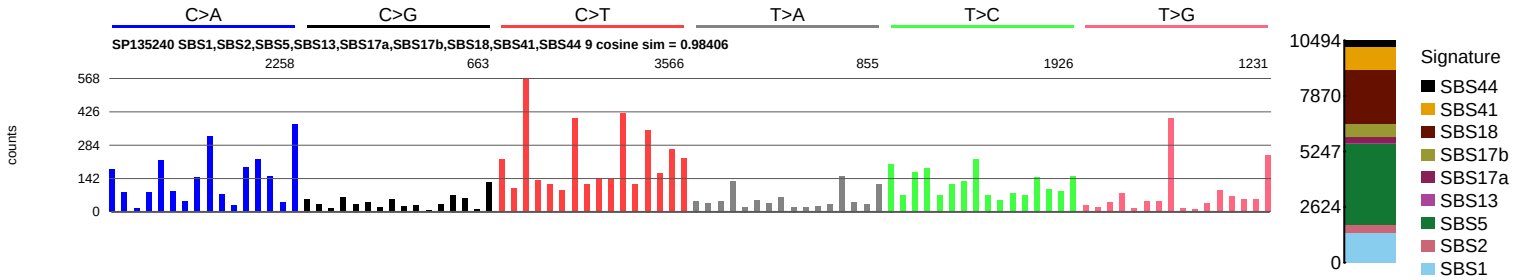
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98409
2259
633
3570
865
1935
1240
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98406
2258
663
3566
855
1926
1231
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
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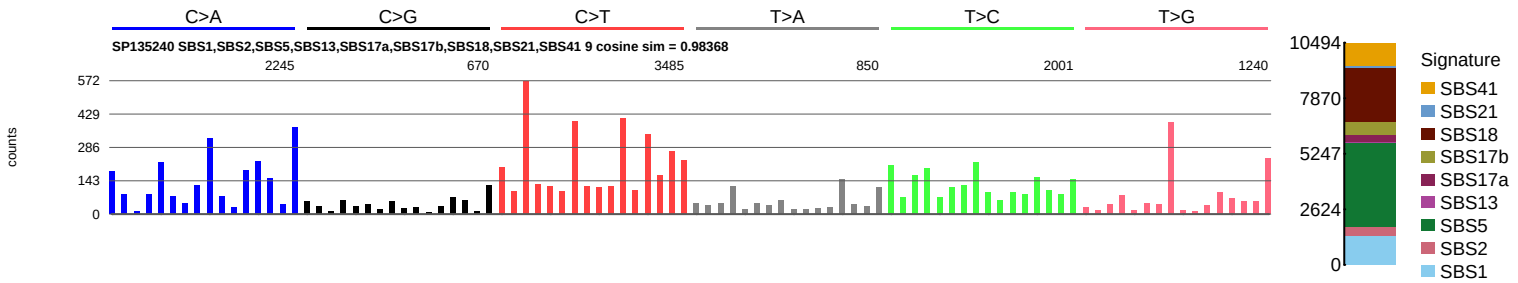
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98368
2245
670
3485
850
2001
1240
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
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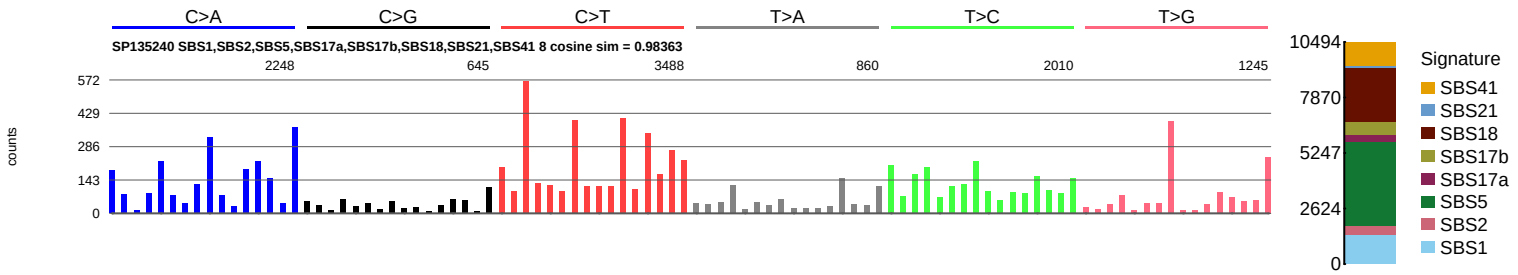
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98363
2248
645
3488
860
2010
1245
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS41
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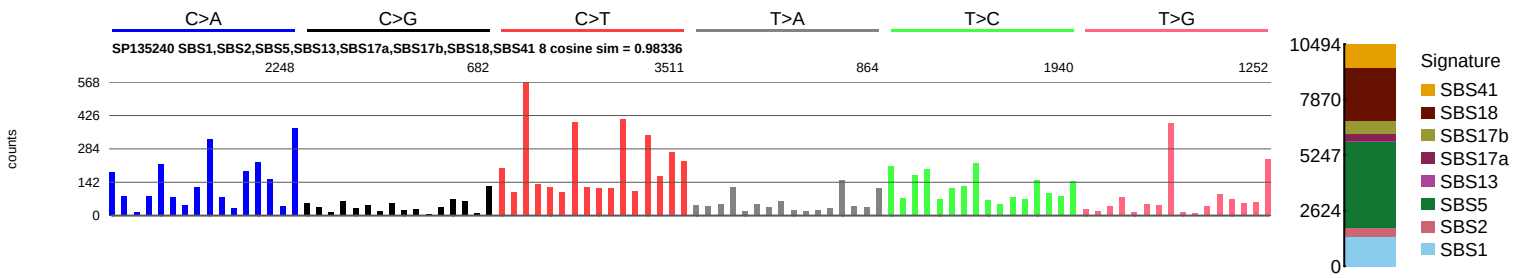
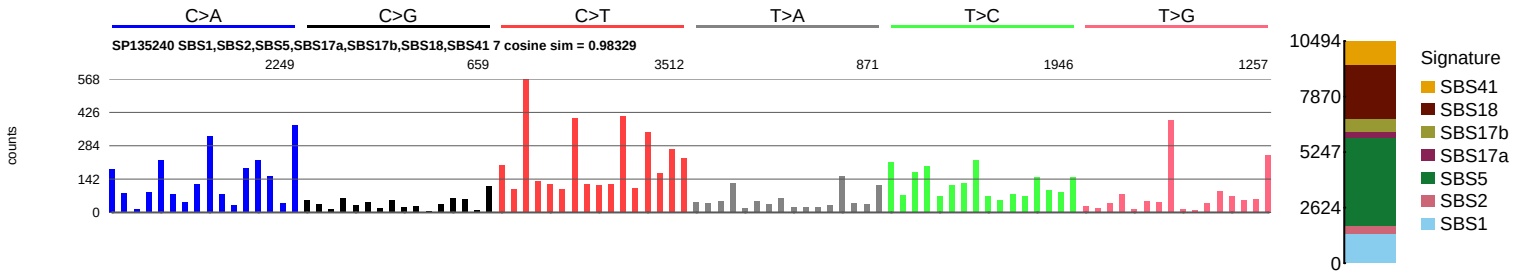
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98336
2248
682
3511
864
1940
1252
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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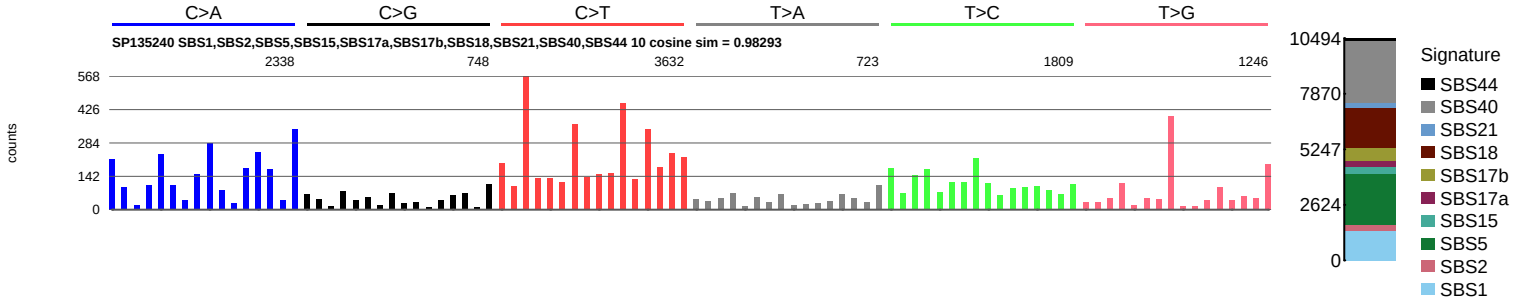
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98329
2249
659
3512
871
1946
1257
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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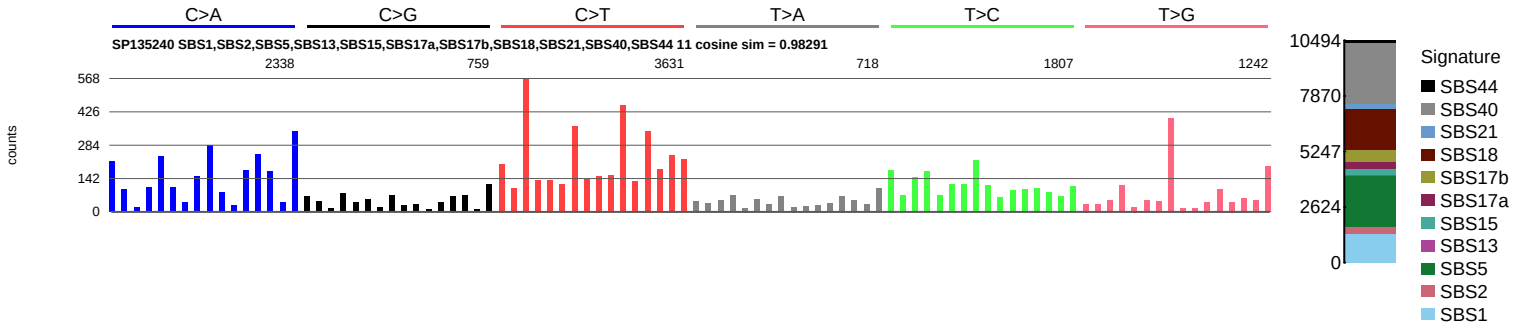
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98293
2338
748
3632
723
1809
1246
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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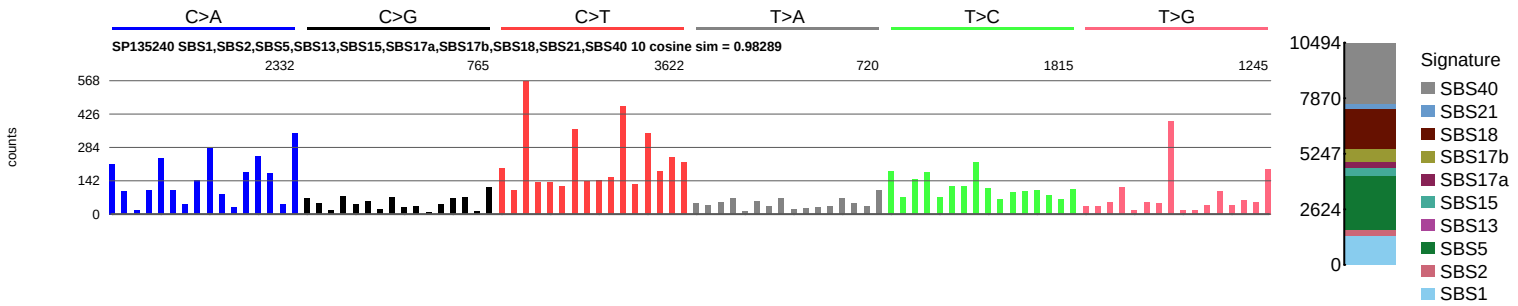
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98291
2338
759
3631
718
1807
1242
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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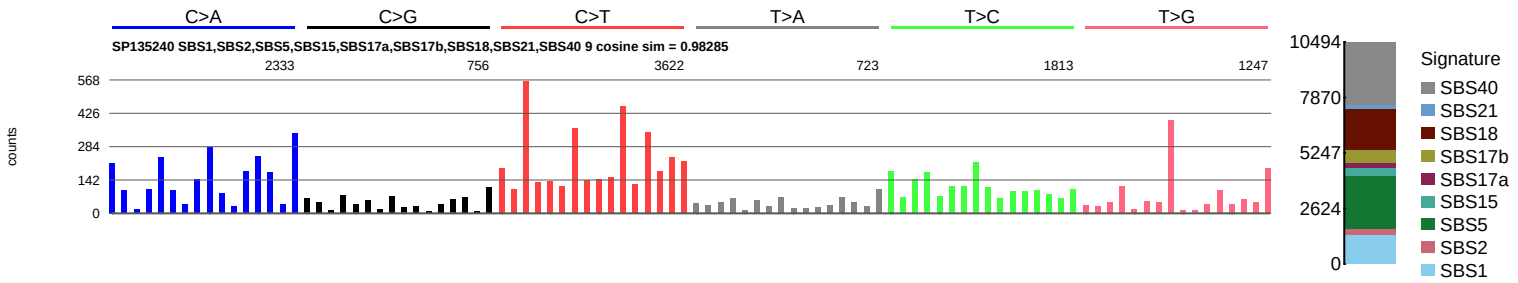
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98289
2332
765
3622
720
1815
1245
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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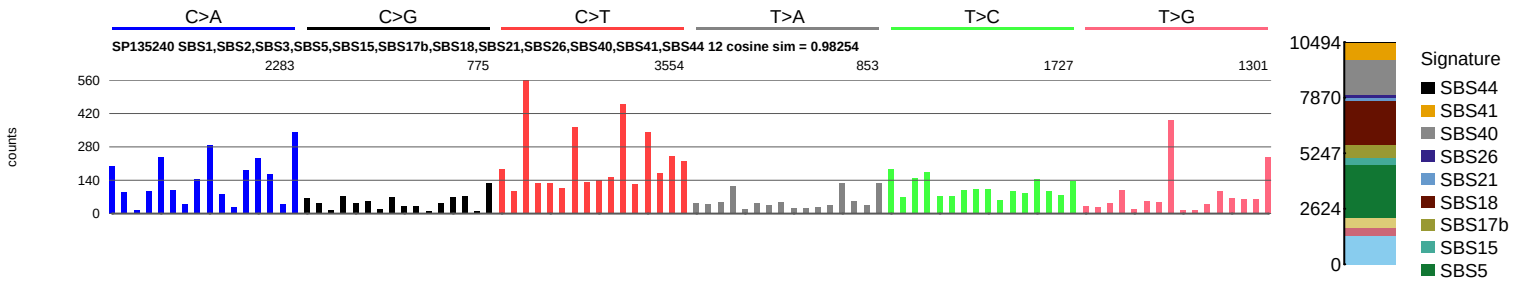
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98285
2333
756
3622
723
1813
1247
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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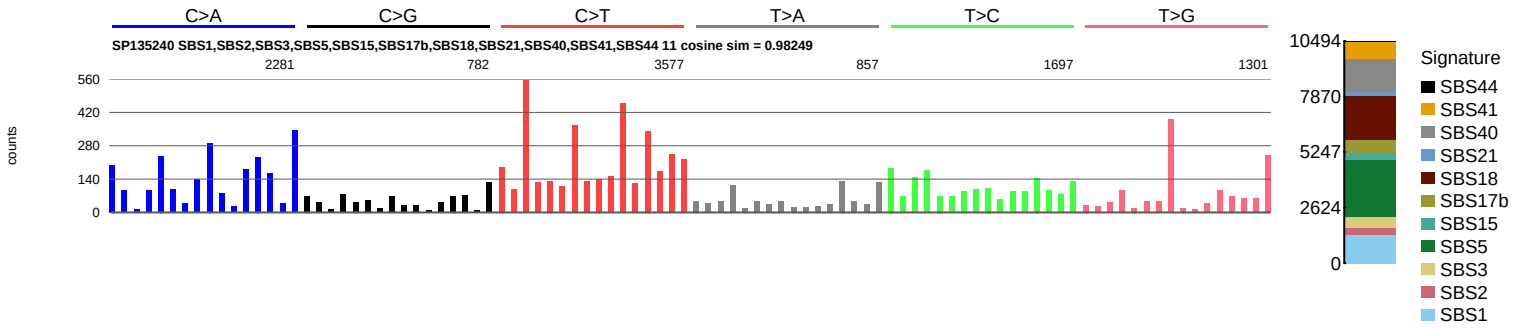
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98254
2283
775
3554
853
1727
1301
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3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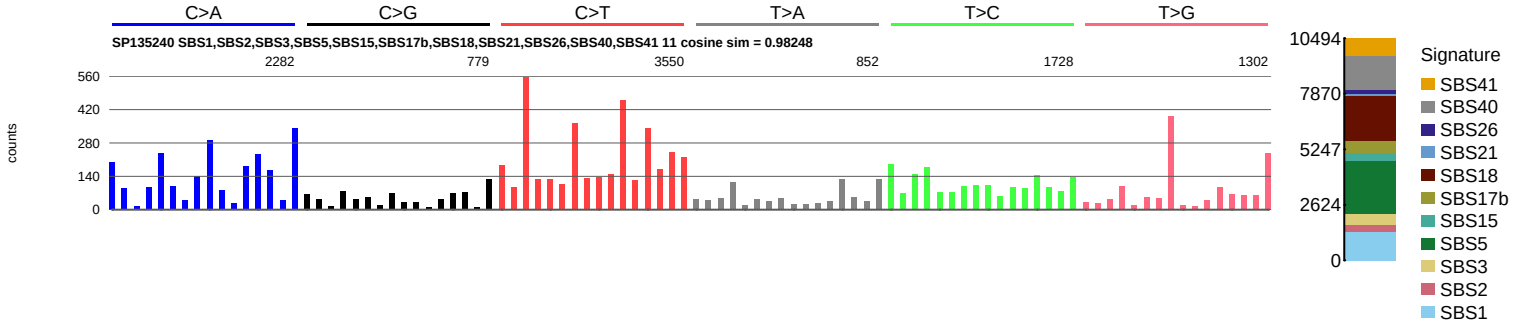
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98249
2281
782
3577
857
1697
1301
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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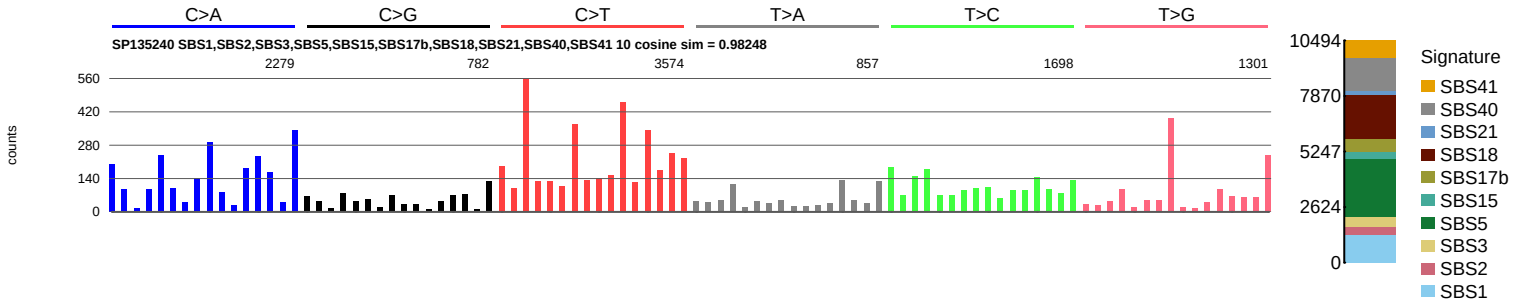
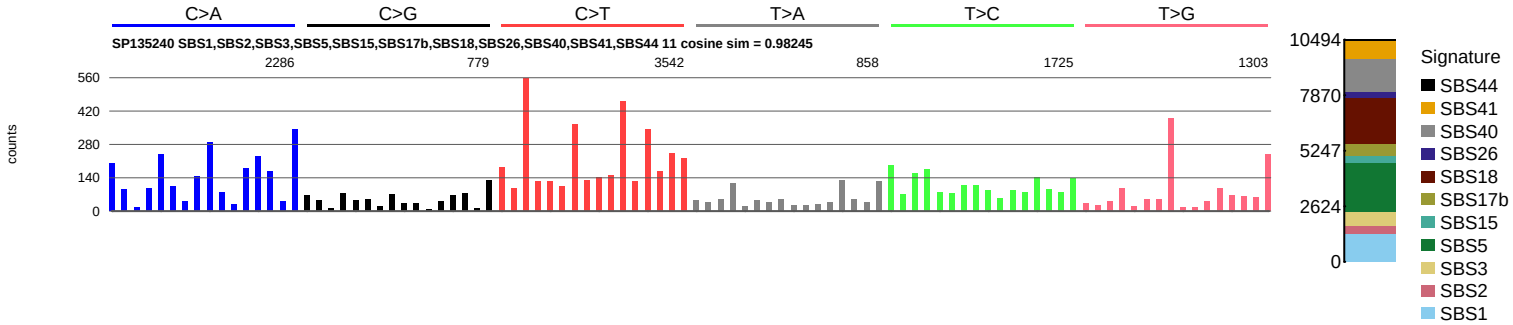
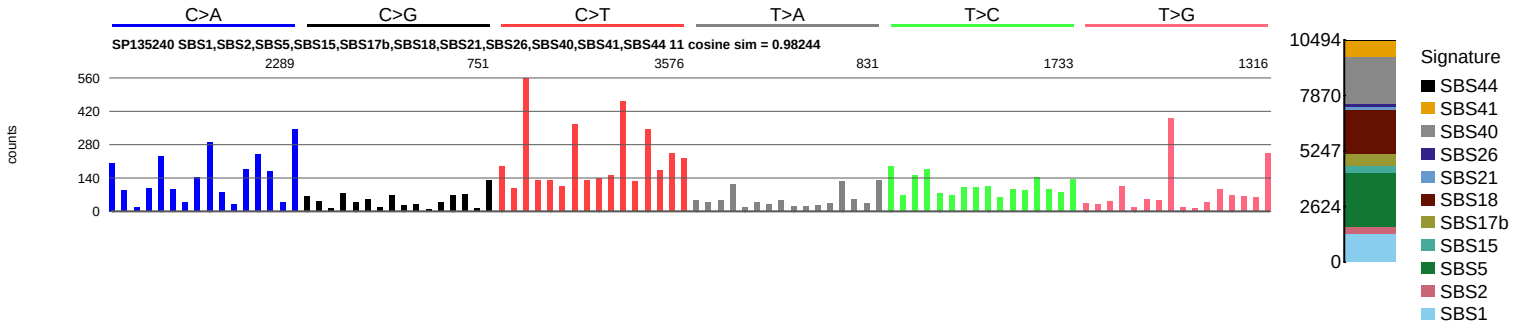
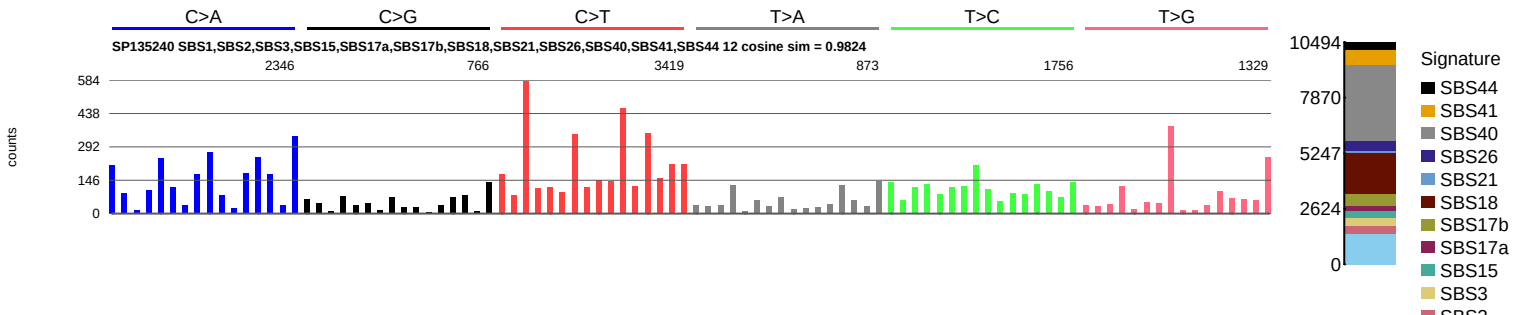
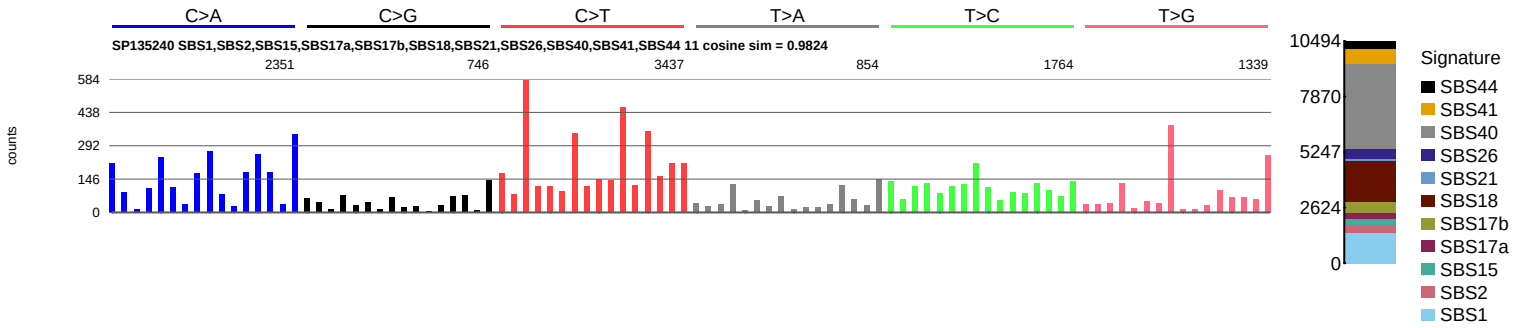
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98248
2282
779
3550
852
1728
1302
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

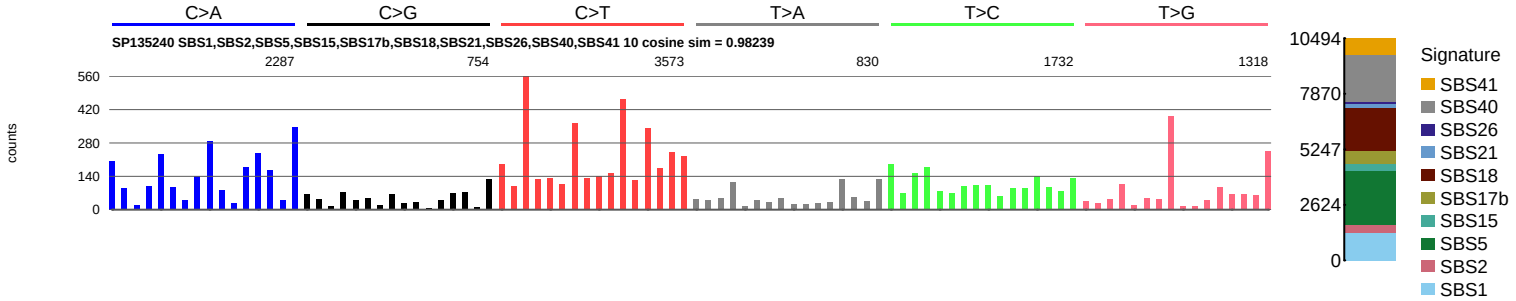

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98248
2279
782
3574
857
1698
1301
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

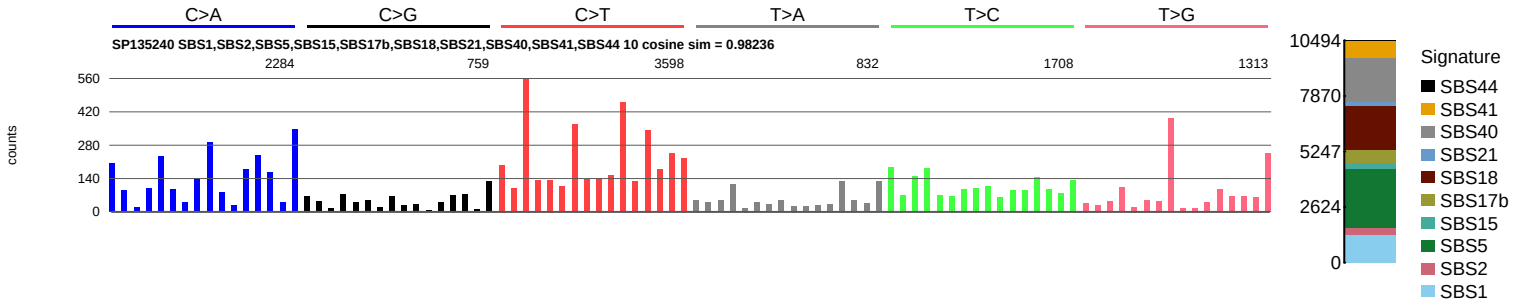

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98245
2286
779
3542
858
1725
1303
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98244
2289
751
3576
831
1733
1316
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.9824
2346
766
3419
873
1756
1329
counts
584
438
292
146
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9824
2351
746
3437
854
1764
1339
584
438
292
146
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98239
2287
754
3573
830
1732
1318
560
420
280
140
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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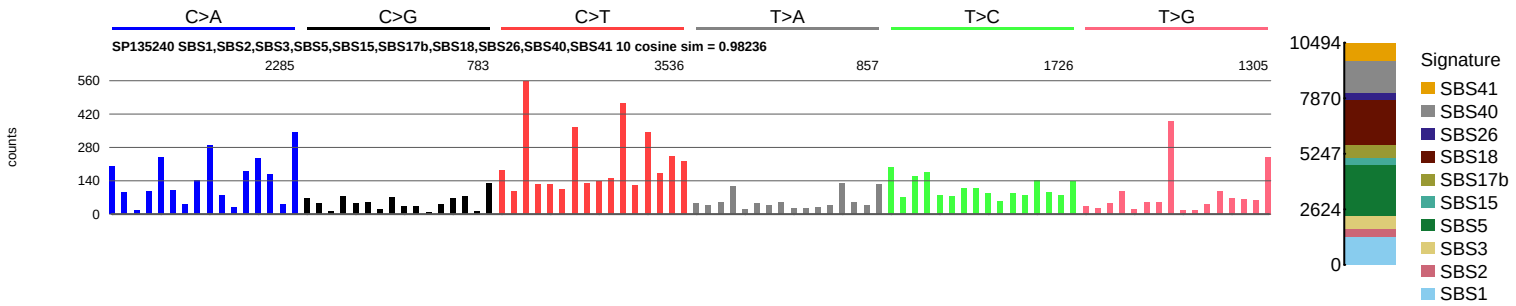
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98236
2284
759
3598
832
1708
1313
560
420
280
140
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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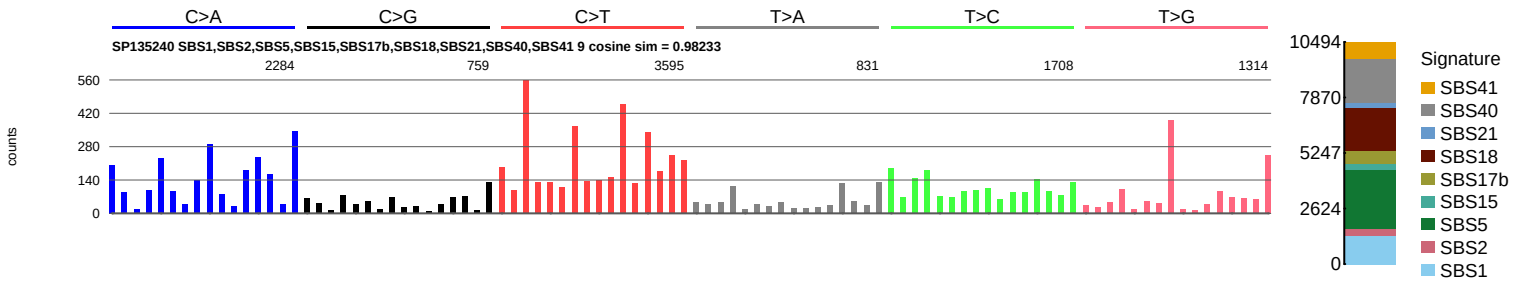
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98236
2285
783
3536
857
1726
1305
560
420
280
140
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98233
2284
759
3595
831
1708
1314
560
420
280
140
0
counts
10494
7870
5247
2624
0
Signature
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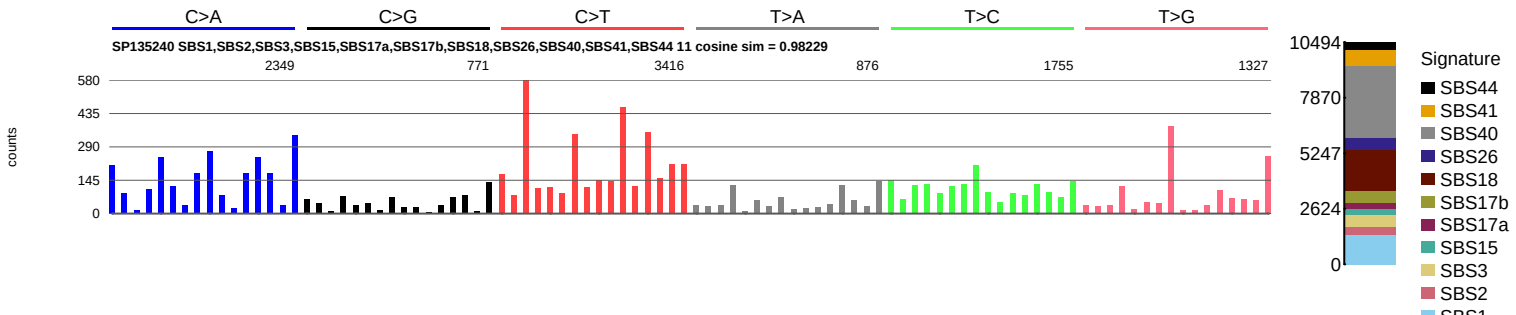
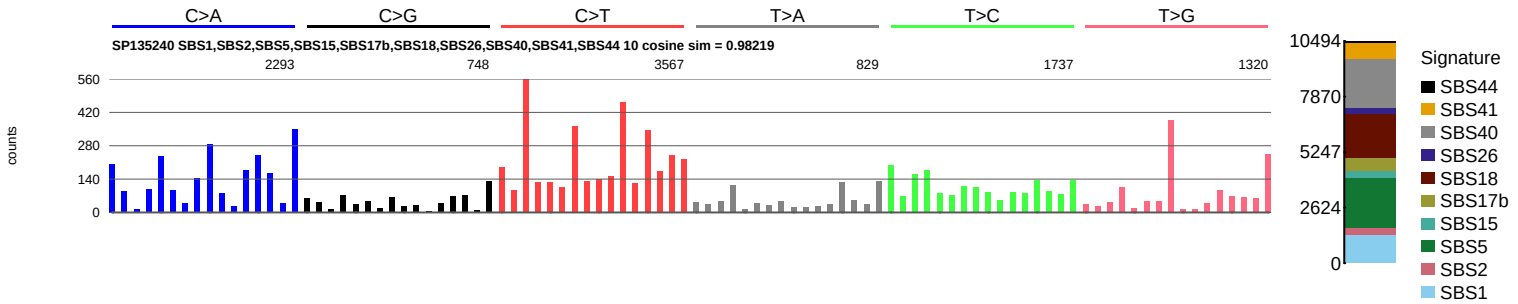
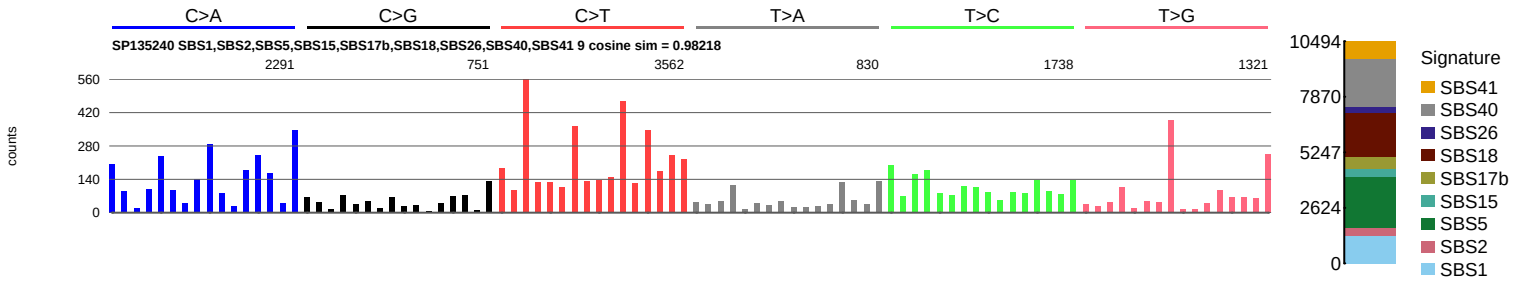
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98229
2349
771
3416
876
1755
1327
580
435
290
145
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98219
2293
748
3567
829
1737
1320
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98218
2291
751
3562
830
1738
1321
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS41
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SBS17b
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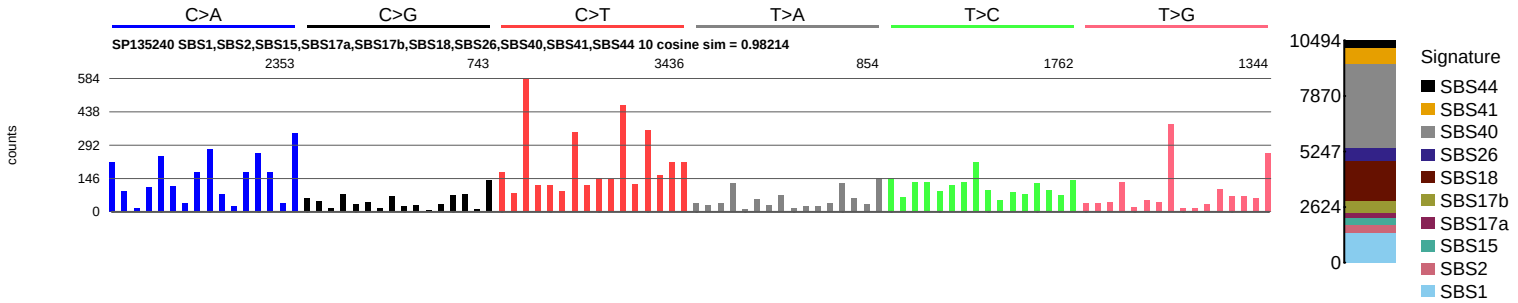
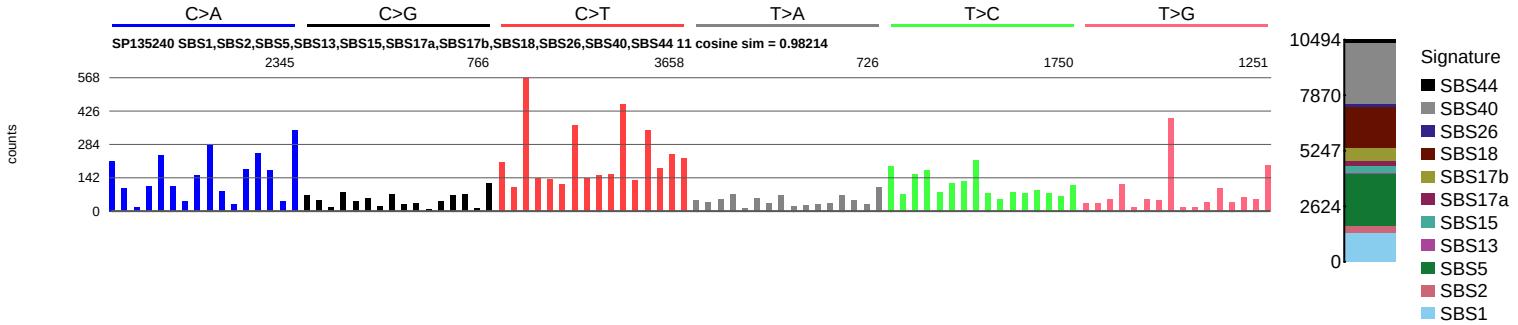
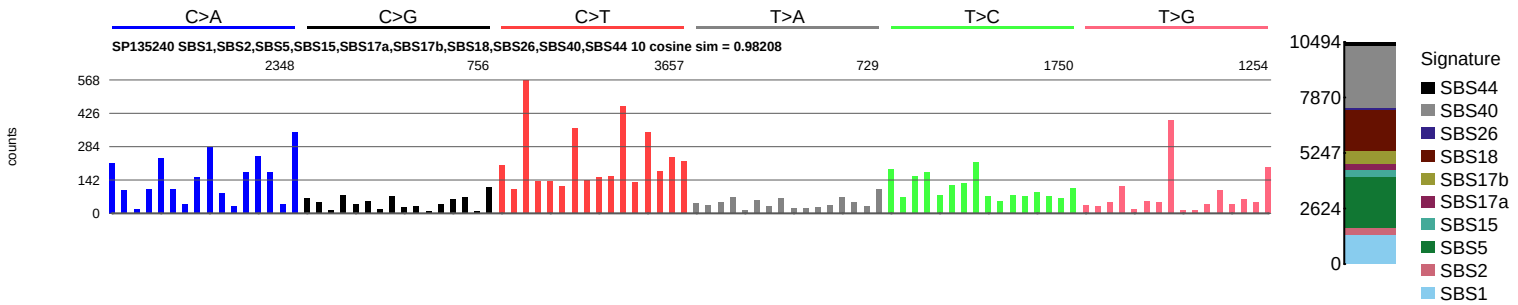
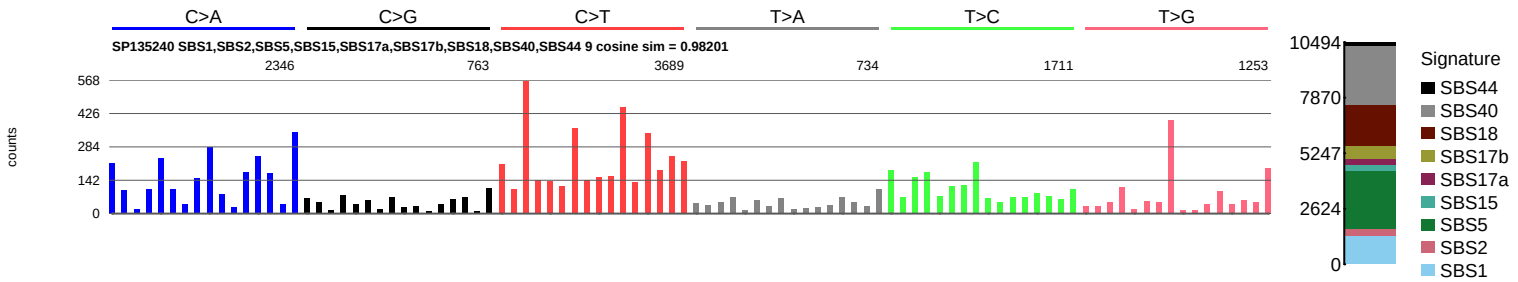
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98214
2353
743
3436
854
1762
1344
counts
584
438
292
146
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98214
2345
766
3658
726
1750
1251
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98208
2348
756
3657
729
1750
1254
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98201
2346
763
3689
734
1711
1253
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
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SBS17a
SBS15
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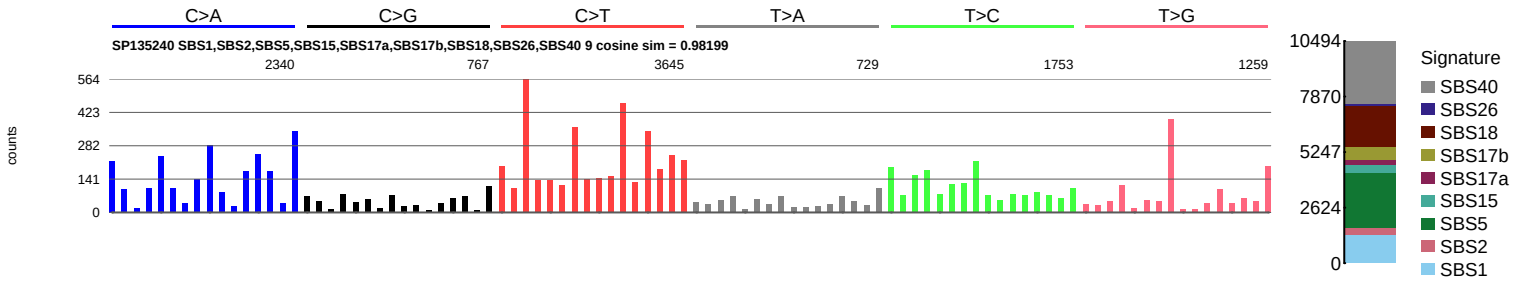
SBS2
SBS1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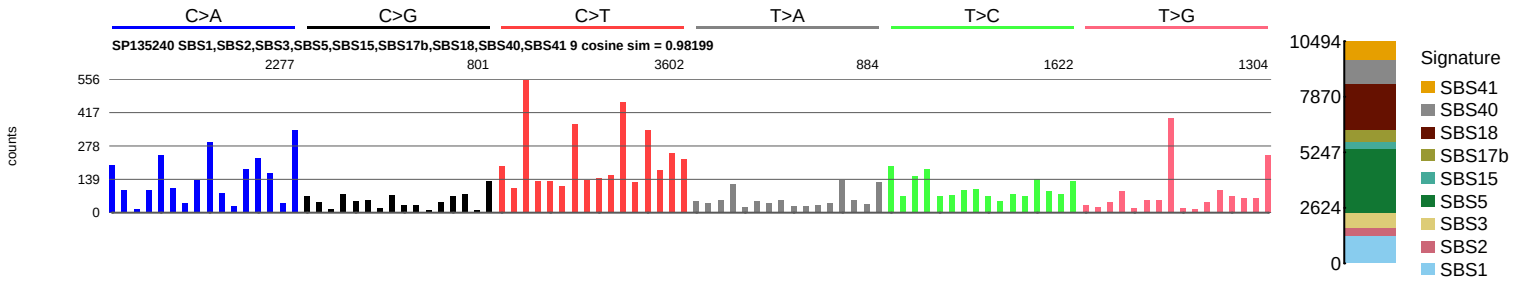
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98199
2340
767
3645
729
1753
1259
counts
564
423
282
141
0
10494
7870
5247
2624
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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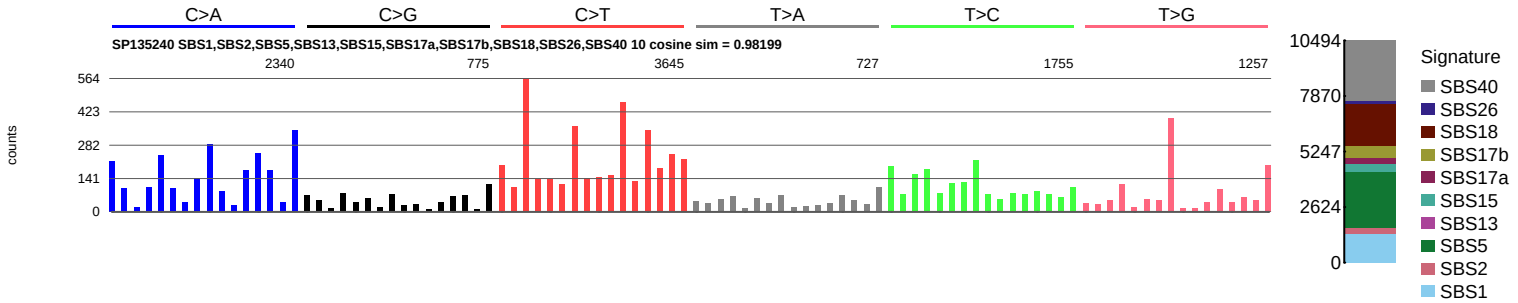
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98199
2277
801
3602
884
1622
1304
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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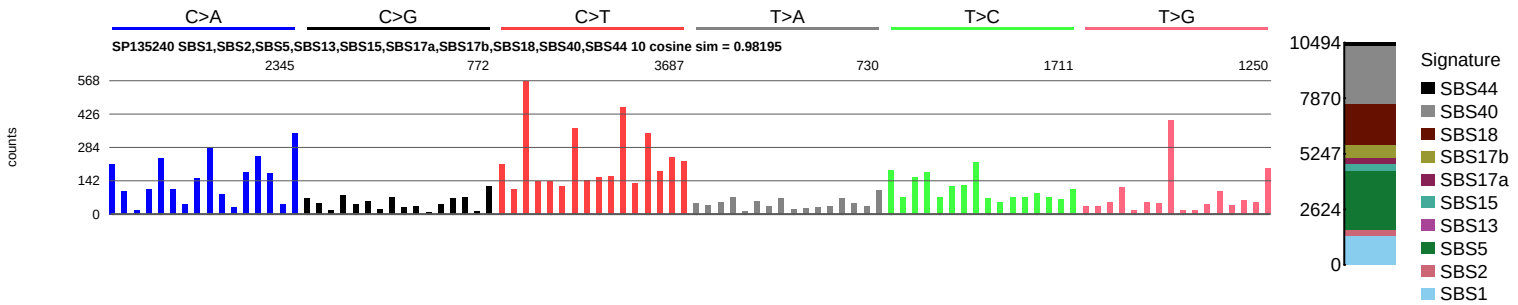
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98199
2340
775
3645
727
1755
1257
counts
564
423
282
141
0
10494
7870
5247
2624
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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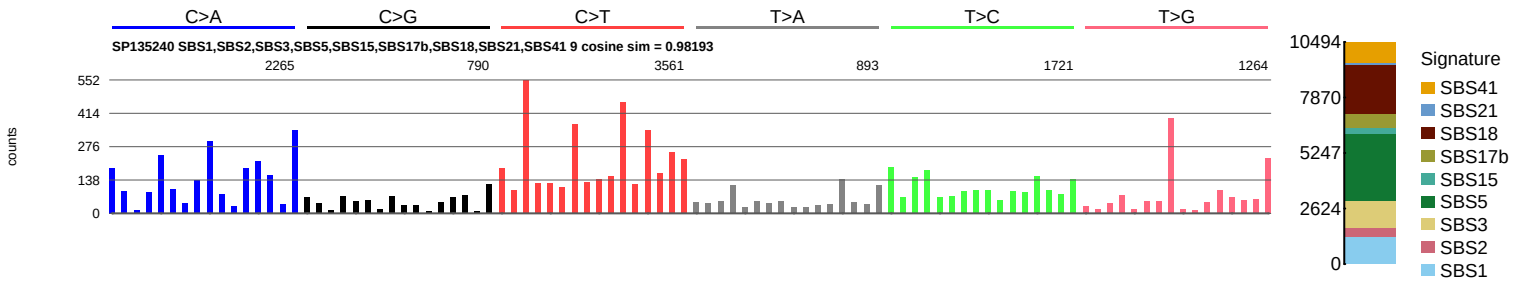
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98195
2345
772
3687
730
1711
1250
counts
568
426
284
142
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98193
2265
790
3561
893
1721
1264
counts
552
414
276
138
0
10494
7870
5247
2624
0
Signature
SBS41
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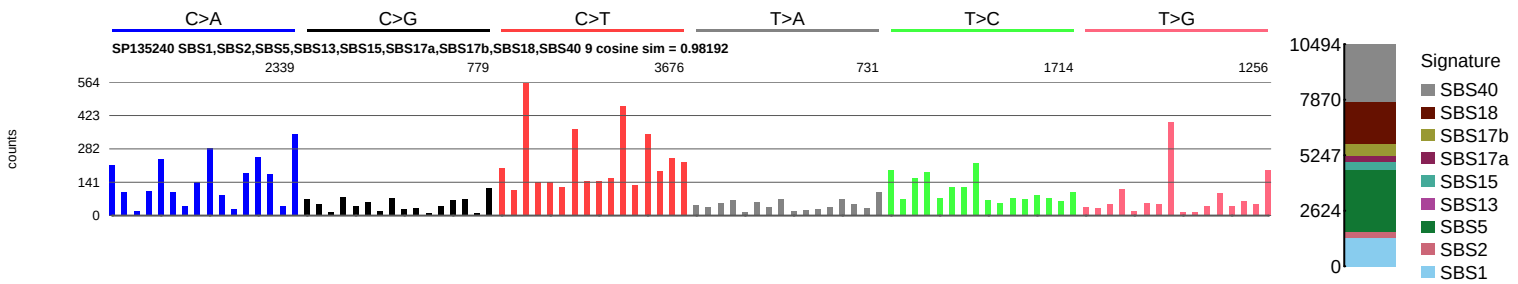
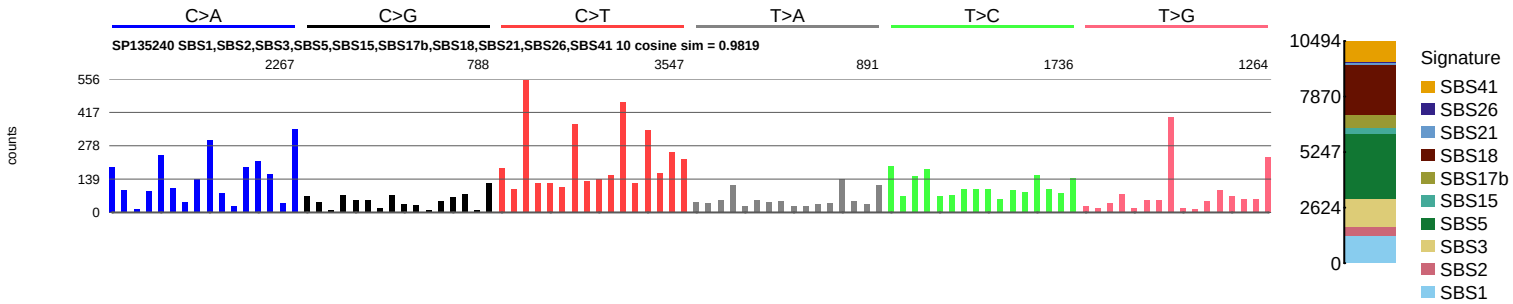
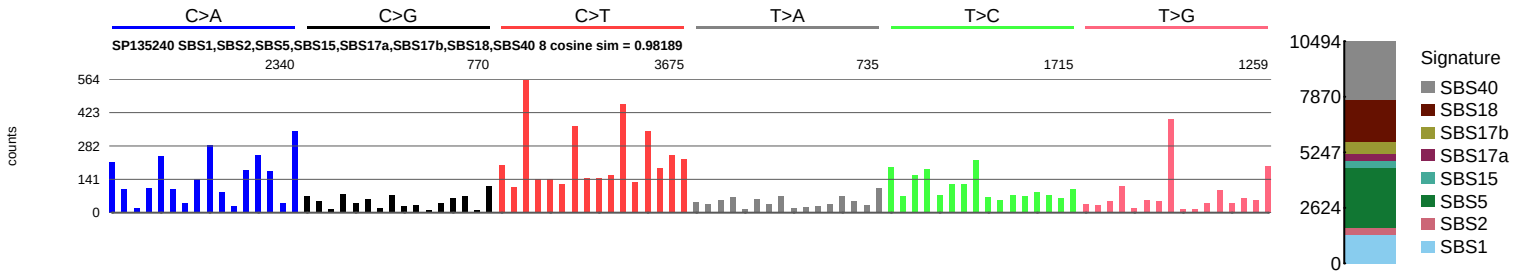
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98192
2339
779
3676
731
1714
1256
counts
564
423
282
141
0
10494
7870
5247
2624
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.9819
2267
788
3547
891
1736
1264
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98189
2340
770
3675
735
1715
1259
counts
564
423
282
141
0
10494
7870
5247
2624
0
Signature
SBS40
SBS18
SBS17b
SBS17a
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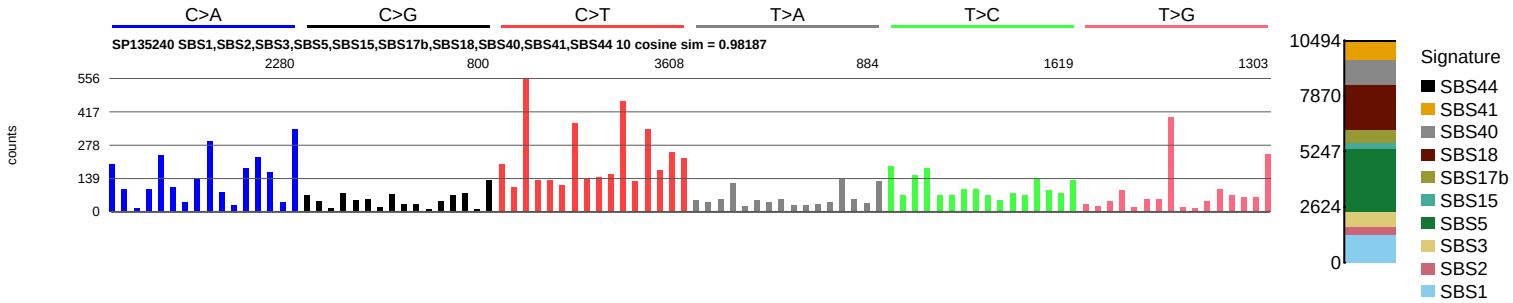
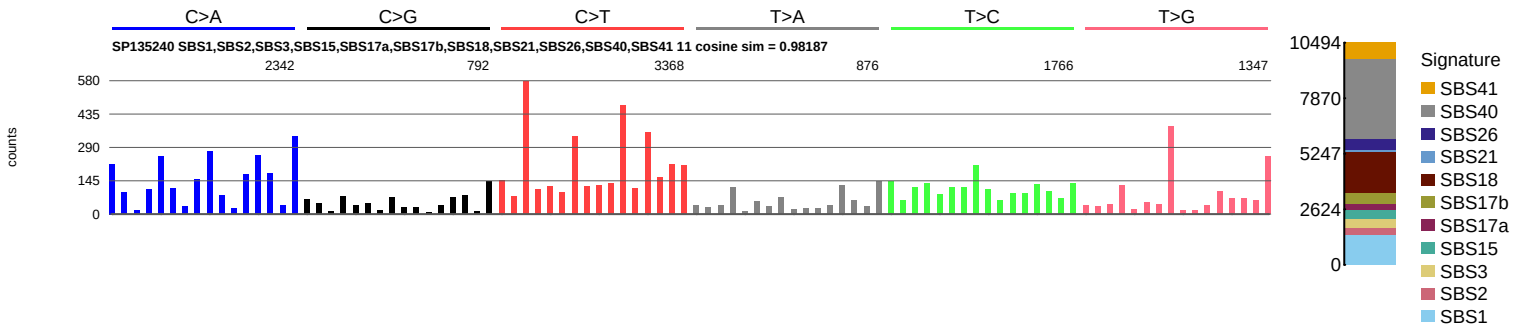
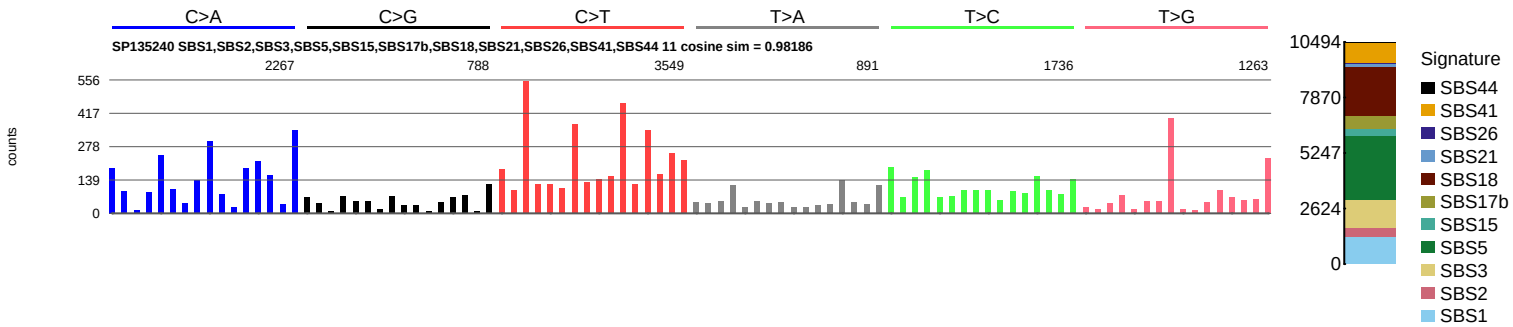
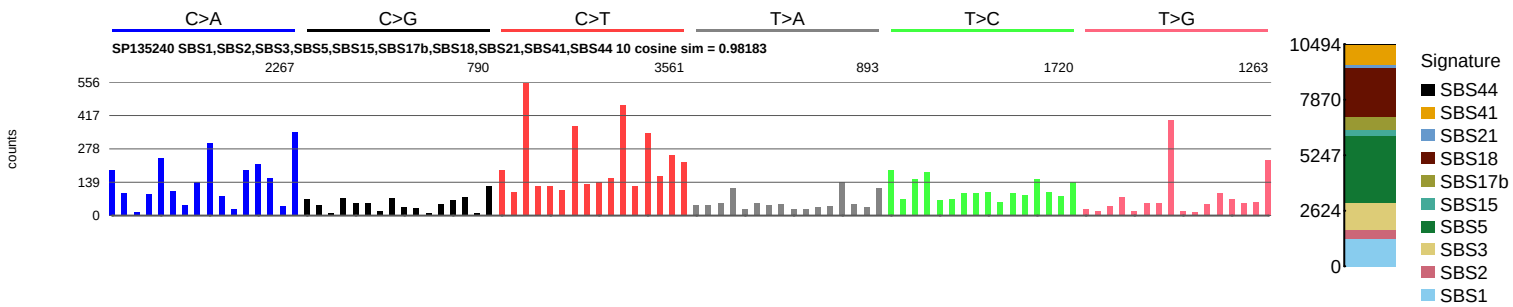
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98187
2280
800
3608
884
1619
1303
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98187
2342
792
3368
876
1766
1347
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98186
2267
788
3549
891
1736
1263
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98183
2267
790
3561
893
1720
1263
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS44
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SBS21
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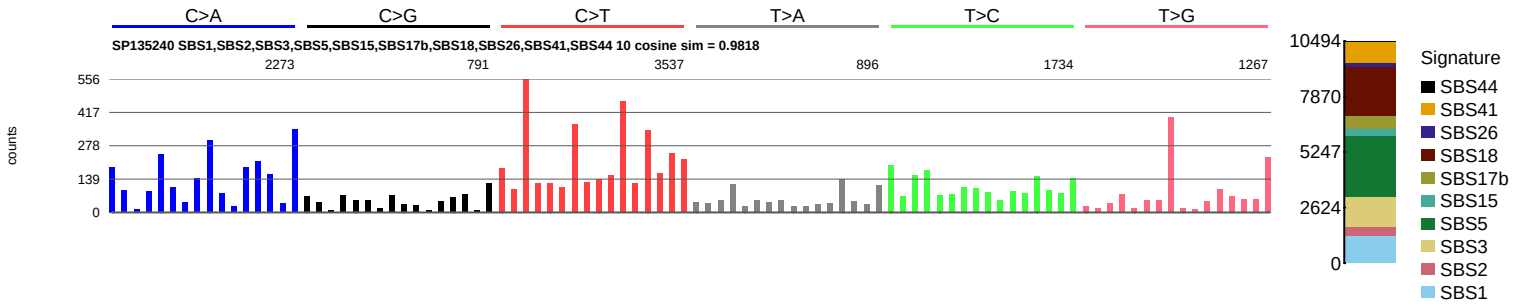
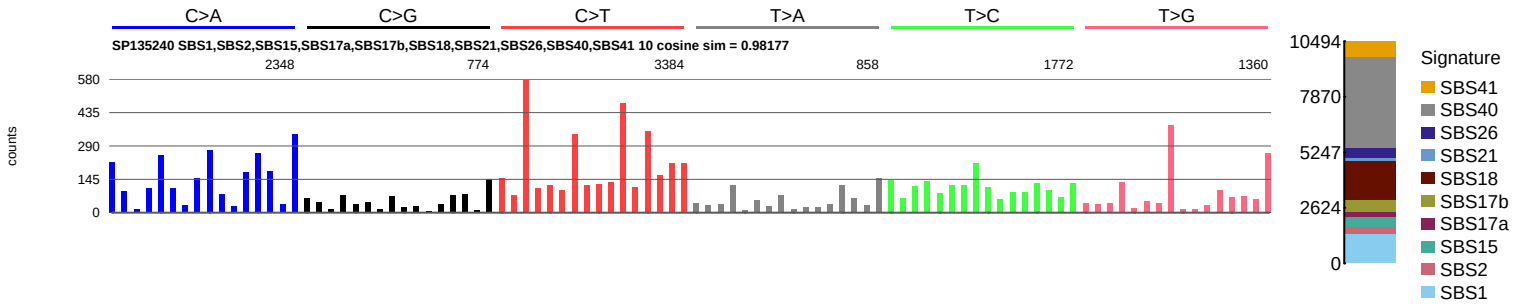
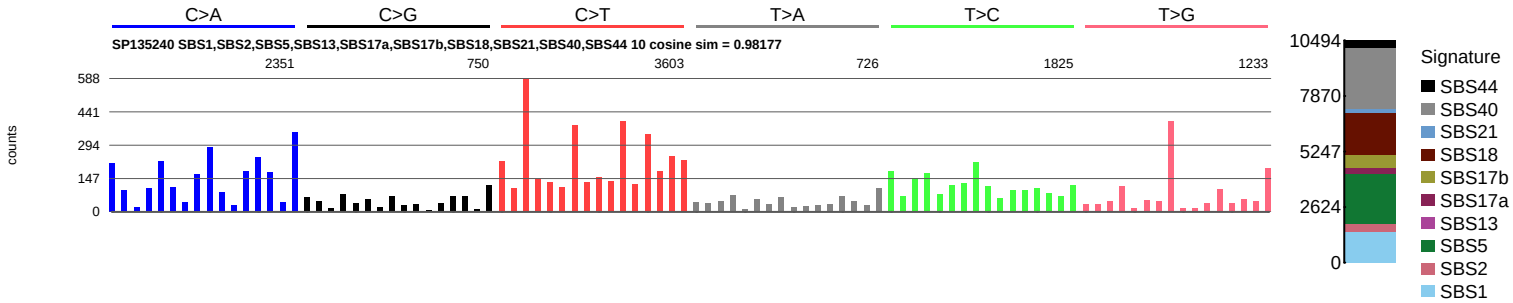
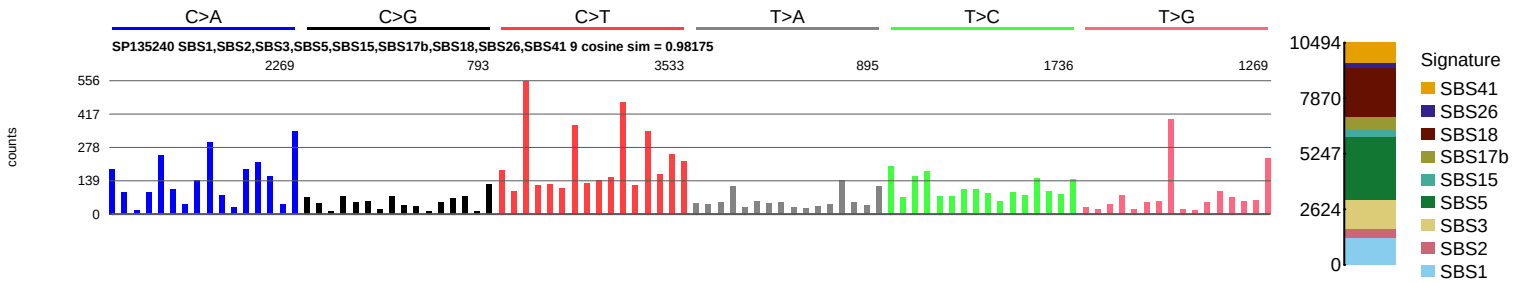
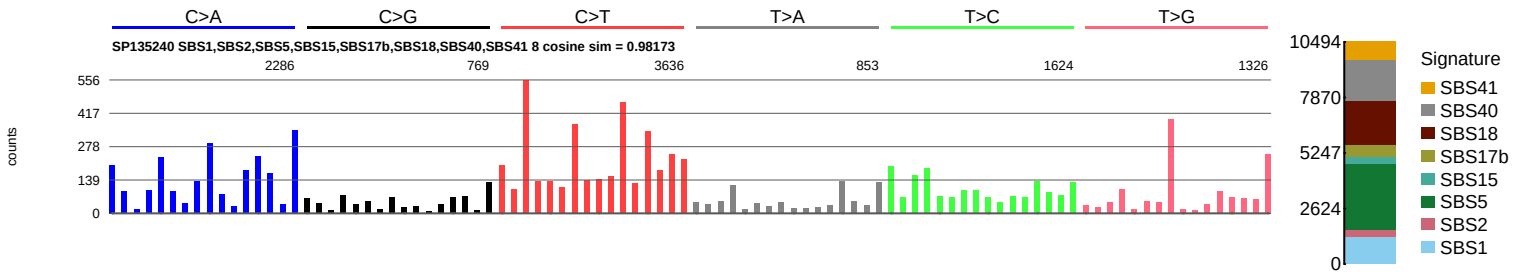
SBS3
SBS2
SBS1

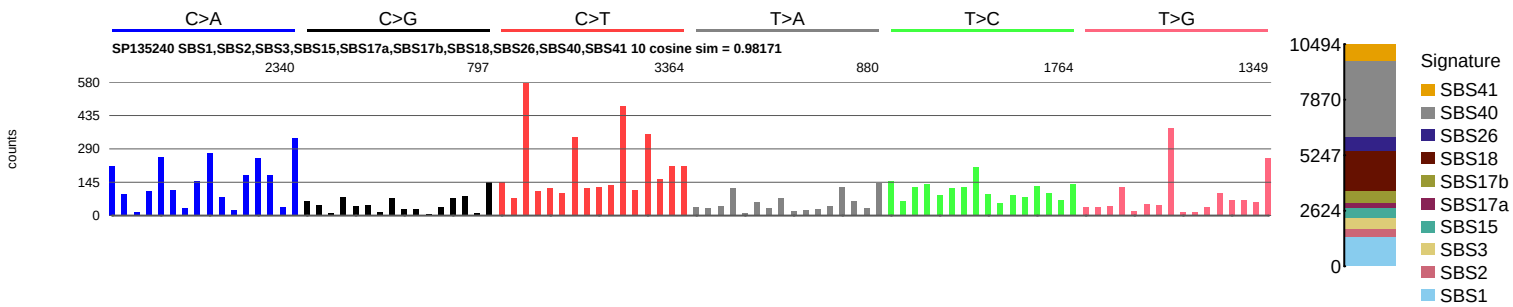

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9818
2273
791
3537
896
1734
1267
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

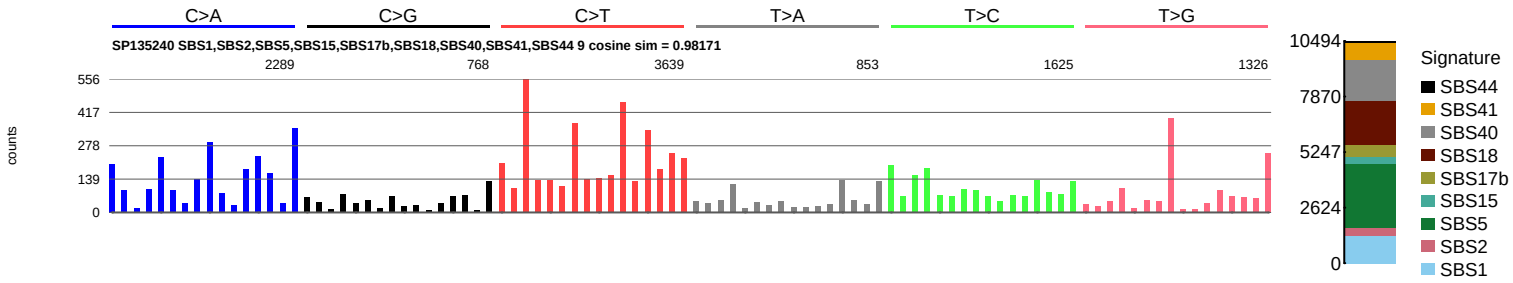

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98177
2348
774
3384
858
1772
1360
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98177
2351
750
3603
726
1825
1233
counts
588
441
294
147
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98175
2269
793
3533
895
1736
1269
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98173
2286
769
3636
853
1624
1326
counts
556
417
278
139
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98171
2340
797
3364
880
1764
1349
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98171
2289
768
3639
853
1625
1326
556
417
278
139
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
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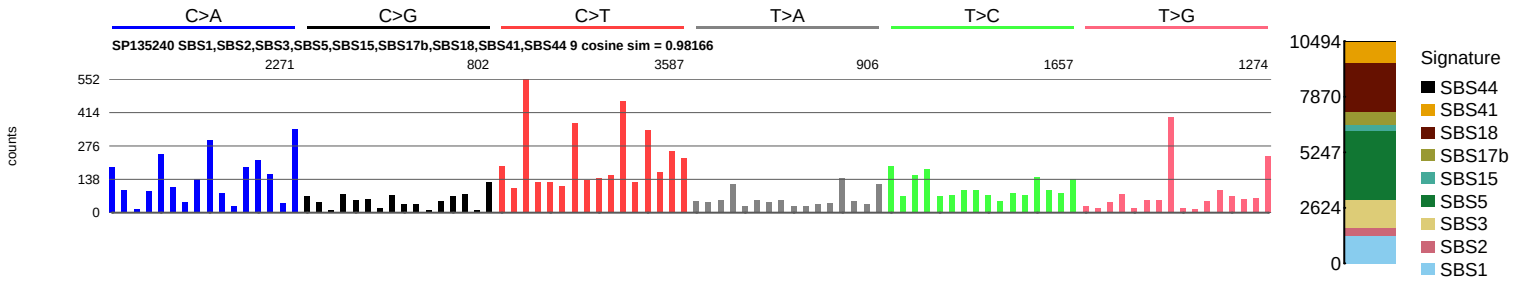
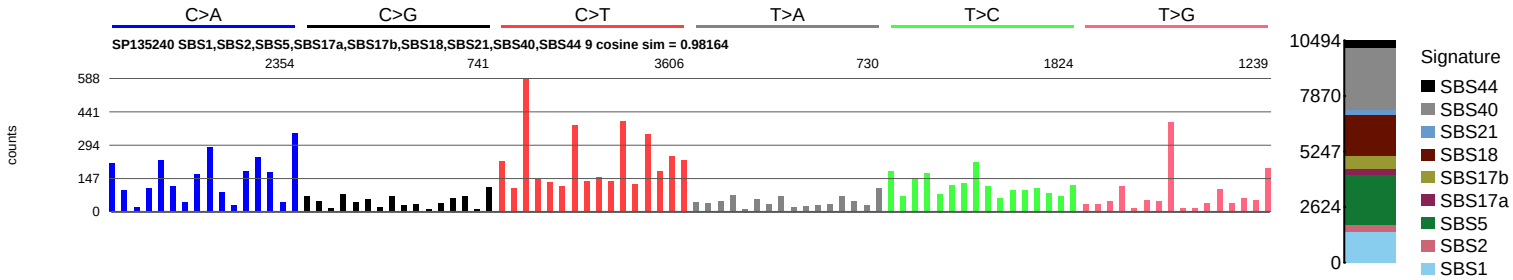
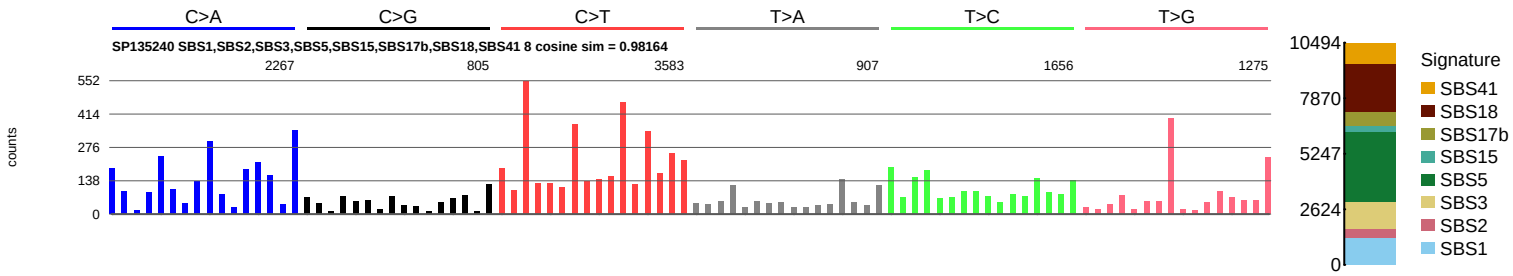
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98166
2271
802
3587
906
1657
1274
552
414
276
138
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98164
2354
741
3606
730
1824
1239
588
441
294
147
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98164
2267
805
3583
907
1656
1275
552
414
276
138
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS18
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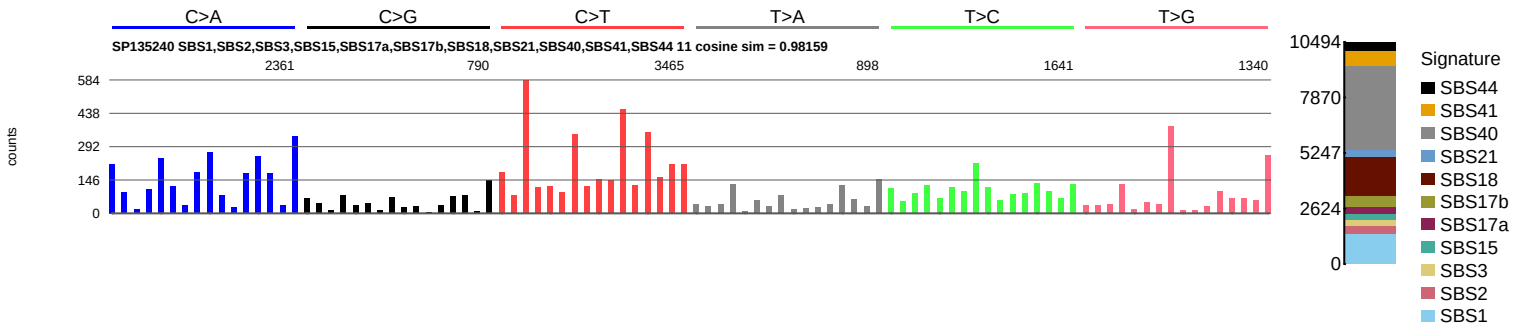
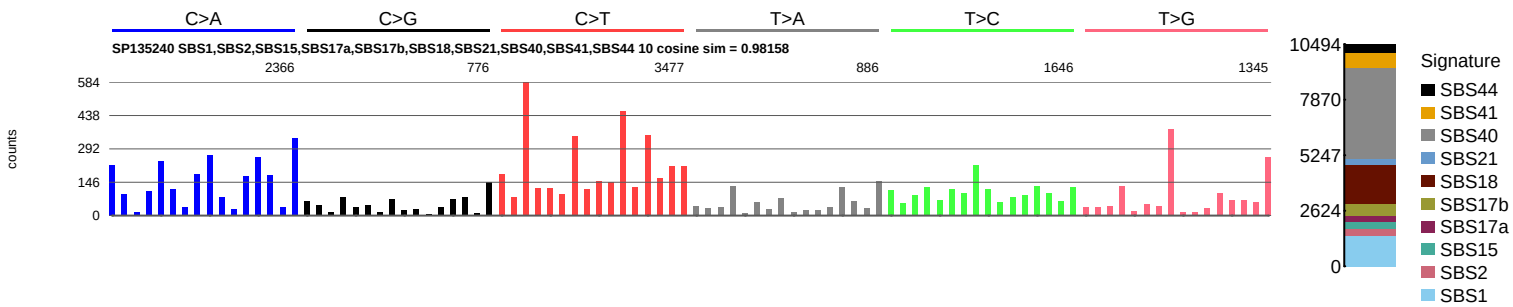
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98159
2361
790
3465
898
1641
1340
584
438
292
146
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98158
2366
776
3477
886
1646
1345
584
438
292
146
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
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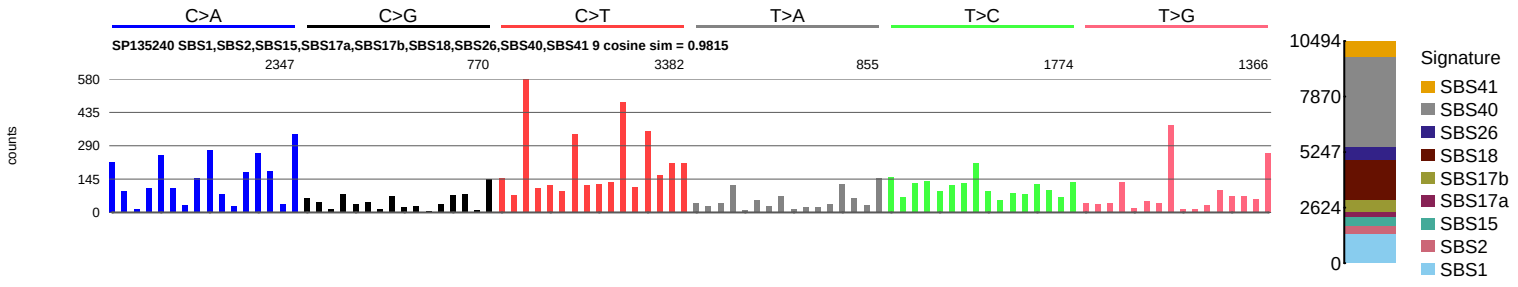
SBS1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SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9815
2347
770
3382
855
1774
1366
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
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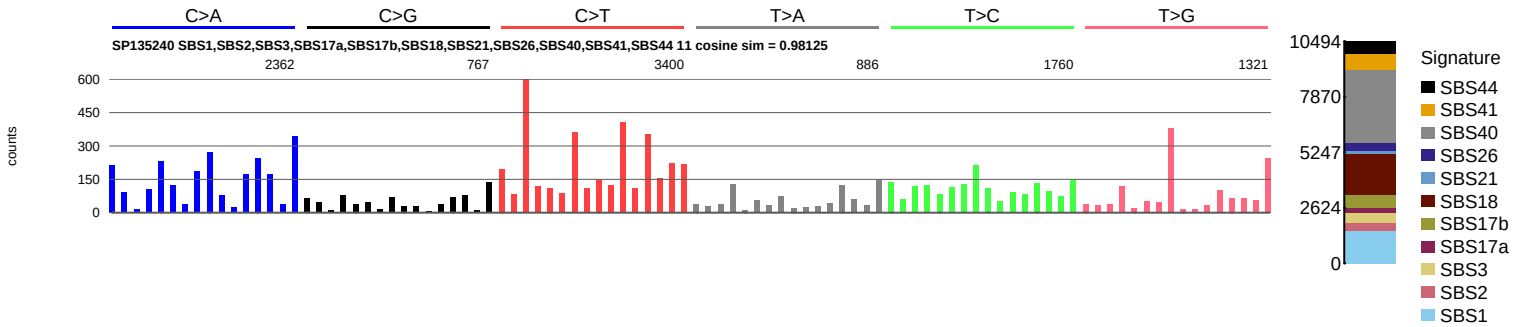
SBS15
SBS2
SBS1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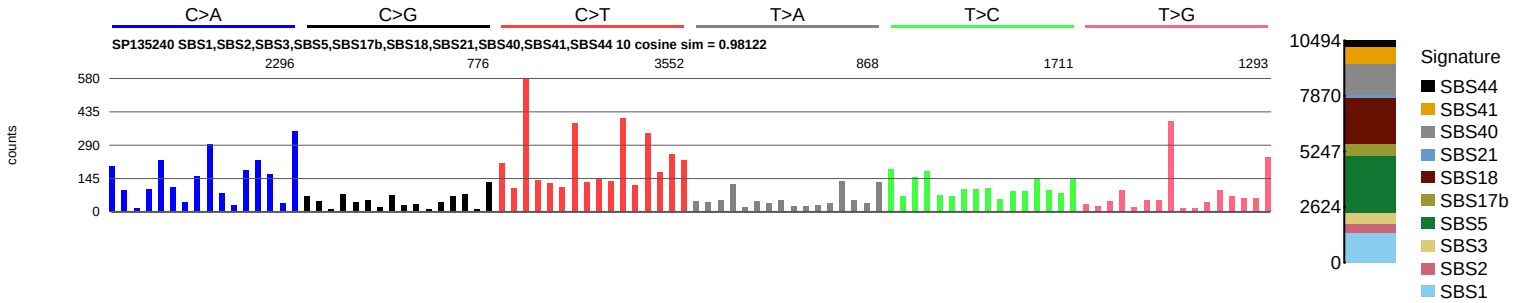
SP135240 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98125
2362
767
3400
886
1760
1321
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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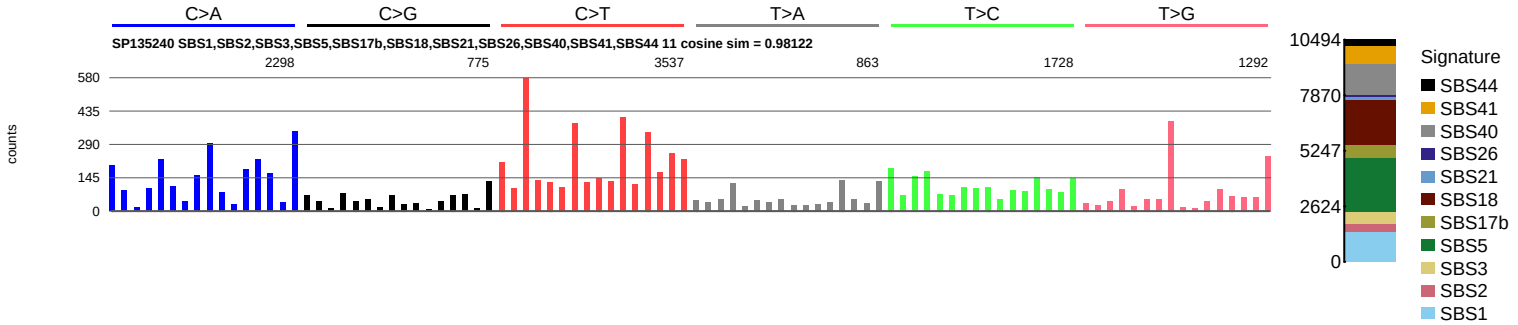
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98122
2296
776
3552
868
1711
1293
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
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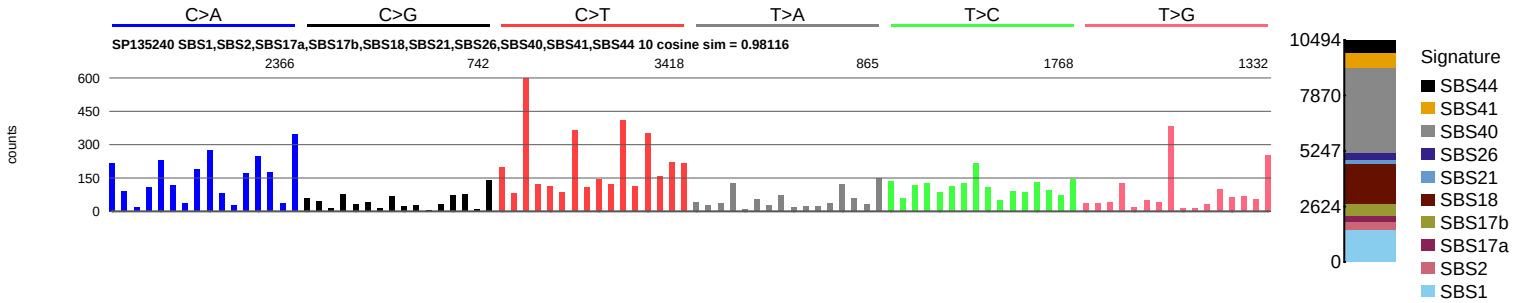
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98122
2298
775
3537
863
1728
1292
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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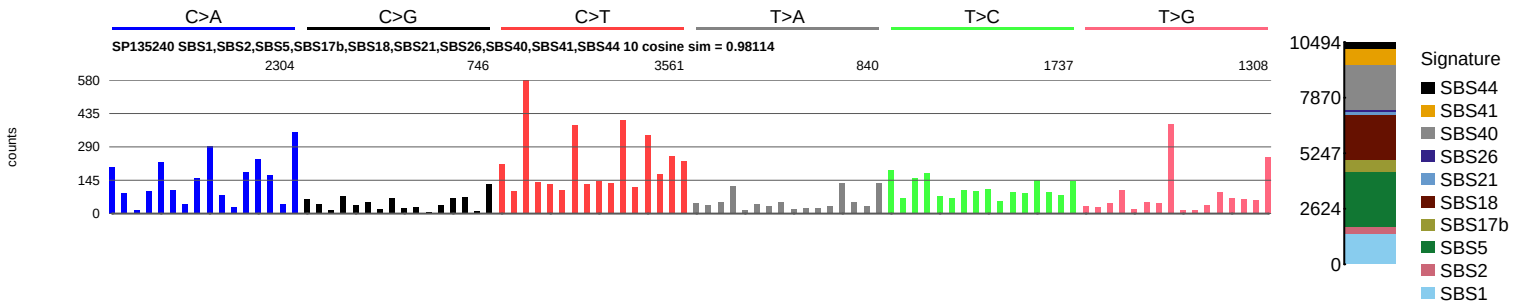
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98116
2366
742
3418
865
1768
1332
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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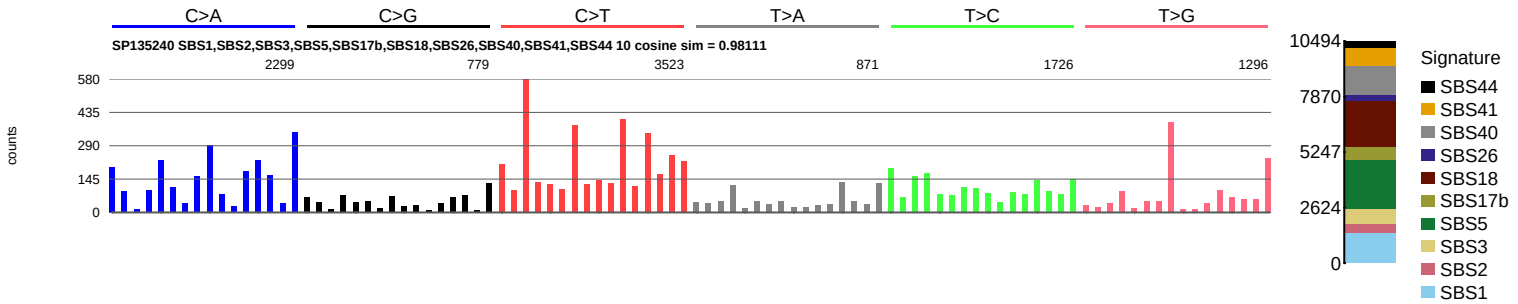
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98114
2304
746
3561
840
1737
1308
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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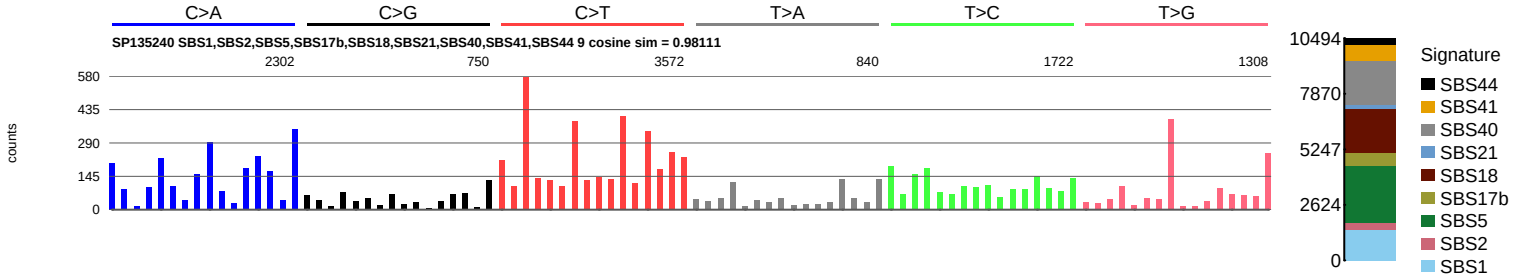
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98111
2299
779
3523
871
1726
1296
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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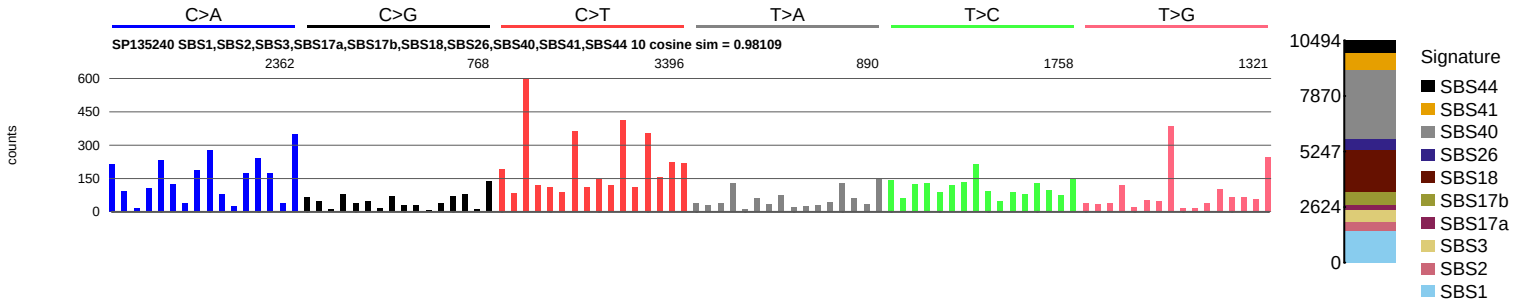
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98111
2302
750
3572
840
1722
1308
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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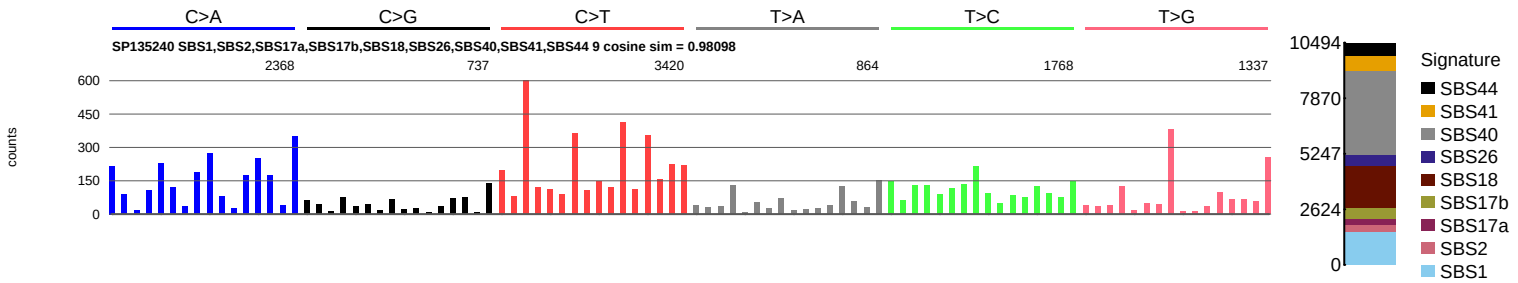
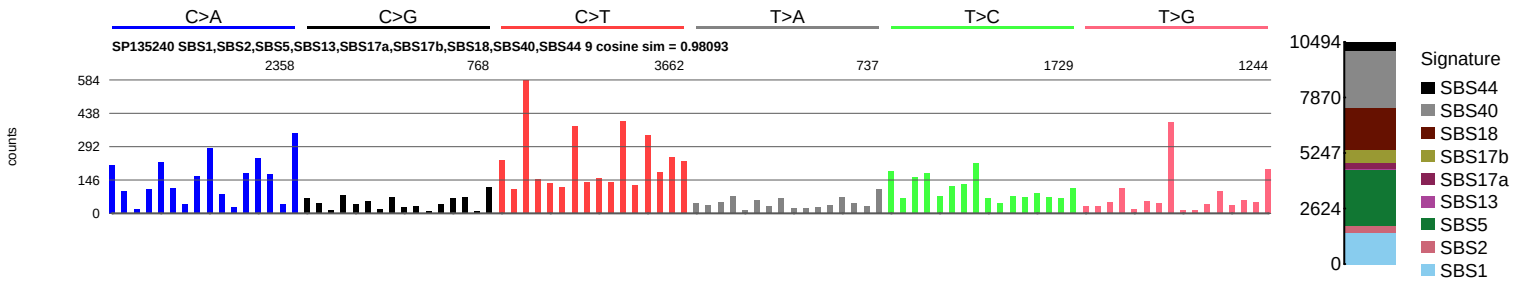
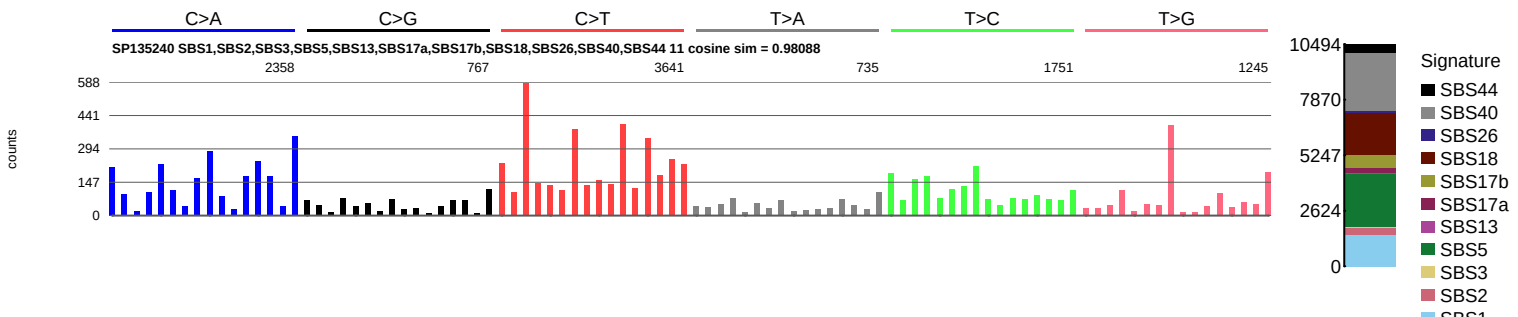
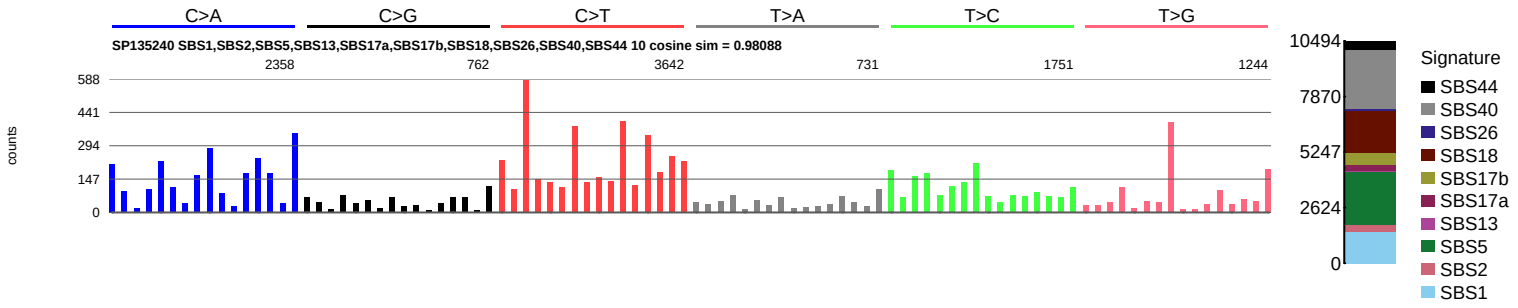
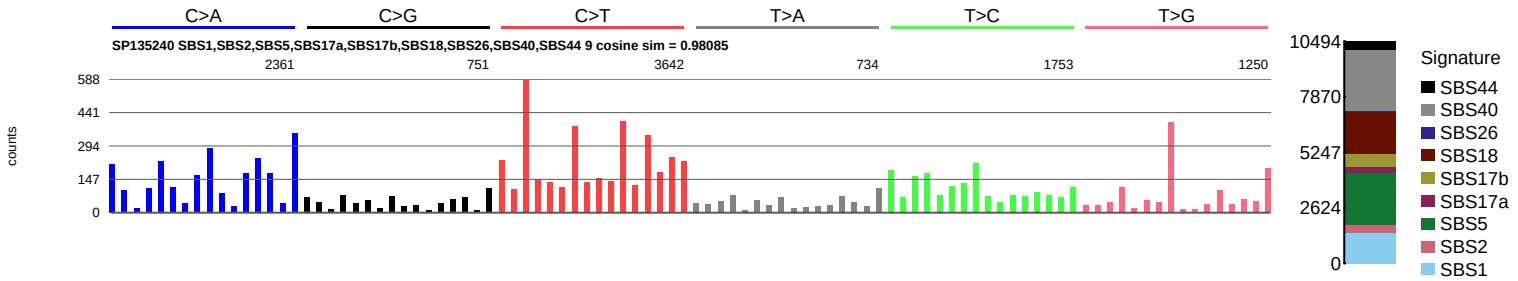
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98109
2362
768
3396
890
1758
1321
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98098
2368
737
3420
864
1768
1337
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98093
2358
768
3662
737
1729
1244
counts
584
438
292
146
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98088
2358
767
3641
735
1751
1245
counts
588
441
294
147
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98088
2358
762
3642
731
1751
1244
counts
588
441
294
147
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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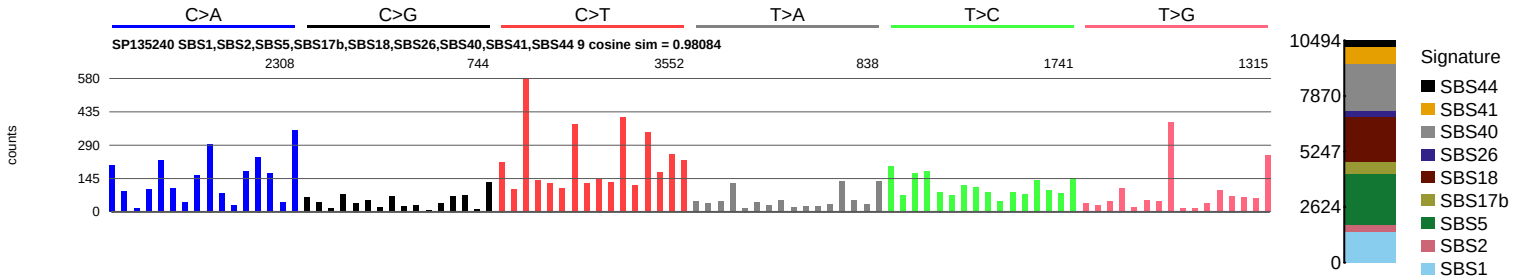
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98085
2361
751
3642
734
1753
1250
counts
588
441
294
147
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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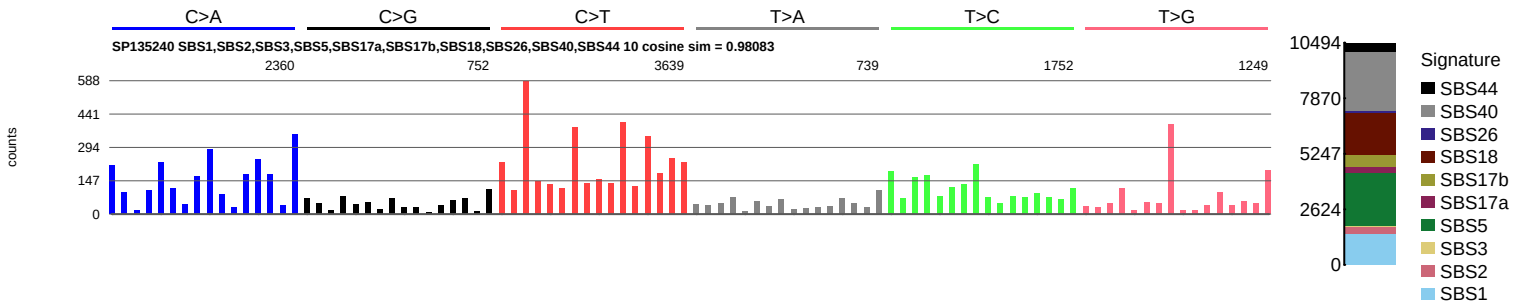
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98084
2308
744
3552
838
1741
1315
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1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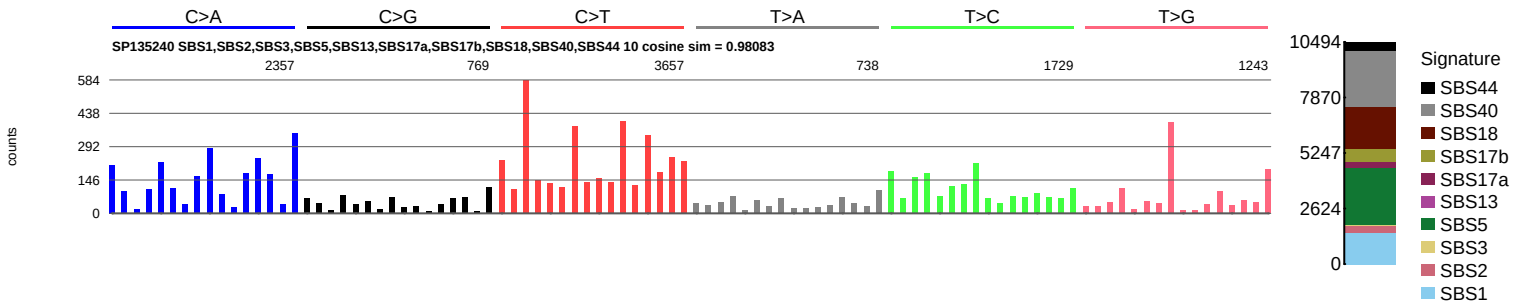
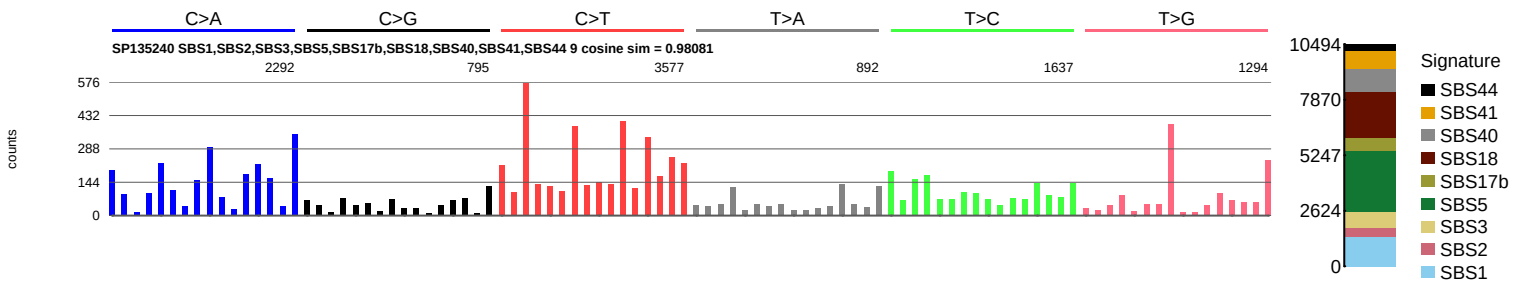
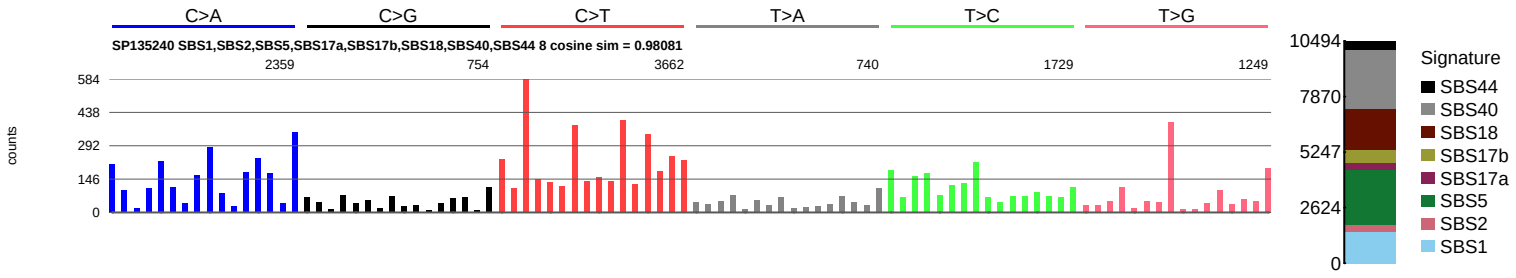
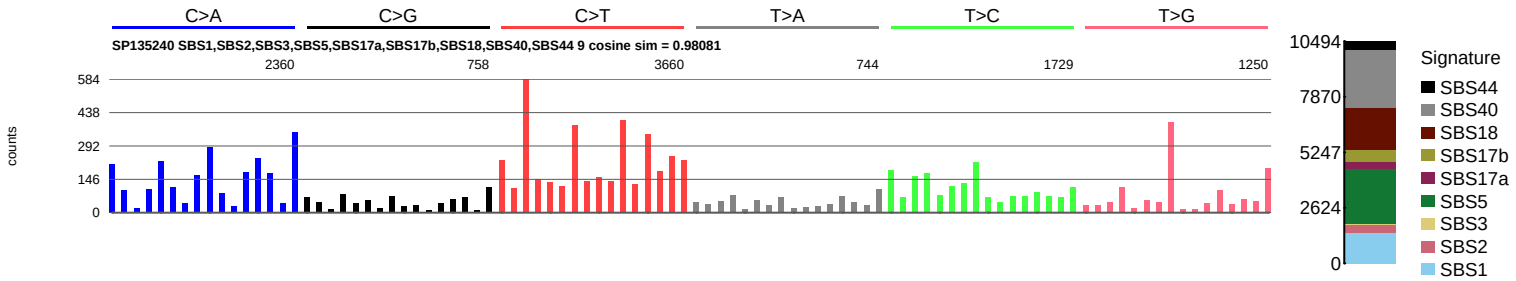
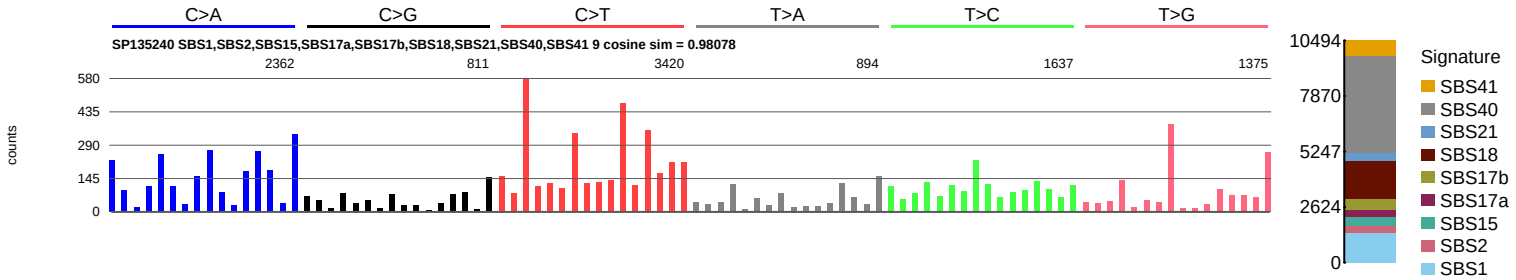
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98083
2360
752
3639
739
1752
1249
counts
588
441
294
147
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

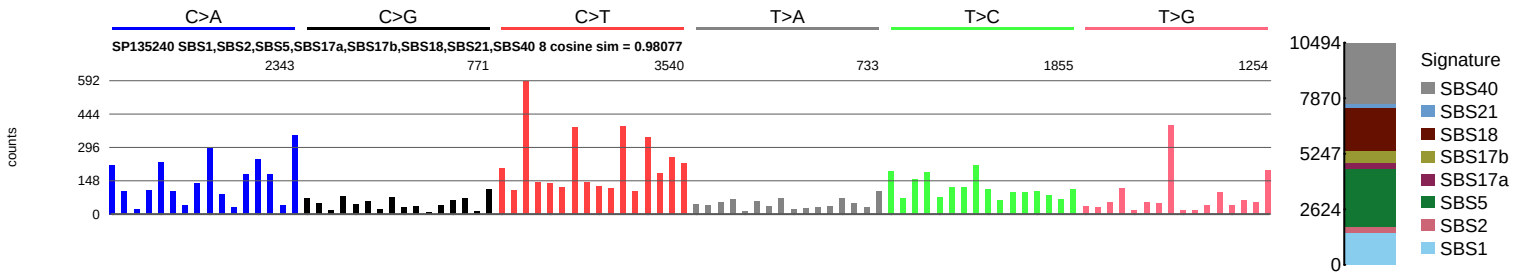

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98083
2357
769
3657
738
1729
1243
counts
584
438
292
146
0
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

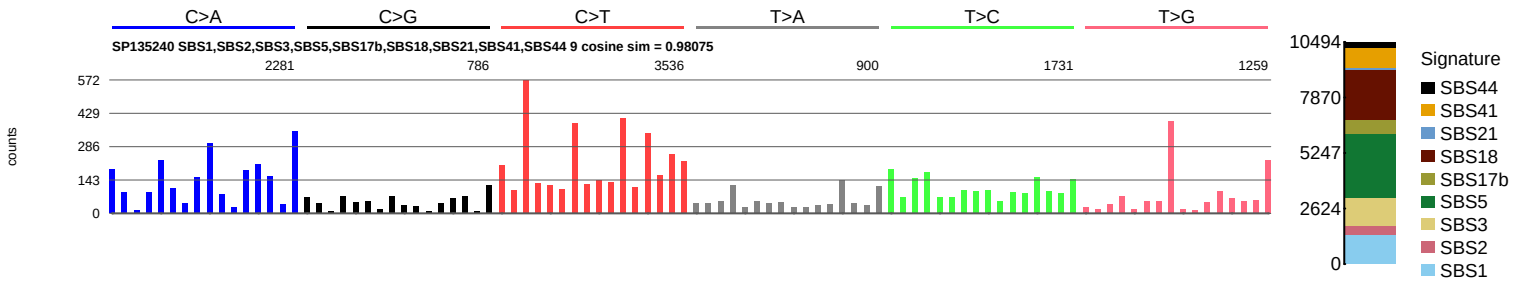

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98081
2292
795
3577
892
1637
1294
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

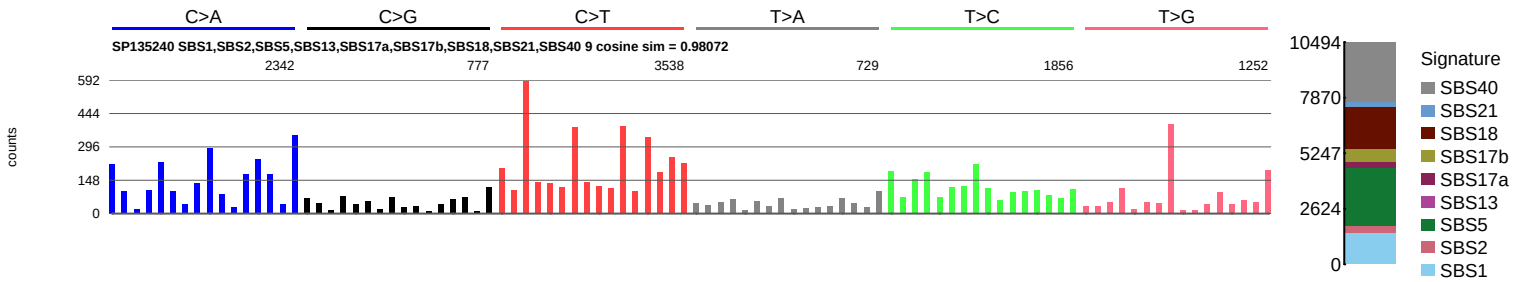

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98081
2359
754
3662
740
1729
1249
584
438
292
146
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

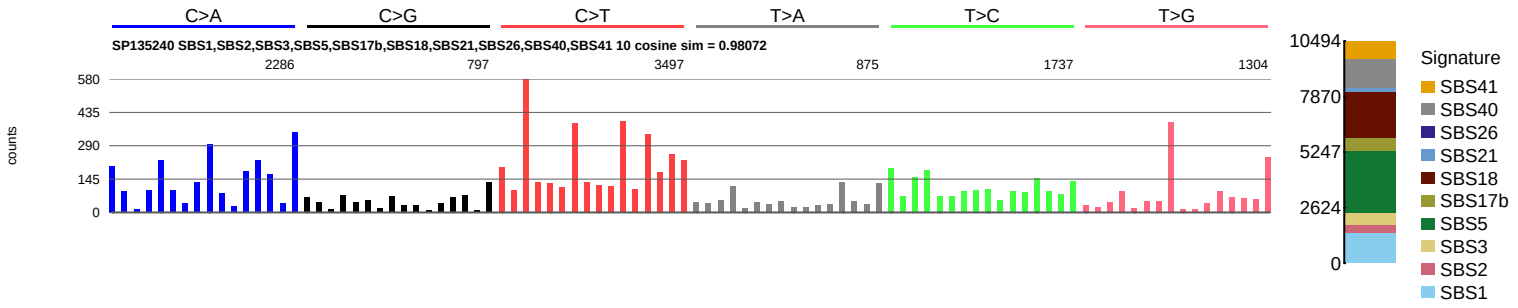

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98081
2360
758
3660
744
1729
1250
584
438
292
146
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98078
2362
811
3420
894
1637
1375
580
435
290
145
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98077
2343
771
3540
733
1855
1254
592
444
296
148
0
counts
10494
7870
5247
2624
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98075
2281
786
3536
900
1731
1259
572
429
286
143
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98072
2342
777
3538
729
1856
1252
592
444
296
148
0
counts
10494
7870
5247
2624
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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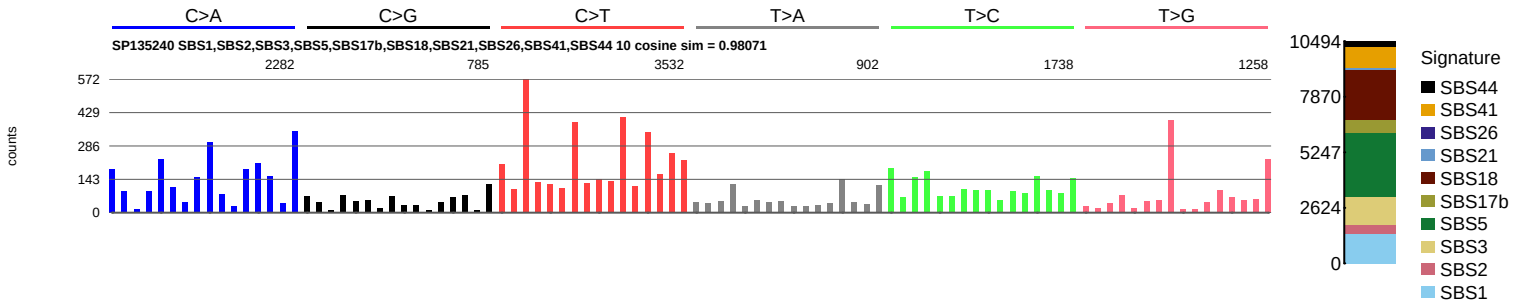
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98072
2286
797
3497
875
1737
1304
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>T
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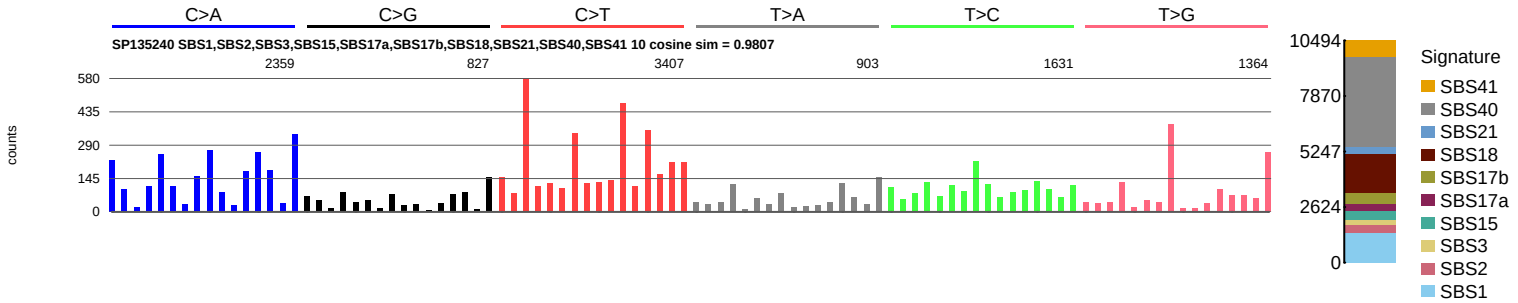
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98071
2282
785
3532
902
1738
1258
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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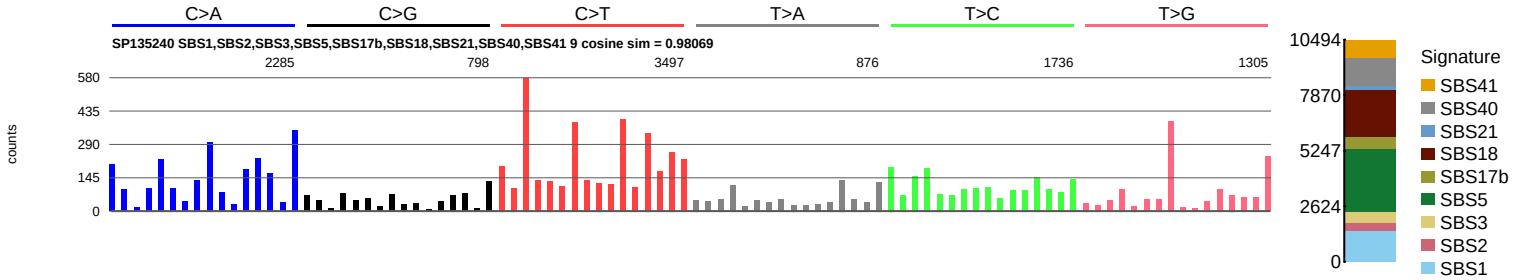
SP135240 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.9807
2359
827
3407
903
1631
1364
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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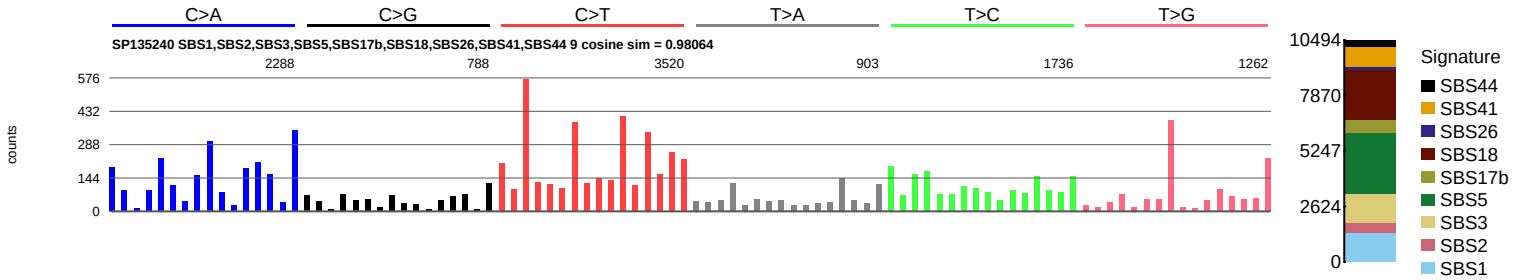
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98069
2285
798
3497
876
1736
1305
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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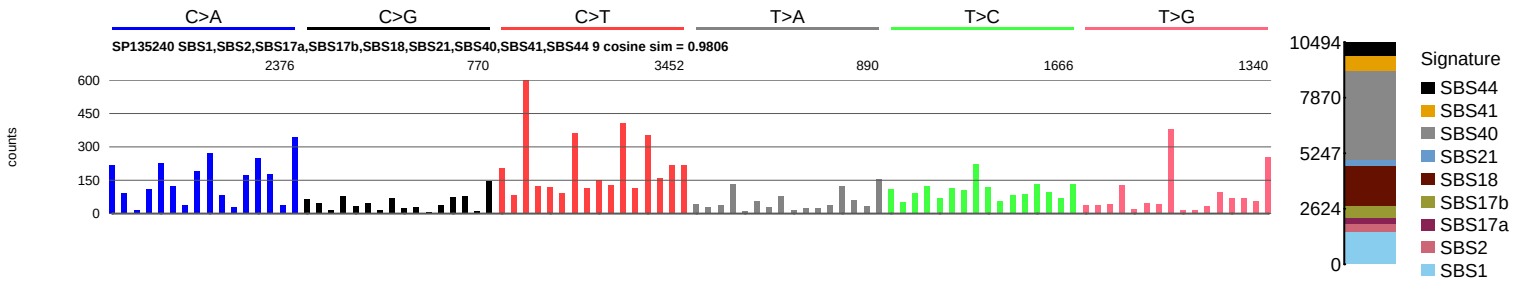
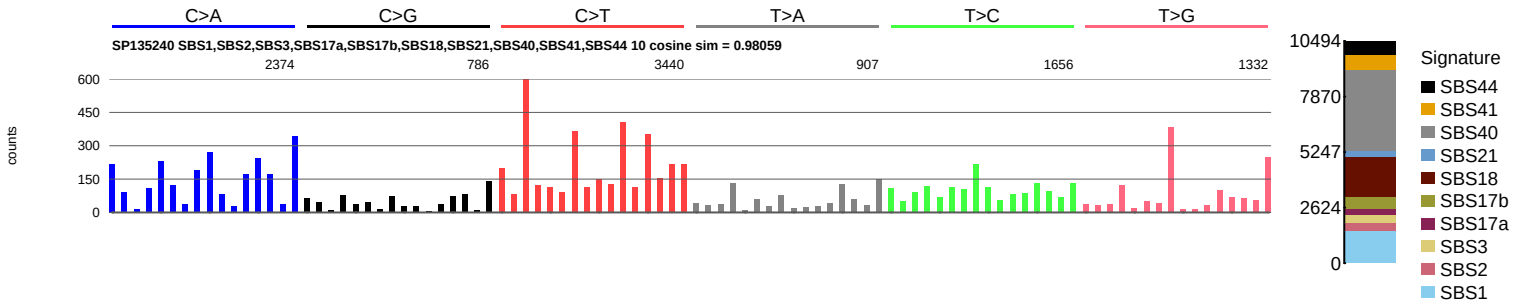
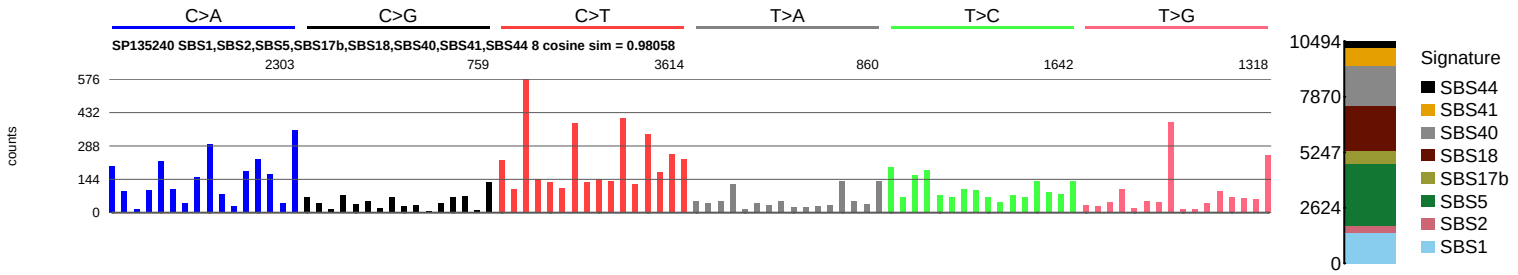
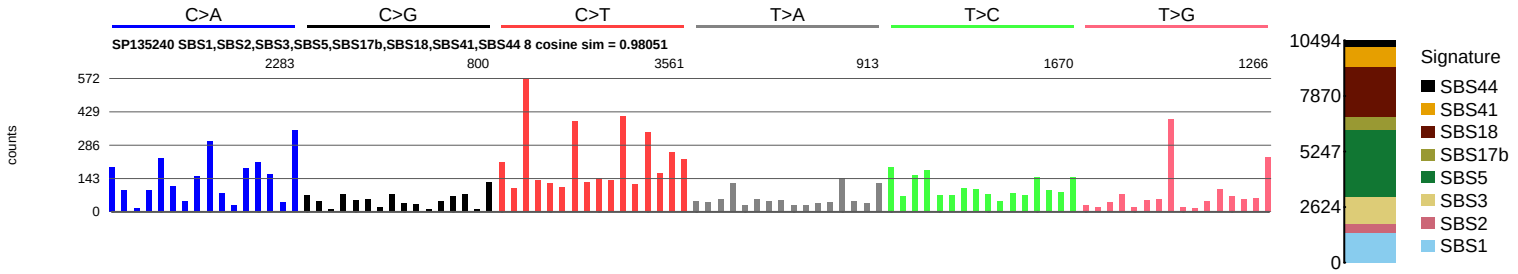
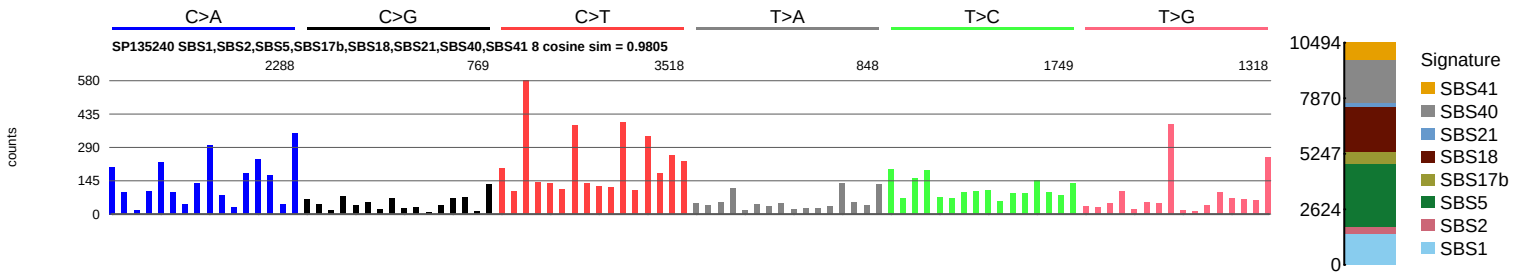
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98064
2288
788
3520
903
1736
1262
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

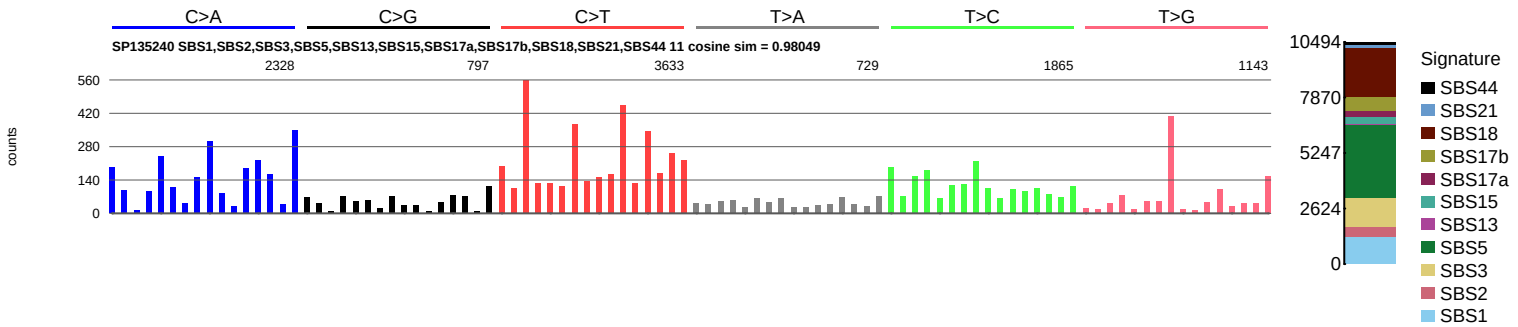

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9806
2376
770
3452
890
1666
1340
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

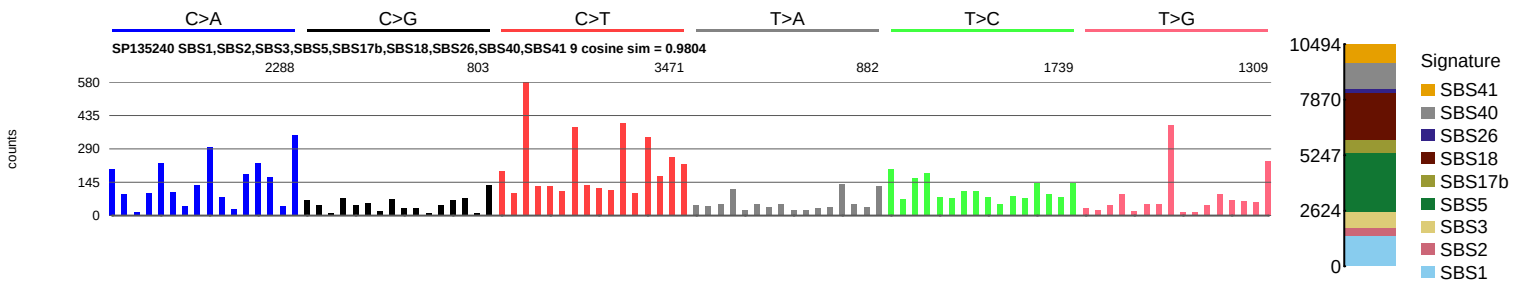

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98059
2374
786
3440
907
1656
1332
counts
600
450
300
150
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

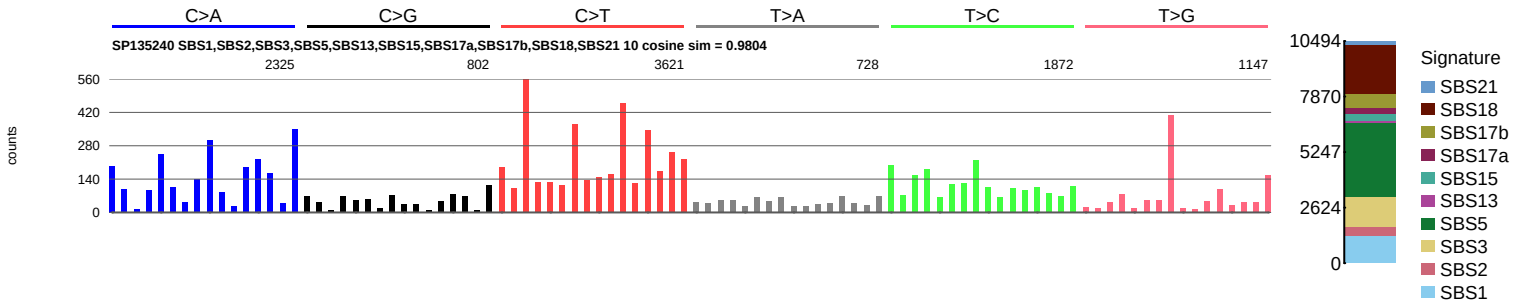

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98058
2303
759
3614
860
1642
1318
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

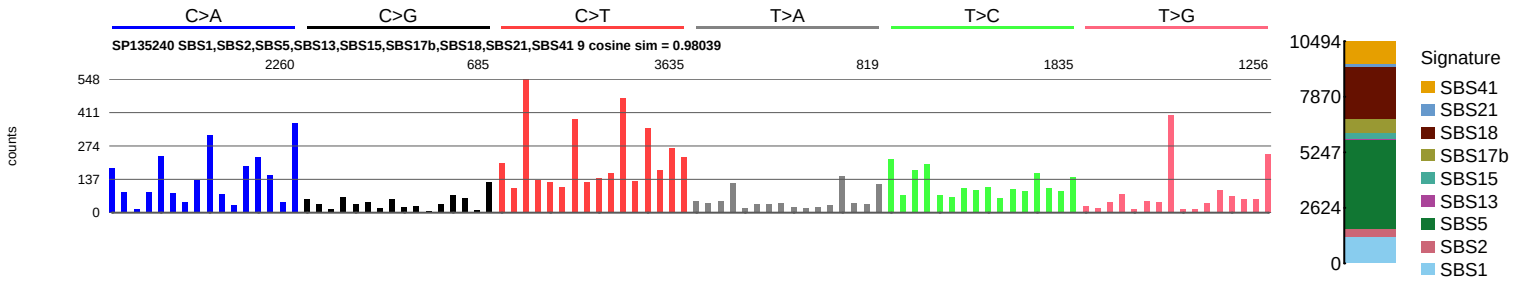

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98051
2283
800
3561
913
1670
1266
counts
572
429
286
143
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

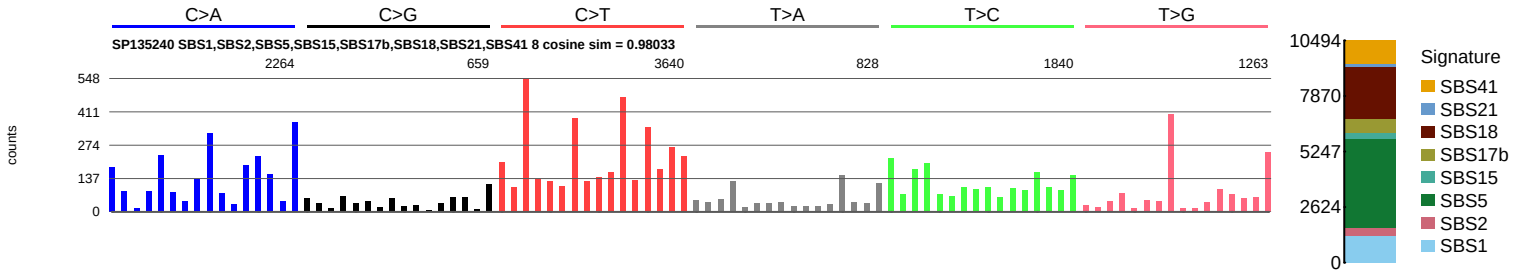

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9805
2288
769
3518
848
1749
1318
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98049
2328
797
3633
729
1865
1143
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.9804
2288
803
3471
882
1739
1309
counts
580
435
290
145
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.9804
2325
802
3621
728
1872
1147
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98039
2260
685
3635
819
1835
1256
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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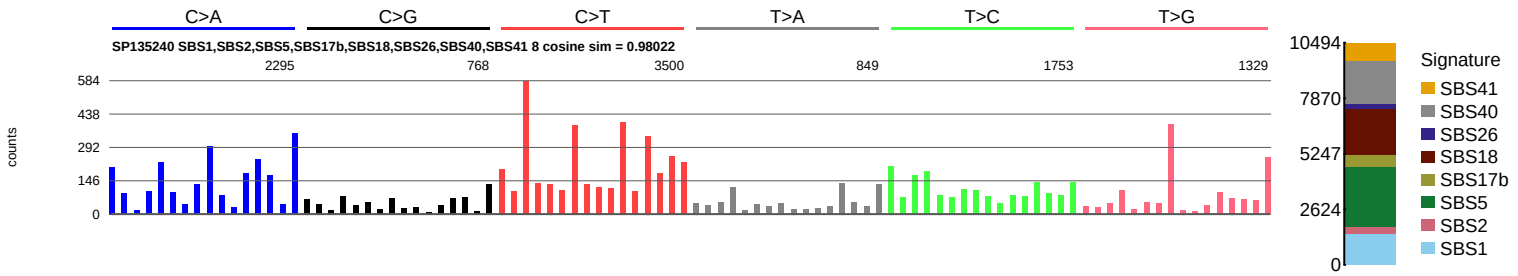
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98033
2264
659
3640
828
1840
1263
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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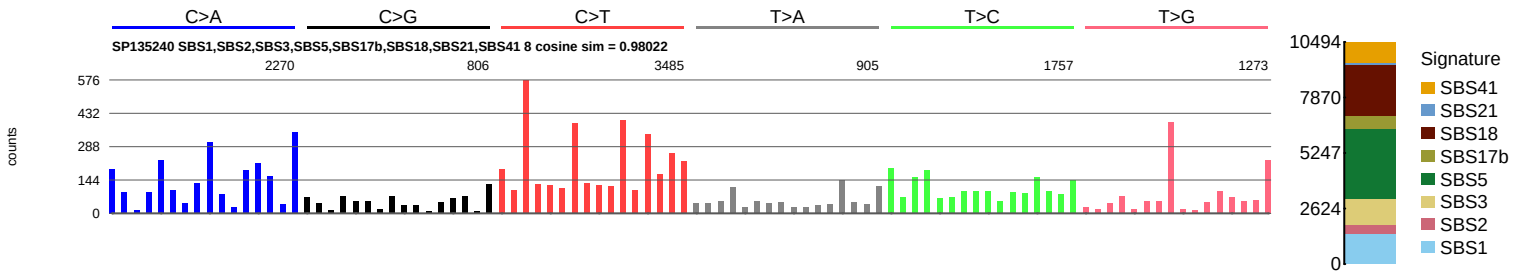
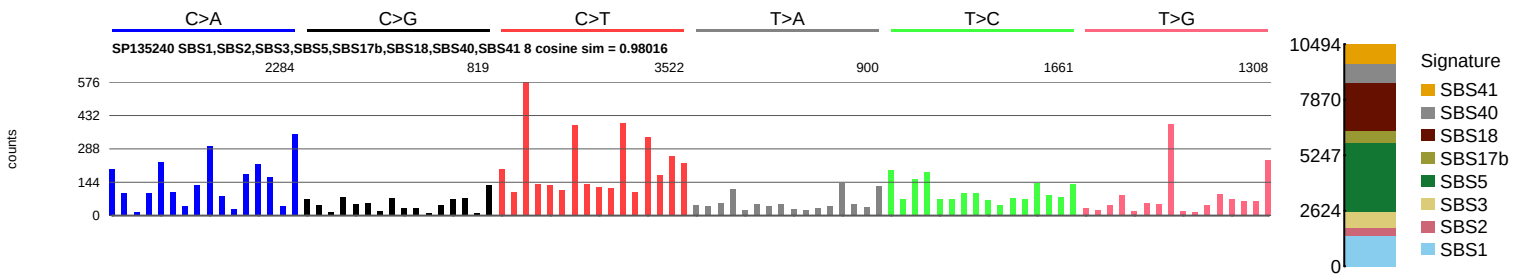
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98022
2295
768
3500
849
1753
1329
counts
584
438
292
146
0
10494
7870
5247
2624
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98022
2270
806
3485
905
1757
1273
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98016
2284
819
3522
900
1661
1308
counts
576
432
288
144
0
10494
7870
5247
2624
0
Signature
SBS41
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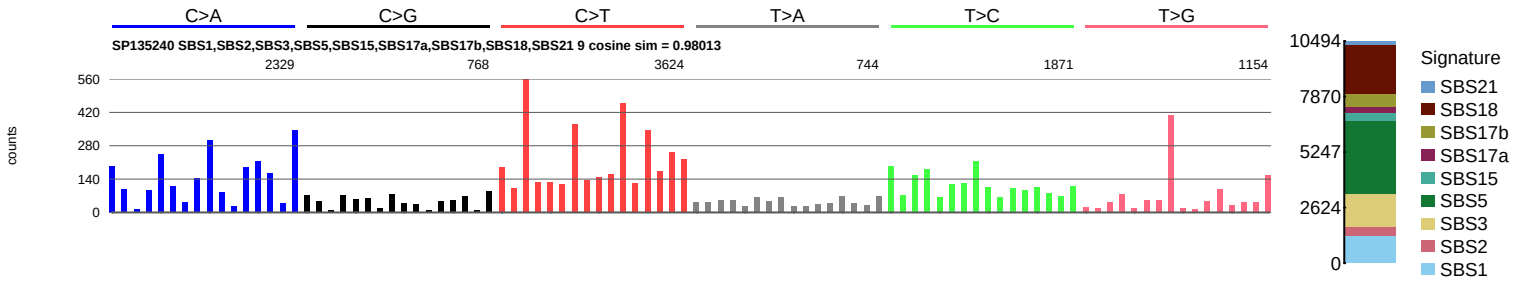
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98013
2329
768
3624
744
1871
1154
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS21
SBS18
SBS17b
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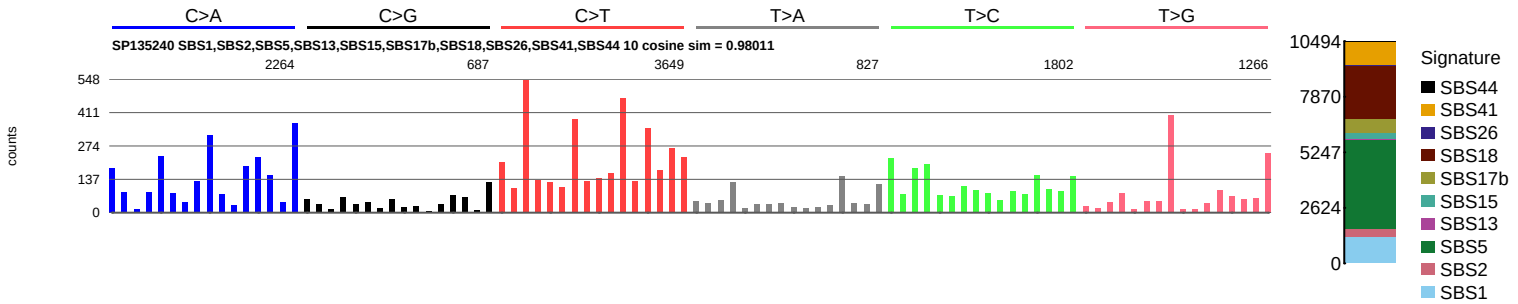
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98011
2264
687
3649
827
1802
1266
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS44
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SBS26
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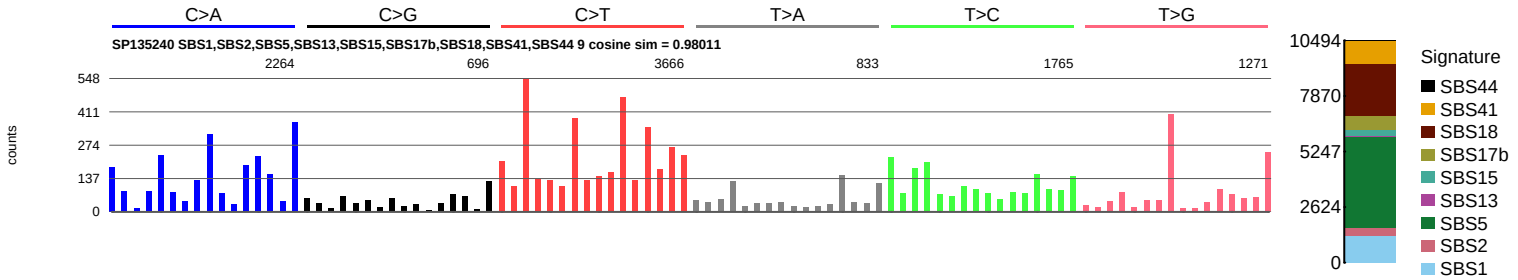
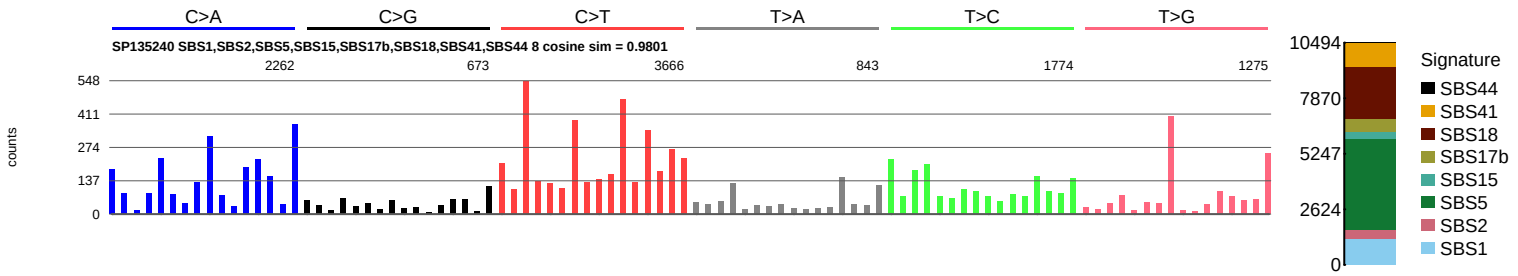
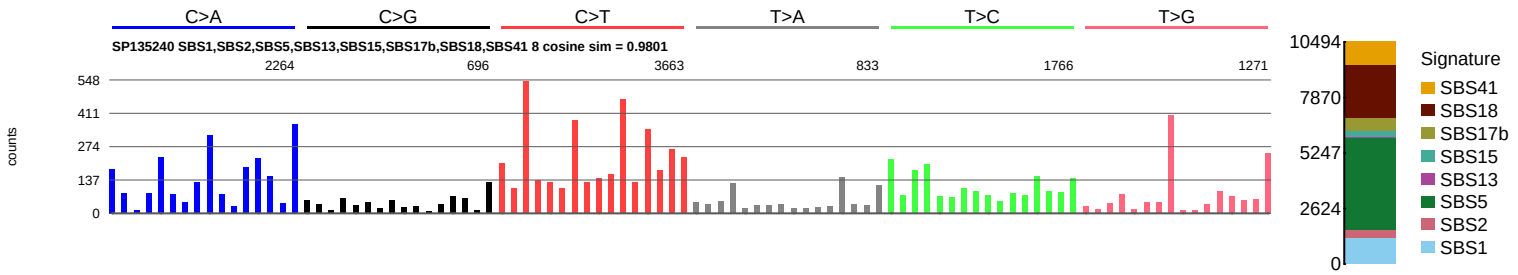
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98011
2264
696
3666
833
1765
1271
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9801
2262
673
3666
843
1774
1275
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.9801
2264
696
3663
833
1766
1271
counts
548
411
274
137
0
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
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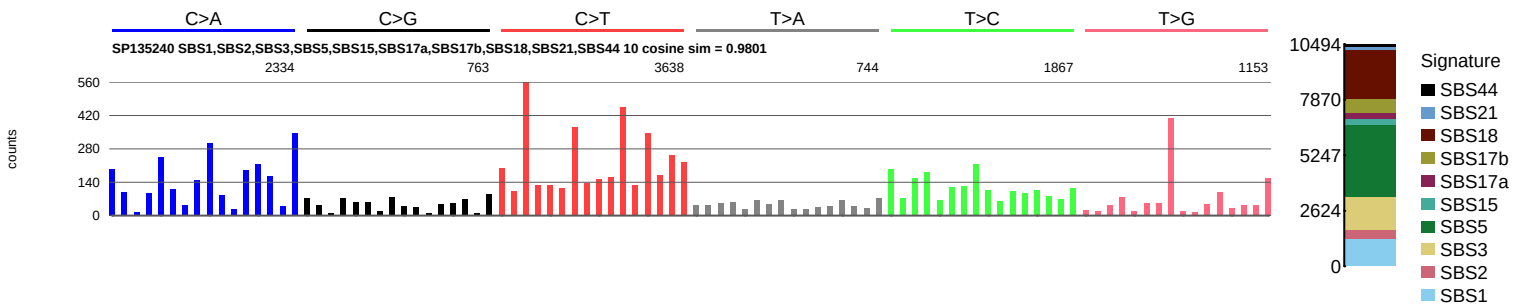
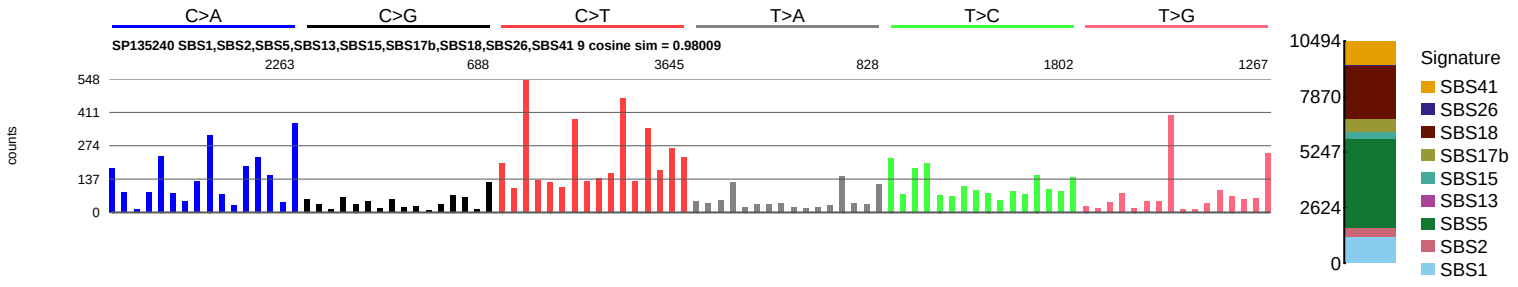
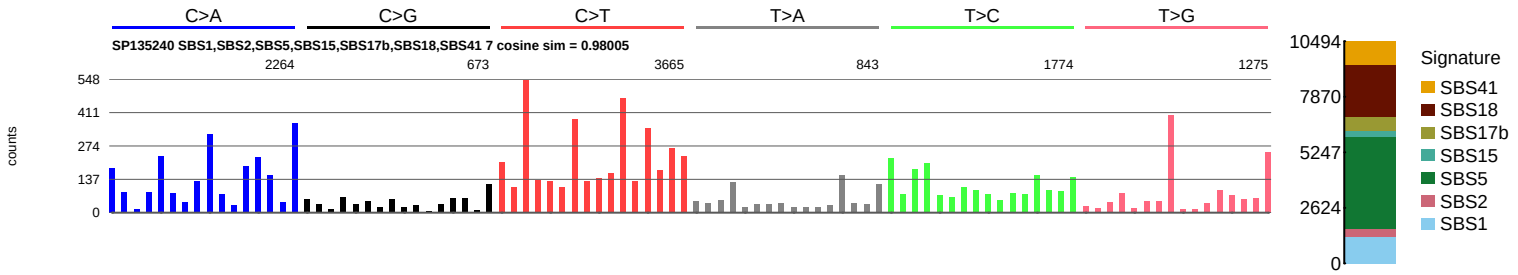
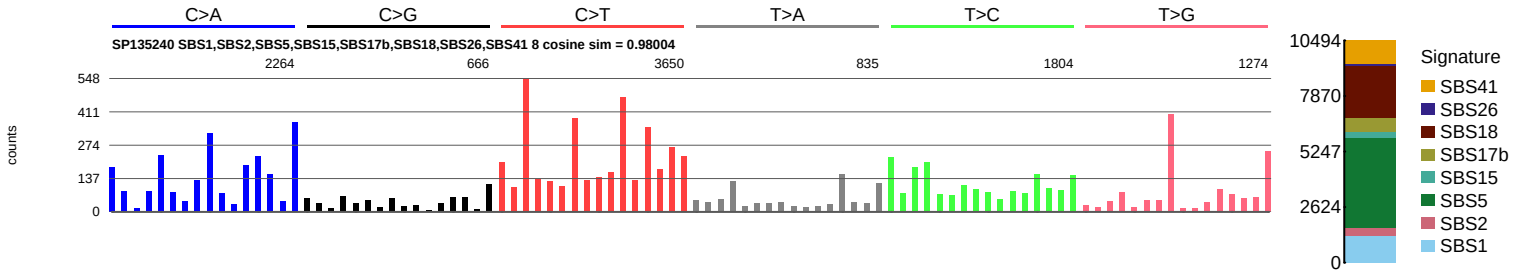
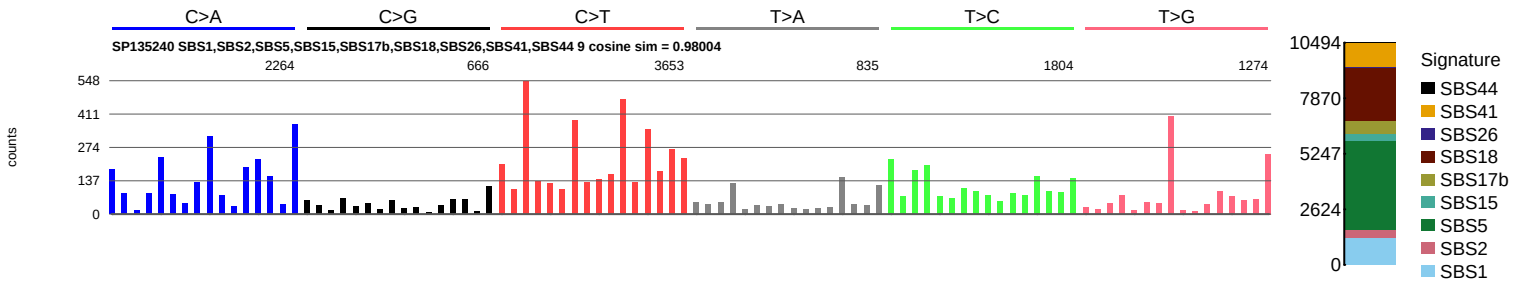
SBS5
SBS2
SBS1

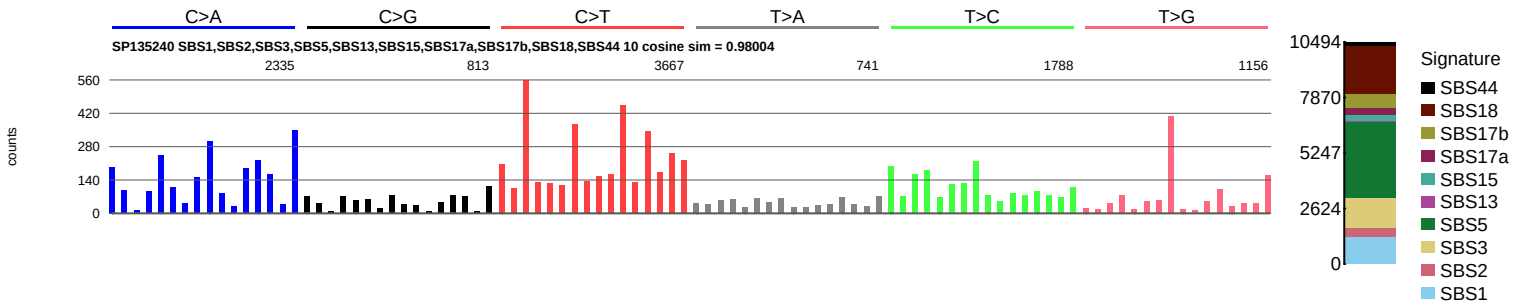

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.9801
2334
763
3638
744
1867
1153
counts
560
420
280
140
0
10494
7870
5247
2624
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

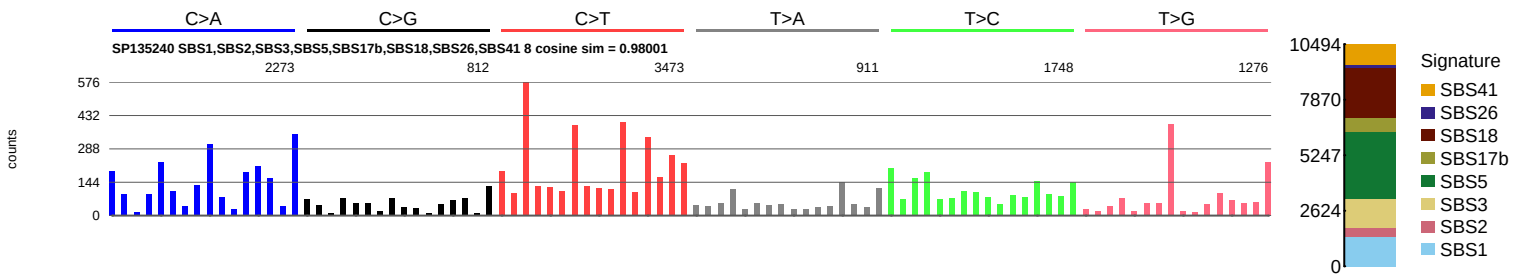

C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98009
2263
688
3645
828
1802
1267
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98005
2264
673
3665
843
1774
1275
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98004
2264
666
3650
835
1804
1274
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98004
2264
666
3653
835
1804
1274
548
411
274
137
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98004
2335
813
3667
741
1788
1156
560
420
280
140
0
counts
10494
7870
5247
2624
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135240 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98001
2273
812
3473
911
1748
1276
576
432
288
144
0
counts
10494
7870
5247
2624
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1